

**STATUS, VARIABILITY AND MANAGEMENT OF DRY ROOT
ROT OF CHICKPEA CAUSED BY *Rhizoctonia bataticola*
[(Taub.) Butler] WITH CHANGING CLIMATE**

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**DEPARTMENT OF PLANT PATHOLOGY
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DECEMBER, 2017

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ROT OF CHICKPEA CAUSED BY *Rhizoctonia bataticola*
[(Taub.) Butler] WITH CHANGING CLIMATE**

*Thesis submitted to the
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In partial fulfillment of the requirements for the
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In

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By

DEEPA

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CERTIFICATE

This is to certify that the thesis entitled “**STATUS, VARIABILITY AND MANAGEMENT OF DRY ROOT ROT OF CHICKPEA CAUSED BY *Rhizoctonia bataticola* [(Taub.) Butler] WITH CHANGING CLIMATE**” submitted by **Ms. DEEPA**, for the degree of **Doctor of Philosophy** in **PLANT PATHOLOGY** to the University of Agricultural Sciences, Raichur is a record of research work done by her during the period of her study in this university under my guidance and the thesis has not previously formed the basis for the award of any degree, diploma, associateship, fellowship or other similar titles.

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AFFECTIONATELY DEDICATED
TO
MY PARENTS AND INDIAN
FARMERS

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With Regardful memories.....

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LIST OF ABBREVIATIONS

DRR	:	Dry root rot
Anon.	:	Anonymous
ARS	:	Agricultural research station
@	:	At the rate of
⁰ C	:	Degree Celsius
CD	:	Critical Difference
cfu	:	Colony forming unit
cm	:	Centimeter/s
CTAB	:	Cetyltrimethyl ammonium bromide
CRD	:	Complete randomized design
DAS	:	Days after sowing
DAI	:	Days after inoculation
<i>et al</i>	:	And others
<i>etc</i>	:	Etcetera
g	:	Gram/s
GIS	:	Geographical information system
GPS	:	Global positioning system
h	:	Hour/s
ha	:	Hectare
<i>i.e.</i>	:	That is
ITS	:	Internal transcribed spacers
Fig.	:	Figure
kg	:	Kilogram/s
l	:	Litre/s

m	:	Metre
m ²	:	Metre square
μg	:	Microgram
μm	:	Micrometer/s
mg	:	Milligram/s
mha	:	million hactares
ml	:	Millilitre/s
mm	:	Millimetre/s
min	:	minutes
mt	:	million tonnes
N	:	Nitrogen
nmol	:	nano mole
NS	:	Non-significant
P	:	Phosphorus
PCR	:	Polymerase chain reaction
PDA	:	Potato Dextrose Agar
pH	:	Proportionate hydrogen ions
ppm	:	Parts per million
Psi	:	Pressure per square inch
%	:	Per cent
-/	:	Per
±	:	Plus or minus
q	:	Quintal/s
RBD	:	Randomized block design
rDNA	:	Ribosomal Deoxyribonucleic acid

RAPD	:	Random amplified fragment length polymorphism
RH	:	Relative humidity
SEm	:	Standard Error of the mean
t	:	Tonne/s
UV	:	Ultraviolet
WA	:	Water agar
WP	:	Wettable powder
<i>viz.</i>	:	Namely
FYM	:	Farm yard manure
CO ₂	:	Carbon dioxide

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Introduction

I. INTRODUCTION

Chickpea (*Cicer arietinum* L.) is one of the most important food legumes being cultivated in almost all over the world including temperate and sub-tropical regions. It originated in South West Asia and is cultivated from ancient times both in Asia and European countries.

Chickpea is grown in over 50 countries with 90 per cent of its area in developing countries. Southern and South-Eastern Asia accounts for 79 per cent of the global chickpea production. In Asia, India is the largest producer of chickpea contributing over 70 per cent of the world production occupying an area of 10.22 million ha with a production of 9.88 million tonnes and with productivity of 920 kg/ha (Anon., 2016). The major chickpea growing states of our country are Madhya Pradesh, Uttar Pradesh, Rajasthan, Maharashtra, Andhra Pradesh and Karnataka. In Karnataka, it is grown on an area of 0.92 million ha with a production of 0.57 million tonnes and the productivity is 622 kg per ha (Anon., 2016). Gulbarga, Bidar, Dharwad, Gadag, Ballari and Raichur are the major chickpea growing districts.

It is an integral part of the cropping system of the farmers all over the country, because this crop fits well in the crop rotation and mixed cropping. It has multipurpose use and ability to grow under the condition of low fertility and varying conditions of soil and climate. Chickpea belongs to Leguminaceae family usually grown after rainy season on conserved soil moisture during winter in the tropics; in spring in the temperate and mediterranean regions. It is the important grain legume grown for protein rich seeds for human consumption and to maintain the soil fertility by its nitrogen fixing capability. Chickpea seeds of *desi* type are brown to black in colour with a rough surface whereas seeds of *kabuli* type are light coloured having smooth surface. The roots usually include a central strong tap root with numerous lateral root branches that spread out in all directions in the upper layers of the soil.

Chickpea seeds contain 23% protein, 64% carbohydrates, 47% starch, 5% fat, 6% crude fiber, 6% soluble sugar and 3% ash minerals such as calcium (202 mg), phosphorous (312 mg), iron (10.2 mg), vitamin C (3.0 mg), calorific value (360 cal), small amounts of B complex, fiber (3.9 g) and moisture (9.8 g). Predominantly chickpea is being consumed as dhal or variety of snack foods, sweets and condiments. Husk and split beans are useful as livestock feed. Acidic liquid from glandular hairs of the plant

contain 94% malic acid and 6% oxalic acid, has medicinal value and used in preparation of vinegar. It is rich in dietary proteins and good for human consumption; moreover, its ability to form nitrogen-fixing nodules *via* interaction with rhizobia adds to its uniqueness (Ferguson *et al.*, 2010).

The crop faces various problems throughout the growing areas, some related to specific regions and some under wider range of climatic conditions. Chickpea cultivation is often subjected to significant yield losses due to insects and diseases ranging from 5-10 per cent in temperate and 50-100% in tropical regions (Van Emden *et al.*, 1988). Currently chickpea is attacked by 172 pathogens *viz.*, 67 fungi, 3 bacteria, 22 viruses and mycoplasma, and 80 nematodes reported from 55 countries. Maximum number of pathogens infecting chickpea had been reported from India while in other countries, it varied from 1 to 40. The pathogens that are most widely distributed are *Ascochyta rabiei* (35 countries), *Fusarium oxysporum* f. sp. *ciceri* (32 countries), *Uromyces ciceris arietini* (25 countries), bean leaf roll virus (23 countries), *Rhizoctonia bataticola* (21 countries), *Botrytis cinerea* (15 countries), *Sclerotinia sclerotiorum* (15 countries) and cucumber mosaic virus (9 countries) (Nene *et al.*, 1996).

Among the diseases of chickpea, the production of chickpea is largely constrained by Fusarium wilt (*Fusarium oxysporum* f. sp. *ciceris*); however, recent reports indicated that dry root rot (DRR) is emerging as a potential threat to chickpea productivity and production (Sharma *et al.*, 2010; Ghosh *et al.*, 2013). As the disease is more prevalent during hot temperature 30 to 35 °C and low soil moisture conditions (Taya *et al.*, 1988; Pande *et al.*, 2010). Dry root rot (DRR) caused by *Rhizoctonia bataticola* (Taub.) Butler [Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid] is a soil and seed borne necrotrophic fungal pathogen that has a global distribution, which can infect more than 284 plant species throughout the world including monocot and dicots (Farr *et al.*, 1995).

Rhizoctonia is a genus of anamorphic fungi in the order Cantharellales and family Ceratobasidiaceae. *Rhizoctonia* species do not produce spores, but are composed of hyphae and sclerotia (hyphal propagules), asexual stage of fungi, saprophytic, but some act as facultative plant pathogens causing commercially important crop diseases. The genus *Rhizoctonia* (means “root killer”) was described in 1815 by the French mycologist Augustin Pyramus de Candolle for plant pathogenic fungi that produce both hyphae and sclerotia. Subsequent authors added over 100 additional names to the genus. A comprehensive survey and redistribution of species name “*Rhizoctonia*” was published in

1994 by Anderson and Stalpus. According to this published information, *R. bataticola* is used as a synonym to *Macrophomina phaseolina* (Tassi) Goid. Currently, *M. phaseolina* is officially recognised as the correct taxonomic name (CMI description of pathogenic fungi and bacteria No.275) with the sclerotial phase known as *R. bataticola* (Holliday and Punithalingam, 1970). Due to the presence of only sclerotial phase in chickpea, the pathogen is referred as *R. bataticola* (Taub.).

The dry root rot in chickpea was first reported from India by Mitra (1931); later, the disease has been reported from most chickpea-growing areas in India and other countries like Iran (Kaiser *et al.*, 1968), the USA (Westerlund *et al.*, 1974) and several countries in Asia and Africa (Nene *et al.*, 1996). The disease was earlier known as “Rhizoctonia wilt” in chickpea; however, later it was named as “dry root rot”. The dry root rot was not of much significance in chickpea earlier; however, it has become a major threat to chickpea production in recent years due to altered weather conditions, particularly on the account of longer drought spells. Higher temperature and soil moisture depletion during crop growth period particularly at pre-harvesting stage is predisposing chickpea to dry root rot (Sharma and Pande, 2013). Recent surveys conducted during 2010–2013 indicated widespread and increased incidence of dry root rot in the central and southern states of India (Ghosh *et al.*, 2013). Disease was found irrespective of soil types, cropping system and cultivars used and incidence ranged from 5 to 50 per cent or more in badly infected soils.

Rhizoctonia of chickpea was observed in India by Haware and Nene (1978) in Hyderabad, Andhra Pradesh, India. Later, Tripathi and Sharma (1983) also reported the appearance of Rhizoctonia root rot in pulses. Periodic surveys on wilt and root rot incidence of chickpea by Sharma *et al.* (1983) in northern region of Madhya Pradesh. In the chickpea field, the onset of the disease appears as scattered drying of the plants. Affected plants are usually straw coloured, but in some cases the lower leaves and stems show brown discolouration. The tap root appears black, rotten and devoid of most of the lateral and fine roots. The dead root become quite brittle and shows shredding of bark. Dark minute sclerotial bodies can be seen on the roots exposed or inside the wood. When the dry stem of the collar region is split vertically, sparse mycelium or minute sclerotia can be seen in the pith (Nene *et al.*, 1991).

Successful survival and adaptation of *R. bataticola* to various geographic environments has been confirmed by several workers demonstrating morphological

(Mayek-Pérez *et al.*, 2001), pathogenic (Mayek-Pérez *et al.*, 2001 and Su *et al.*, 2001) and genetic diversity (Vandemarket *et al.*, 2000; Mayek-Pérez *et al.*, 2001; Su *et al.*, 2001; Almeida *et al.*, 2003; Jana *et al.*, 2003 and Aboshosha *et al.*, 2007) in various crops. Techniques such as RAPD markers are useful for measuring genetic relationships and variations within and among populations of *R. bataticola* in different countries like Australia (Fuhlbohms, 1997), Brazil (Almeida *et al.*, 2003) and India (Jana *et al.*, 2003).

The necrotrophic fungus *R. bataticola* exists in an anamorph (sclerotial) stage in soil and on crop residues. Under *in vitro* conditions, the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later produce typical black sclerotia. The fungus may produce abundant aerial mycelium, as high as to touch the cover of the culture plates or mycelium is found to be completely or partially suppressed (Sharma *et al.*, 2012a). The characteristic features of *R. bataticola* are right angle branching of the mycelium and constriction of the branch near the point of origin. The sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes (Sharma *et al.*, 2012a). Pathogenic variability in the target production area is a prerequisite for identifying genotypes with a stable resistance to the variable pathogen populations.

Environmental conditions like temperature, soil moisture and carbon dioxide play an important role in the viability and growth of *R. bataticola* (Khan, 2007). *Rhizoctonia bataticola* is able to produce microsclerotia under relatively low water conditions while viability of microsclerotia is drastically reduced at high water potentials (Olaya and Abawi, 1996). Temperature and soil moisture are the two important weather parameters influencing the dry root rot infection, colonization and development in chickpea. Better understanding the role of temperature and soil moisture will help in standardization of dry root rot resistance screening techniques which will assist breeders in developing breeding strategy for dry root rot resistance over broader geographical areas.

In Karnataka, systematic work on important aspects of the dry root rot and its pathogen has not been done. Realizing the potentiality of dry root rot disease in causing economic losses, it was thought necessary to initiate systematic studies, on different aspects of the disease as well as pathogen. It is necessary to conduct a survey of the

disease, so that its distribution and extent of its spread can be understood and “hot spots” can be located.

Variability is the property of an organism to change its characters from one generation to the other. *Rhizoctonia bataticola* pathogen is a major hindrance to chickpea production in tropics and subtropics, needs information based on the fungal features and variability in cultural, morphological and molecular. Therefore, there is need to study and characterize the fungus based on cultural, morphological and molecular variability.

In the changing climatic scenario, studies on impact of climatic factors on chickpea dry root rot is limited and moreover the crop is largely grown in rainfed environments and climate variability such as high temperature within the rainfed ecologies leading to varying intensities. Therefore, studies on effect of environmental variables on dry root rot incidence are felt necessary.

Growing of resistant cultivars against dry root rot disease is a cost effective, sustainable strategy and ideally fits into integrated disease management. Unfortunately, high degree of resistance to soil borne disease is not available among cultivable germplasm. Hence, it is of extremely useful to identify potential resistant sources to dry root rot and it is precise at this juncture to screen germplasm to identify resistance sources.

As *R. bataticola* has a broad host range and survives in soil for longer periods in the form of sclerotia. The sclerotia can survive for more than ten months even in the absence of host plants. Therefore, over use or over dependence on chemical control or any other single control method is not sufficient to manage dry root rot. A systematic control approach integrating all disease management options may produce better pathogen management and it is broad-based, ecologically sound by combining all the available disease management approaches with each method compensating the deficiencies of others (Kumar *et al.*, 2009). Therefore, it is necessary to develop integrated disease management strategies to manage of dry root rot.

Considering the severity and loss caused by *R. bataticola*, it was thought necessary to initiate systematic studies on emerging and destructive dry root rot disease problem occurring in different parts of Karnataka. Keeping these points in view, thorough study of different aspects on pathogen and disease are planned with following objectives.

1. To survey for incidence of dry root rot of chickpea in North Karnataka
2. Studies on cultural, morphological and molecular variability of pathogen
3. Effect of climate change variables (temperature, moisture and carbon dioxide) on dry root rot disease
4. Identification of dry root rot resistant sources of chickpea
5. Integrated management of dry root rot of chickpea

Review of Literature

II. REVIEW OF LITERATURE

A brief review of literature pertaining to occurrence and distribution of dry root rot disease, symptomatology, causal organism responsible for the disease and its cultural, morphological, pathological and molecular variability among the isolates, effect of temperature, moisture and carbon dioxide (CO₂) on the development of the pathogen and disease development in plants are presented in this chapter. Wherever the literature on particular aspect of *R. bataticola* on chickpea was scanty, it was amply supplemented with and supported by other pathogens and crops.

2.1 Occurrence and distribution of dry root rot disease

Rhizoctonia is the causal organism of dry root rot of chickpea and was first observed in India by Haware and Nene (1978) in Hyderabad, Andhra Pradesh, India. Later, Tripathi and Sharma (1983) also reported the appearance of rhizoctonia root rot in pulses. An association of *Fusarium* sp. and *Rhizoctonia* sp. with wilted chickpea was first reported by Narasimhan (1929). Subsequently, it was reported from Punjab by Luthra (1938), Madhya Pradesh (Sharma and Khare, 1969), West Bengal (Biswas and Guptha, 1981) and Haryana (Tripathi and Sharma, 1983). It has been reported from Egypt, Iran, Kenya, Lebanon, Mexico, Myanmar, Pakistan, Spain, Sri Lanka, Sudan, Syria, Tanzania, Turkey, Uganda, USA and Zambia. It causes considerable yield losses that vary from 5-50% and may cause 100% losses in susceptible cultivars under favourable conditions (Nene *et al.*, 1996).

The incidence of *R. bataticola* on *Cicer arietinum* was higher in sandy than clay soil. Low soil moisture was conducive to disease development (Taya *et al.*, 1988). Pandey and Singh (1990) conducted the survey in Allahabad region of Uttar Pradesh and reported that the chickpea was infected by *R. bataticola* in 83% of villages with less than 25% disease intensity.

Cook *et al.* (1973) reported an increase in microsclerotia in a field planted on soybeans for three years from 80, 120 to 149 microsclerotia per gram of soil. High levels of *M. phaseolina* colonization are positively correlated with poor seed quality and lower yields. After harvesting, roots loaded with microsclerotia start decaying and release the microsclerotia into the soil (Olaya and Abawi, 1996). Inoculum survival in the soil depends on soil moisture and the soil microbial community (Collins *et al.*, 1991).

M. phaseolina poses greater problem in pigeonpea at high temperature (>30 °C) and dry weather. This disease was more prevalent in vertisols than alfisols. Rain after a prolonged dry spell predisposed plants to disease early (Bajpal *et al.*, 1999).

Rajkrishan *et al.* (1999) studied the effect of edaphic factors and moisture regime on the incidence of *Macrophomina* root rot in sesame. Among the various soil types, sandy soil supported maximum disease (78.33%) as compared with clay soil (51.56%). Similarly, more disease incidence of 48.33 per cent was observed when pots were irrigated after the interval of seven days as compared to pots receiving irrigation every day. *R. bataticola* isolates of chickpea from several locations in Myanmar collected by Than *et al.* (1991) reported the possible occurrence of more than one type of isolate in a same field and the existence of closely related isolates at different places.

2.1.1 Disease symptomatology

According to Dastur (1935) only mature plants were affected, showing bronzing of the leaves on one or more of the lower branches. The colour of leaves later changed to yellow and then brown. The affected branches and leaf stalks of diseased plants were stiff and turned upwards and leaflets stand more or less vertically and shed prematurely. The terminal part of the tap root was black or brown and shriveled.

Singh *et al.* (1990) reported that roots inoculated with *R. bataticola* (*M. phaseolina*) caused disintegration of cortical tissues with mycelial and sclerotial bodies plugging the xylem vessels.

The dried plants due to disease generally appeared during flowering and podding time. These were scattered in the field. The seedlings can also get infected. Drooping of petioles and leaflets was confined to those at top of the plant. Leaves and stem of affected plant usually straw coloured but in some cases, the lower portion of the stem was brown. When the plant was uprooted, lower portion of the tap root remains in the soil. It was devoid of lateral and finer roots. Dark minute sclerotial bodies can be seen on the roots exposed inside the wood (Nene *et al.*, 1991).

2.1.2 Pathogen

Rhizoctonia bataticola (Taub.) Butler as a plant pathogen was recognized by Halsted (1890). Taubenhause (1913) gave the name of the genus as *Sclerotium* because of absence of spores and the species name as *bataticola* because it was pathogenic to *Ipomea*

batatus (L.) Lam. Briton Jones (1925) transferred the fungus to the genus *Rhizoctonia* based on the identification of cultures by Butler (1918).

Ashby (1927) accepted *Macrophomina* and rejected the binomial *M. phaseoli* and proposed a new binomial *M. phaseolina* as the pycnidial stage of *R. bataticola* on the basis that *M. phaseoli* was the earliest applicable binomial. He was not aware that Tassi had earlier described *M. phaseolina*. Haigh (1930) suggested that *R. bataticola* be used for sclerotia isolate and pycnidial strains should be called *M. phaseolina*. Goidanich (1947) examined the original material of Tassi and compared it with *M. phaseoli*, *M. corchori*, *M. cajani*, *M. sesami*, *M. philippinensis*, *Dothiorella cajani* and *D. phaseoli*. He confirmed all of them were identical. He corrected the mistake made by Ashby (1927). According to the International Code of Botanical Nomenclature, the binomial *M. phaseolina* was the valid name for the pycnidial stage of *R. bataticola*.

Mycelia width varied from approximately 2-11 µm and distance between two consecutive septa measured 46 µm. However, the most important character regarding taxonomy and classification were the production size and composition of microsclerotia (Reichert and Hellinger, 1947).

Sclerotia within plant parts were black, smooth, hard and varied in size from 100 µm-1mm while in culture, it varied from 50-300 µm. These descriptions were given by Commonwealth Mycological Institute (CMI), Kew, England. During the sclerotial formation, 50–200 individual hyphal cells aggregate to give multicellular bodies called microsclerotia. The microsclerotia were black and variable in size from 50–150 µm depending on the available nutrients of the substrate on which the propagules were produced (Short and Wyllie, 1978).

Sharma *et al.* (2012a) reported that the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later produce typical black sclerotia. The fungus may produce abundant aerial mycelium, as high as to touch the cover of the culture plates, or mycelium is found to be completely or partially suppressed.

Growth and microsclerotial bodies production studied on PDA indicated that the mycelium was pale white in color in the initial stages of the growth but, later turned to

dark brown to black as and when microsclerotial bodies started to form. Hypha showed right angled branching. The microsclerotia varied in shape and size with growth pattern of scattered to clustered Manjunatha (2014).

2.1.3 Pathogenisity test

Forty isolates of *R. bataticola* causing dry root rot of chickpea collected from four major chickpea growing states viz., Madhya Pradesh, Chhattisgarh, Gujarat and Maharashtra of central India were screened for virulence of the isolates on susceptible chickpea variety BG 212 by blotter paper technique. The highly virulent isolates from Madhya Pradesh, Chhattisgarh, and Maharashtra were fast growing isolates and exhibited largest sclerotia. The sclerotial size and number of sclerotia of *R. bataticola* isolates had positive correlation with virulence (Gupta *et al.*, 2012).

Lekhraj *et al.* (2012) conducted pathogenisity test of *M. phaseolina* isolates collected on infected chickpea from 9 various parts of Rajasthan, Hissar and New Delhi. Pathogenicity of these isolates tested on 40 day old chickpea cultivar C-235 showed there was no direct relationship between cultural characters and virulence of the isolates. The isolate BKN had scanty, suppressed growth was highly pathogenic. Isolates SGN, NHR, DUR, HAU, DEL and HNG having different types of colony growth were moderately pathogenic and the isolates CHR and UPR with different mycelium characters were less pathogenic. Similarly, there was no apparent correlation between the virulence and growth rate of the fungal colonies.

Fifty isolates of *R. bataticola* collected on chickpea from different agro climatic zones in India were evaluated for the degree variability in pathogenicity test. All the isolates proved pathogenic on chickpea cultivar BG 212. The dry root rot disease severity ranged from 4 to 9 (1-9 severity rating scale). The effect of isolates on severity was significant. The pathogenic and non pathogenic isolates were not concentrated in any one particular state/region. Among the 94 isolates of *R. bataticola* collected from different agro ecological zones, 8 isolates from Andhra Pradesh and Madhya Pradesh showed less aggressive on chickpea cultivar BG 212 (Sharma *et al.*, 2012).

2.2 To survey for incidence of dry root rot of chickpea in Northern Karnataka

Periodic surveys on wilt and dry root rot incidence of chickpea by Sharma *et al.* (1983) in northern region of Madhya Pradesh for two consecutive years showed that

15.90-37.99% and 3.58-20.63% of wilt and root rot incidence, respectively in the districts surveyed. The highest wilt and root rot incidence was 27.79 and 14.17% and the lowest was 20.81 and 8.76%, respectively.

The incidence of dry root rot in chickpea caused by *R. bataticola* was studied in 20 locations in Uttar Pradesh and Uttaranchal, India during February to March of 1998-2000 and reported disease was widespread in Uttar Pradesh. It was severe in bold seeded cultivars (Prajapati *et al.*, 2003).

Veerendra Kumar (2004) noticed an incidence of 9.97% dry root rot of chickpea in Gulbarga followed by Raichur (5.16%) and Bidar (4.28%). The highest incidence was 12.24% in Chittapur of Gulbarga and least was in one of the locations of Raichur (3.21%). Survey conducted by Shubha and Gurha (2006) in different blocks of Bhundelkhand region of Uttar Pradesh found infection of *R. bataticola* on chickpea ranged from 5-22%. Chickpea fields in Shimoga, Raichur and Bangalore districts in Karnataka were surveyed by Gurha and Trivedi (2008) for the prevalence of dry root rot. *R. bataticola* was the predominant pathogen infecting 60 to 70% of the plants in the fields of Gulbarga, Manchalapur, Ekhaspur, Raichur and Tengri.

Intensive roving survey conducted by Manjunatha *et al.* (2011) during November to December, 2008 in the farmers fields at Raichur, Gulbarga and Bidar districts in Karnataka. The maximum dry root rot incidence in chickpea was noticed in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18%) in Bidar. The overall incidence of dry root rot ranged from 1-19 per cent across the districts. This was mainly attributed due to variation in soil type (black and red soil) and cultivars grown (Annegiri and JG 11) under rain fed condition.

The survey for the occurrence and severity of chickpea dry root rot was made during crop season 2010-11 in Jammu and Kashmir. The highest incidence was up to 40% in Shangus and lowest up to 4.11% in Naina of Jammu and Kashmir. The incidence of dry root rot in chickpea caused by *R. bataticola* was observed in late October to mid November. The intensity of the disease was high in the month of February and March during late flowering and podding stage (Khan *et al.*, 2012).

A survey was conducted in 2010-2011 *Rabi* cropping season to obtain information on the distribution and incidence of chickpea diseases with respect to soil type, cultivar

used, seed treatment in Central and Southern parts of India (Andhra Pradesh, Karnataka, Madhya Pradesh and Chhattisgarh). Local cultivars Annigeri (19.23%) followed by Harbora (12.82%) were most frequently grown by the farmers. However, among the improved varieties JG 11, JG 130, JG 16, JG 74 and Jawahar were most commonly grown covering 34.61 per cent area. Dry root rot disease was found in all the sites surveyed where the incidence ranged from 8.9-10.30% irrespective of cultivar type and location. The local variety occupied 48% area in Karnataka while in Andhra Pradesh was 39%. Disease incidence was lower on improved (9%) as compared to local varieties (14%). Sixty three percent of the farmers practiced seed treatment with fungicides. The disease incidence in seed treated fields ranged from 6.63-12.40% compared with untreated 9.22-20.02% (Ghosh *et al.*, 2013).

Recent surveys conducted during 2011–2013 indicated widespread and increased incidence of dry root rot in the central and southern states of India (Ghosh *et al.*, 2013). Disease was found irrespective of soil types, cropping system and cultivars used and incidence ranged from 5 to 50% or more in badly infected soils. Soils where soybean or susceptible hosts are planted successively year after year, *M. phaseolina* populations tend to increase.

2.3 Studies on cultural, morphological and molecular variability of pathogen

2.3.1 Cultural variability among *R. bataticola*

Lokesha and Benagi (2004) reported among four isolates of *M. phaseolina* in pigeon pea, maximum growth was recorded in ICRISAT isolate (82.7 mm) which was significantly superior over other isolates, followed by Bidar isolate (78.35 mm). The radial growth was found maximum after 24 hours of incubation on PDA (40.3 mm) followed by potato sucrose agar (39.0 mm) and it completely covered the PDA plate by 48 hours of incubation (Chowdary and Govindaiah, 2007).

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their cultural characters. After 48 hours of incubation, highest radial growth in Rb1 (Bangalore) followed Rb4 (Faridkot) was observed. Rb3 (New Delhi) and Rb5 (Faridkot) were next best with statistically similar

colony diameters (83.5 mm). Least and similar colony diameter of 51.3 mm was recorded in Rb7 and Rb10 (Kanpur) isolates (Aghakhani and Dubey, 2009). Thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions were highly variable in their cultural characters. Among these, twelve isolates were fast in growth, another twelve were moderate and the remaining six isolates were slow in growth (Manjunatha and Naik, 2011).

Dhingra and Sinclair (1973) collected nine isolates of *M. phaseolina* from various parts of United States reported variation in growth rate and colony characters. Further, the growth rate was correlated with pathogenicity.

Iqbal and Mukhtar (2014) collected 65 isolates of charcoal rot caused by *M. phaseolina* in mung bean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan were studied for cultural variability. Significant differences were noticed among 65 isolates with sixteen isolate rated as fast growing, 11 isolates as slow growing and the rest as medium growing. Maximum colony diameter of 87.17 and 86.67 mm was observed in isolate MP-7 (Dera Ghazi Khan) and MP-26 (Layyah) proved to be the fast growing, while isolates MP-8, MP-29 and MP-30 showed minimum radial growth and rated as slow growing.

Dhingra and Sinclair (1978) noticed fluffy growth in stem isolates, partially fluffy growth in root isolates and appressed growth in all other plant part isolates of *M. phaseolina* on soybean. Thirty six isolates of *M. phaseolina* of mungbean collected from different region of North, South, North East and Central India and observed MP-1 isolate had cottony mycelial growth while MP-2 isolate showed 5 μ m appressed mycelial growth (Devi and Singh, 1998).

The diversity pattern of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans were analyzed based on presence or absence of aerial mycelium. They observed 15 isolates had intermediate and 15 isolates had poor aerial mycelium. Eight from Mexican isolates and 7 isolates of non Mexican isolates had intermediate aerial growth (Fernandez *et al.*, 2006).

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and

Chattisgarh) of India were highly variable in colony texture. The majority of nineteen isolates produced suppressed mycelium whereas 4 isolates in which three from Kanpur and one from Rohtak produced aerial mycelium (Aghakhani and Dubey, 2009).

Edraki and Banishashemi (2010) studied sixty isolates of *M. phaseolina* collected from different parts of Iran on various crops like soyabean, cucumber and sesame. Isolates were grouped in to four phenotypes viz., fluffy with abundant sclerotia, fluffy with few sclerotia, partially fluffy and appressed growth.

Thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions were highly variable in their colony texture. These isolates were categorized into three groups. The first group comprised of all isolates from Bidar (except RB7) exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur, Gulbarga (except RG3 and RG7) and an isolate from Bidar (RB7) exhibited brownish black with flat colony and sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown colour sclerotia formed at the center of the culture (Manjunatha and Naik, 2011).

Variability among the eleven isolates of *R. bataticola* from different pulse crops (redgram, greengram, cowpea, soybean, blackgram) isolates were categorized into three groups viz., linear, fluffy and linear at the end with fluffy growth at the centre. All isolates were more aggressive on the original host from which it was isolated, which was shown by the variability in pathogenic characters (Sundravadana *et al.*, 2012).

Thirty six isolates of *M. phaseolina* of mungbean collected from different region of North, South, North East and Central India were studied for colony colour. MP-1 isolate had pale grey colonies while brown in MP-2 (Devi and Singh, 1998). The diversity patterns of 30 isolates of *M. phaseolina* of beans 15 from Mexico and 15 from other countries were analyzed based on morphological characteristics such as colour of colony (grey and black) and observed 22 isolates were grey, 4 isolates were white and 4 isolates were black in colour (Fernandez *et al.*, 2006).

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi,

Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in pigmentation of isolates. Pigmentation varied from white to dull white, creamy, grey and black. Black was prominent among the isolates (Aghakhani and Dubey, 2009).

2.3.2 Morphological variability of *R. bataticola*

Work has been done to elucidate the variability in morphology, physiology, pathogenicity and genotypes of *M. phaseolina* in different hosts [(Dhingra and Sinclair 1973 (castor); Sobti & Sharma 1992 (groundnut); Ratnoo *et al.* 1997 (cowpea); Atiq *et al.* 2001 (sunflower); Okwulehie 2001 (groundnut); Suriachandraselvan and Seetharaman 2003 (sunflower); Sharma *et al.* 2004 (pearl millet, sesame, horsegram and mothbean); Fernandez *et al.* (2006) (bean); Ndiaye 2007 (cowpea)].

The pathogenic variability in *R. bataticola* is assumed to be due to mutation, hyphal fusion and mitotic recombination. However, not much has been reported on *R. bataticola* of chickpea. Recently, cultural, morphological and molecular variations in 94 isolates of *R. bataticola* collected from various chickpea grown in various agro-ecological zones of India (Sharma *et al.*, 2012a). The isolates varied in cultural and morphological characters like colony colour, growth pattern, growth rate and sclerotial initiation and production. Significant relationships between sclerotial initiations, sclerotial intensity and disease severity between isolates were found. Positive correlation between pathogenic isolates and sclerotial production (Hooda and Grover, 1988) was found in mung bean; however, Manici *et al.* (1992) found no such correlation in sunflower. Than *et al.* (1991) studied the relationship among *R. bataticola* isolates based on colony fusion type with 37 isolates in chickpea and found existence of more than one isolate at different places.

Gupta *et al.* (2012) observed variation in size of hyphal width of 40 *R. bataticola* isolates from chickpea. Eighteen highly virulent isolates were classified under very thin (5.2-6.5 μm), 17 highly virulent and 2 virulent isolates under thin (6.6 -7.9 μm) and only three highly virulent isolates from Chhattisgarh and Gujrat exhibited (8.0 - 9.3 μm) thick hyphal width. Sobti and Sharma (1992) reported that the width of hyphae varied from 4.16-8.48 μm in *R. bataticola* isolates from chickpea. Thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India were studied for morphological variability. Typical right-angled branching of

mycelium was found in isolate MP-1 and it was acute to right angled in isolates MP-2 and MP-3 (Devi and Singh, 1998).

Hildebrand *et al.* (1945) observed the sclerotia were black, smooth varying from spherical to oblong to irregular shaped. The sclerotial shape of any isolate varied with substrate on which fungus was grown. Sclerotial diameter of *R. bataticola* in gram varied from 98.1-119.2 μm (Ghosh and Sen, 1973).

Morphological variability among thirty six isolates of *M. phaseolina* from mung bean collected from different region of North, South, North East and Central India were studied for by Devi and Singh (1998) and observed bigger sclerotia in isolate MP-2 (400 x 280 μm). Seven isolates of *M. phaseolina* on maize collected from Ludhiana, Hyderabad, Delhi, Arabhavi, Udaipur, Bangalore and Coimbatore by Shekhar *et al.* (2006) observed that Hyderabad isolate produced highest number of sclerotia (180.3 sclerotia) of bigger size (95.7 μm) compared to other isolates. On the basis of sclerotial morphology, isolates were divided into oblong shape having a regular edges and round with regular edges. On the basis of disease expression, Hyderabad isolate was more virulent compared to other isolates.

Purkayastha *et al.* (2004) observed variability in time taken for production of sclerotia of 55 isolates of *R. bataticola* from clusterbean. Significant differences were noticed among 55 isolates with sixteen isolate produced sclerotia in one day, thirteen isolates in three days after inoculation and the rest of isolates taken five days to produce sclerotia.

The diversity pattern of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans were analyzed based on morphological characteristics such as shape (round, ovoid and irregular) and production of microsclerotia. Microsclerotia were higher in most Mexican isolates than in non-Mexican isolates. However, no significant differences between Mexican and non-Mexican isolates were found in length, width and length/width ratios of microsclerotia (Fernandez *et al.*, 2006).

Macrophomina phaseolina isolation was done from T9, PU 30, TPU 4, UK 17, UPU97-10 cultivars of black gram by Suryawanshi *et al.* (2008) from stem and leaves. The typically dark black to brown colony observed on PDA medium. The mycelium was

superficial, hyaline to brown, septate and tree like in form. The sclerotia produced were round, oblong and irregular in shape.

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their morphological characters. The sclerotia formed in different isolates were dark brown to black and size varied from 40 - 600 μm . The isolates from Bangalore (Rb1) and Faridkot (Rb 5) produced the largest size sclerotia ranged from 200 – 600 μm and 100 – 400 μm respectively. Dumka (Rb19) isolate produced small sized sclerotia which ranged from 43.9- 81 μm . They observed most virulent isolate (Rb1 from Bangalore, Karnataka) was fast growing and produced largest sclerotia. The isolates were highly variable in virulence (Aghakhani and Dubey, 2009).

Thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions variable in their characters were categorized into three groups. The first group comprised of all isolates from Bidar except RB7 and exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur and Gulbarga except RG3 and RG7 and also an isolate from Bidar RB7 and showed brownish black with flat sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown coloured sclerotia formed at the center of the culture (Manjunatha and Naik, 2011). From 14 Iranian provinces, fifty two *M. phaseolina* isolates were recovered from 24 host plant species. These isolates were studied for average size and time taken for microsclerotia formation. Isolates from Khorasan and Isfahan Kashan had the largest microsclerotia while isolates Yazd and Kerman had the smallest microsclerotia (Mahdizadeh *et al.*, 2011).

Variations in the 35 isolates of *M. phaseolina* collected from the major sesame producing regions in central China comprising of Hubei, Henan, Anhui, Jiangxi province were studied by Linhai *et al.* (2011). Sclerotia quantity and size had high variations. Pingyu and Xiaogan produced no sclerotia after 24 h of culture and sclerotium size of 7 days old culture varied between 0.10 mm (Wuxue) and 0.16 mm (Jiayu) with average of 0.13 mm.

Forty isolates of *R. bataticola* from chickpea were studied by Gupta *et al.* (2012), they observed sclerotial sizes varied from 72.7- 117.2 x 57.1-106.5 μm . Based on sclerotial size, the isolates were divided into three groups viz. minute size (72.7 - 87.5 x 57.1-73.0 μm), medium size (87.6 - 103.2 x 73.5 - 90.0 μm) and bold (103.3 -117.2 x 90.1-106.5 μm) size of sclerotia groups. Bold size of sclerotia was observed in six highly virulent isolates belong to Madhya Pradesh and Maharashtra. Rest of the thirty two highly virulent and two virulent isolates placed in minute and medium size of sclerotial groups. The isolate from Betul, Madhya Pradesh produced maximum number of sclerotia (22.3 sclerotia per microscopic field) and isolate from Dhule, Maharashtra exhibited minimum number of sclerotia (8.1 sclerotia per microscopic field). The shape of sclerotia ranged from oblong (12 isolates) to round (28 isolates).

Eleven isolates of *R. bataticola* from different pulse crops viz., redgram, greengram, cowpea, soybean, blackgram showed variability in sclerotial intensity, shape and intensity of pigment synthesis. All isolates were more aggressive on the original host from which it was isolated than other hosts (Sundravadana *et al.*, 2012). Isolates of *M. phaseolina* were collected from semi arid regions of Maharashtra, Andhra Pradesh and Karnataka states of India. Variability in shape, length, width and length/width in morphological characters of microsclerotia were studied and observed AP 21 isolate varied in length/width ratio (483.3/200.0 μm) while no significant differences were observed in others (Prasad *et al.*, 2014).

Iqbal and Mukhtar (2014) collected 65 specimens of charcoal rot caused by *M. phaseolina* in mungbean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan. They classified nine isolates under large sized, 26 under small sized and remaining 30 isolates as medium sized sclerotia. Maximum sclerotial size was observed in isolates MP-20 and MP-3 showing 29.00 and 27.33 μm diameter respectively, while the isolates MP-39 and MP-28 were found to be the smallest in size. The average sclerotial size of isolates ranged from 17.00 to 29.00 μm .

Maruthi (2016) isolated 24 isolates from pigeonpea plants observed for microsclerotial production where 18 isolates have produced microsclerotia ranges from 12.33 to 57.66 per microscopic field. While six isolates viz., Rb-8, Rb-11, Rb-12, Rb-16, Rb-17 and Rb-19 have not produced any microsclerotia under PDA.

2.3.3 Molecular variability in *R. bataticola*

Aghakhani and Dubey (2009b) determined genetic diversity among six isolates of *R. bataticola* representing five different ITS groups with respect of their nucleotide sequences of ITS I, 5.8S and ITS II regions. The ITS of Karnataka isolate (Rb1) was largest (670 bp), whereas it was smallest (499 bp) in Haryana (Rb21) isolate. After sequencing the ITS sequences of the isolates were deposited to Gene-Bank nucleotide database. The phylogeny tree constructed from the nucleotide sequence, similarity of these six isolates along with 27 other ITS sequences of fungi which showed 96–100 per cent similarity during blast analysis clustered them into three major groups. The isolates Rb16 (Jharkhand), Rb21 (Haryana) and Rb26 (Madhya Pradesh) clustered in one group and showed 96-100 per cent ITS nucleotide sequences similarity with the nucleotide sequences of *M. phaseolina* (*R. bataticola*) available in the Gene Bank.

Fifty isolates of *R. bataticola* collected from different agro climatic zones in India were evaluated for the degree of subdivision in isolates level by Sharma *et al.* (2012). Genetic characteristics were analyzed based on the sequence of rDNA- ITS region. The phylogenetic tree based on rDNA-ITS analysis showed maximum number of *R. bataticola* isolates were very diverse and did not depend on geographical origin. Both pathological and molecular data correlated with each other supported that the *R. bataticola* present in India were very diverse and independent to their origin.

The RAPD technology has quickly gained wide spread acceptance and application because it has provided a relatively simple tool for genetic analysis in biological systems. However, RAPD is the best assay when the nucleotide sequence is not known. Unlike other PCR protocols, which utilize two primers of defined sequence, RAPD detects nucleotide polymorphism using only one primer of an arbitrary nucleotide sequence. DNA amplification approaches using arbitrary primers have been employed to a large number of genetic studies including the estimation of genetic relationships (Wilde *et al.*, 1992).

Jana *et al.* (2003) showed single RAPD primer A13 could be used to identify and discriminate several isolates of *M. phaseolina* and *Fusarium* sp. obtained from 20 hosts including soybean, cotton, chickpea and safflower. Universal rice primers were used to cluster 40 isolates of *M. phaseolina* from soybean, cotton and chickpea (Jana *et al.*, 2005).

R. bataticola isolates of maize from Ludhiana, Hyderabad, Delhi, Arabhavi, Udaipur, Bangalore and Coimbatore analyzed through RAPD marker for genetic diversity by Shekhar *et al.* (2006). The UPGMA cluster analysis for 706 loci score permitted for identification of three main clusters. Similarity matrix and Jaccard's similarity coefficient between the isolates indicated that the maximum genetic variation was among isolates of Arabhavi and Coimbatore with 70.8 per cent followed by Ludhiana and Coimbatore with 69.5 per cent. The most closely related isolates were Hyderabad and Delhi with an affinity percentage of 65.5 per cent followed by Udaipur and Bangalore isolates with 62.9 per cent similarity.

Thirty isolates of *R. bataticola* from chickpea were studied for molecular variability by Manjunatha (2009) using three primers OPO-10, OPO-12 and OPN-12. There were four isolates in Group-I, 16 isolates in Group-II and 10 isolates in Group-III. However, in Group-II, three isolates (RG-18, RG-19 and RG-20) exhibited 100 per cent similarity. RB-29, RB-25 and RG-16 isolates from Bidar and Gulbarga respectively showed genetically diversity among thirty isolates.

Bayraktar and Dolar (2009) observed intra and inter specific polymorphism among fungal pathogens that cause root rot on chickpea by using 30 RAPD primers. UPGMA cluster analysis of RAPD datasets using Dice's coefficient differentiated all fungal isolates from each other and revealed considerable genetic variability between the isolates.

Genetic diversity of 23 isolates from chickpea and 4 from other host crops (mungbean, urdbean and groundnut) of *R. bataticola* representing 11 different states of India were studied by Aghakhani and Dubey (2009) using RAPD. The clusters generated by RAPD grouped all the isolates into six categories at 40 per cent genetic similarity. High level of diversity was observed among the isolates of different as well as same state. Some of the RAPD primers viz., OPN 4, OPN 12 and OPN 20 clearly distinguished majority of the isolates into the area specific groups based on variability in virulence.

Fourteen isolates of *M. phaseolina* collected from different groundnut growing regions in India were studied for their virulence and genetic diversity using PCR-RAPD markers. A total of 85 amplicon levels were obtained with 7 primers with an average of 12.1 bands per primer were available for analysis, of which 83 were polymorphic

(97.74%). It was possible to discriminate all the isolates with any of the 7 primers employed. The UPGMA clustering of data indicated the isolates shared varied levels of genetic similarity within a range of 0.069 to 0.65 similarity coefficient index. Isolates Bikaner, Churu, Delhi, Hissar which were clustered in one group happened to be more virulent reflecting correlation of similarity with pathogenicity to some extent but there was practically no direct correlation between the virulence and RAPD genetic markers of all the isolates (Kumar and Gaur, 2011).

The genomic DNA of 21 isolates of *R. bataticola* of soybean from Jabalpur were subjected for diversity analysis with 8 randomly selected decamer primers which amplified 64 RAPD marker loci. Out of these 64 bands, 29 bands (45.3%) were polymorphic. Similarity coefficient values for 21 accessions of *R. bataticola* in RAPDs marker was 0.69–0.98 (Pancheswar *et al.* 2012). Genetic diversity of fifty isolates of *R. bataticola* was studied by using 10 RAPD primers and generated 158 polymorphic bands. Those were subjected to hierarchical cluster analysis based on the principle of UPGMA and a dendrogram was generated. The similarity coefficient ranged from 0.63 to 0.92 indicating that no any two or more isolates were 100% similar. The highest similarity coefficient (0.92) was between isolates Rb 2 and Rb 4. All the isolates of Madhya Pradesh and Karnataka fell under same group IA. The Maharashtra isolates were distributed across all the three groups and were found in IA, IIA and III, while the Andhra Pradesh isolates were also fell under all the three groups, but were found in IA, IB, IIB and III (Srinivas, 2016).

2.4 Effect of climate change variables (temperature, moisture and carbon dioxide) on dry root rot disease

2.4.1 Effect of temperature on *R. bataticola*

High day temperature above 30 °C and dry soil conditions at flowering and podding stages rapidly increase the severity of the dry root rot disease on chickpea (Gurha *et al.* 2003). Singh and Sharma (2002) observed that deficit soil moisture favours the severe disease development on pulse crops. Under hot and dry environmental conditions, many economically important crops are predisposed to the infection and colonisation of *R. bataticola* and caused drastic yield losses on chickpea (Thripathi and Sharma 1983), soybean (Pearson *et al.* 1984) and sunflower (Nawaz, 2007). The growth and sclerotial production of *R. bataticola* causing root rot in mulberry at different temperatures *in vitro*

were studied by Chowdary and Govindaiah (2007) recorded maximum growth of 90 mm at 30 and 35 °C and minimum of 40 mm at 20 °C. Sclerotial production was observed in all the temperatures except at 20°C.

R. bataticola on chickpea grew over a wide range of temperatures viz., 10 °C to 45 °C. The optimum temperature for its growth was found to be 30 °C while the next best was 35 °C. Statistically the growth of the pathogen gradually decreased both below and above the optimum temperature (30 °C). Minimum growth was recorded at 10 °C (Khan *et al.*, 2012).

Variability in *R. bataticola* isolates from roots, leaf and seeds of pulses viz., blackgram, greengram, cowpea, soybean, and redgram were studied at different temperatures. It was observed that 15 °C, 20 °C and 25°C had recorded mean mycelial growth of 26.83, 34.23 and 48.51 mm respectively. The maximum mycelial growth of 74.76 and 74.20 mm was recorded at 30 and 35 °C, respectively. As the temperature increased from 10 to 30°C, the mycelial growth gradually increased and at 35°C there was gradual decrease. Temperature was the specific limitation for the growth and its pathogenicity (Sundravada *et al.*, 2012).

Lotfalinezhad *et al.* (2013) collected forty three isolates of *M. phaseolina* on different hosts (canola, cotton, melon, olive, pine, potato, safflower, soybean, sunflower, tomato and watermelon) from North, Northwest and Southwest of Iran. The optimum temperature for growth was 25 °C for 33 isolates and 33 °C for 10 isolates, when these isolates were incubated at 5, 10, 15, 20, 25, 30, 33, 35 and 40 °C. Isolates growth rate varied considerably at all temperatures but maximum variability occurred at 20 and 33 °C between isolates. They concluded environmental condition of the North and Northwest areas of Iran were similar but differ from those of the Southwest. High optimum temperature was detected for growth of southwest isolates as they were from warm area in contrast to other isolates.

Influence of seven temperatures regimes (15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C and 45 °C) were tested on growth of *R. bataticola* isolates representing Madhya Pradesh, Karnataka, Maharashtra, Andhra Pradesh and Telangana states of India. The maximum mycelial growth was observed at 35 °C followed by 30 °C in all the isolates (Srinivas *et al.*, 2017).

2.4.2 Effect of temperature on disease development

Dry root rot is becoming more intense in typically tropical humid areas. Increasing incidence of dry root rot at various locations over years suggests a strong influence of rising temperatures on *R. bataticola* (Savary *et al.*, 2011). Data collected in India from 2000 to 2010 showed higher incidence of dry root rot in chickpea varieties that are resistant to Fusarium wilt in years when temperatures exceed 33°C at flowering to podding stages (Pande *et al.*, 2010 and Sharma *et al.*, 2010).

Agarwal *et al.* (1973) reported that charcoal rot in soyabean found to occur only at temperatures between 25 to 40 °C. In chickpea, seed exudates increased when incubated at 35 °C than 15 and 25 °C was reported by Singh and Mehrotra (1982). They also observed that increased seed exudation was a major factor contributed to increased pre-emergence damping off of gram seedling by *R. bataticola* at high temperatures. Ratnoo *et al.* (1997) reported the infection and development of ashy grey stem blight in cowpea caused by *M. phaseolina* was most favoured by high temperature 25 – 40 °C.

Severity of dry root rot in chickpea with respect to temperature was studied by Sharma and Pande (2013) under controlled environment. Out of five temperature regimes viz., 15 °C, 20 °C, 25 °C, 30 °C and 35 °C, optimum temperature for disease severity was 35 °C with maximum disease severity of 9 on 1-9 severity rating scale. This was followed by 30 °C, 25 °C, 20 °C and 15 °C. The disease severity was very low (2 - 3 rating on 1-9 scale) at 20 and 25 °C. The control plants did not show any symptoms at their respective temperatures. They confirmed that high temperature predisposed chickpea to *R. bataticola* infection, colonization and development.

Influence of seven temperatures regimes (15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C and 45 °C) were tested on growth of *R. bataticola* isolates representing Madhya Pradesh, Karnataka, Maharashtra, Andhra Pradesh and Telangana states of India. The optimum temperature for dry root rot severity rating was at 35 °C (8.5) followed by 30 °C (7.9) followed by 25 °C (7.0). Among the isolates, Telangana isolate was virulent and caused maximum disease severity. (Srinivas *et al.*, 2017).

2.4.3 Effect of moisture on disease development

When cotton plants grown in soil temperatures of 20 – 40 °C were subjected to soil water stress, the severity of *M. phaseolina* was much greater in moisture stress

condition than those provided with sufficient soil water (Ghaffar and Erwin, 1969). Dry root rot was observed in chickpea when exposed to temperature more than 30 °C. The disease was aggravated by dry soil conditions especially at flowering stage leading plants to sudden drying (Singh and Mehrotra, 1982).

Colonization of *M. phaseolina* on chickpea roots was high when subjected to moisture stress conditions compared to unstressed plants. Soils artificially inoculated produced poor growth of plants both in low and high moisture compared to natural soil. Root colonization was greater at 40-50% moisture holding capacity as compared to 10-20% soil moisture holding capacity (Husain and Ghaffar, 1995). Ratnoo *et al.* (1997) reported the infection and development of ashy grey stem blight in cowpea caused by *M. phaseolina* was most favoured by higher temperature 25-40 °C and low in flooded soil compared to dry soil of 40-60% soil moisture.

Role of edaphic factors on the incidence of dry root rot of sesame caused by *R. bataticola* was studied by Rajkrishan *et al.* (1999). Among various soil types, sandy soil supported maximum disease incidence (78.33%) as compared with clay soil (51.56%). More disease incidence of 48.33% was observed when pots were irrigated after the interval of seven days as compared to pots receiving irrigation every day. Wokocho (2000) observed the disease incidence of dry root rot of soybean was significantly low when the soil moisture was 60 - 70% but it was 5.0% at 10 - 20% soil moisture.

Patel and Anahosur (2001) conducted an experiment with chickpea cultivars ICC 4951 (susceptible) and Bheema (resistant) in earthen pots filled up with artificially infected soil with *R. bataticola* at four levels of soil moisture viz., 25, 50, 75 and 100%. They observed *R. bataticola* infection was maximum at 25 per cent and minimum at 100% soil moisture. ICC 4951 developed higher disease than Bheema at 25% soil moisture. Both cultivars showed maximum disease incidence at the lowest soil moisture level compared to high moisture.

Role of soil moisture on disease severity of dry root rot in chickpea was studied by Sharma and Pande (2013), under controlled environment. The plants exposed to 40% moisture stress showed higher mortality as compared to 60, 80 and 100%. It was observed that 40% soil moisture was insufficient for the normal growth of the plants as the plants grown in control showed the physiological stress (wilting of the plants due to lack of moisture). At 60% soil moisture, no physiological stress was found in control plants and

dry root rot severity was maximum 35 days after imposing moisture stress. Disease progressed slowly at 80 and 100% soil moisture had least disease severity.

Srinivas (2016) studied on the effect of soil moisture on dry root rot of chickpea using different moisture levels in black and red soils. Result of experiment shown that, in black soil the disease severity rating was 9.0, 8.9, 8.1, 6.9, 5.3, 3.8 and 2.3 while in red soil, it was 9.0, 8.9, 8.7, 7.9, 6.5, 4.7 and 3.0 at 40, 50, 60, 70, 80, 90 and 100% soil moisture respectively. The disease severity decreased as the soil moisture increased in both the types of soil. There was significant difference between them.

2.4.3 Effect of carbon dioxide on disease

Cheng *et al.* (2009) conducted a pot experiment in four controlled environment chambers with two levels of CO₂ (ambient, 380 ppm; and elevated, 680 ppm) and two levels of night temperature (22 and 32 °C). The authors observed that the whole plant and stem dry weight were decreased by both elevated CO₂ and high night temperature. And also several studies have shown that increased temperature may reduce the mass of grains even under high levels of CO₂ (Horie *et al.*, 2000), while increasing floral sterility (Jagadish *et al.*, 2007).

Baker (2004), studied the effect of elevated CO₂ (700 µmol mol⁻¹) under different temperature regimes (24, 28, 32, 36 and 40 °C), found no increase in grains with enriched CO₂ atmosphere.

2.5 Identification of dry root rot resistant sources of chickpea

Pande *et al.* (2004) tested 47 lines of chickpea against the dry root rot, among those one germplasm accession (ICC 14395), a cultivar (ICCV 2) and an advanced (ICCX830203-BH-BH-11H) breeding line were resistant to dry root rot and the remaining 22 lines moderately resistant, 19 susceptible and 3 highly susceptible.

Four seventy germplasm lines of chickpea were tested against the dry root rot, among those KG-86 KWR-4, KWR-108 and KWR-277 are found as a resistant genotype (Mishra *et al.*, 2005). Jayalakshmi *et al.* (2008) tested 12 genotypes against root rot disease 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot disease. Chaturvedi and Dua (2009) reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot.

Soregaon and Ravikumar (2010) screened fifteen diversified genotypes against fusarium wilt of chickpea with two RAPD markers. The primer pair CS-27F/ CS-27R, termed allele specific associated primer (ASAP) linked to the allele for susceptibility at H₁ locus and RAPD marker A07C linked to H₂ locus of susceptibility for fusarium wilt. The CS 27700 marker linked to H₁ locus was absent in all other genotypes except JG 62 and Karikadle. Other marker A07C 417 linked to H₂ locus was present in all genotypes except WR 315 and GBS 964. This indicated that both markers CS 27 and A07C were present in susceptible genotypes *i.e.* JG 62 and Karikadle. On the contrary both the markers were absent in resistant genotypes *i.e.* WR 315 and GBS 964. Presence of any one marker in other genotypes indicates susceptible but late wilting.

Gupta *et al.* (2012) screened one seventy chickpea accessions against dry root rot disease, out of which 68 genotypes exhibited resistant reaction (<10% mortality), of which 26 were the promising lines namely (JG1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas) from JNKVV, Jabalpur. These lines were further evaluated for their performance in sick field for three consecutive years which revealed that six lines *viz.*, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas as resistant exhibiting <10 per cent mortality, however 14 lines showed moderately resistant reaction.

2.6 Integrated management of dry root rot of chickpea

Species of *Trichoderma* and *Gliocladium* have exhibited promise in glasshouse and in field trials for the control of chickpea root rot, collar rot and wilt pathogens (Mukhopadhyay *et al.*, 1992; Mukhopadhyay, 1995 and Tewari and Mukhopadhyay, 2001).

A variety of soil micro-organisms have been demonstrated in the control of fungal plant pathogens including *R. bataticola*. *Trichoderma harzianum*, *Bacillus subtilis* and *Pseudomonas fluorescens* have been used as biological control agents against *R. bataticola* in French bean and chickpea (Mukherjee and Tripathi, 2000).

Saikia *et al.* (2003) tested the efficiency of *P. fluorescens* with or without Salicylic acid amendment in chickpea against *Fusarium* wilt infection. The application of

P. fluorescens (pf4-92) with SA recorded highest protection of chickpea seedlings against wilting.

Biologically and chemically treated chickpea seeds observed after one month showed that vitavax (carboxin @ 1 g/kg seed) did not significantly affect the spore viability of *Gliocladium virens* (Gv). Application of carboxymethyl cellulose (CMC) with *G. virens* powder in combination with vitavax provided maximum protection (81.9%) to the crop against chickpea root rot and collar rot pathogens in glasshouse. Chickpea seeds treated with Gv powder+ CMC+vitavax significantly increased seedling emergence (47.9%); final plant stand (85.8%) and grain yield (79.7%) which was statistically at par with the treatment Gv powder+vitavax and Gv suspension + vitavax in a sick plot (Tewari and Mukhopadhyay, 2003).

Singh *et al.* (2003) the efficacy of *Trichoderma harzianum*, *T. viride*, *T. hamatum*, *Gliocladium virens*, *Pseudomonas fluorescens* and *Bacillus subtilis* in controlling *R. bataticola* causing dry root rot in chickpea was determined *in vitro* and in field experiment conducted in Kanpur, Uttar Pradesh, India 2001-02. *T. harzianum* recorded the highest control of the pathogen both *in vitro* and *in vivo*. In fungicidal trails on management of dry root rot of chickpea caused by *R. bataticola*, Carbendizim (0.2%) and Etaconazole (0.1%) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 and 18.2% highest grain yield of 192 and 18.9 q/ha respectively (Vijay Mohan *et al.*, 2006).

Paramasivan *et al.* (2007) tested the fungal and bacterial antagonists against dry root rot pathogen in which *T. virens* (Isolate 4) resulted in maximum growth inhibition followed by *T. harzianum* and least growth inhibition by *P. fluorescens*. The per cent inhibition by all other fungal and bacterial antagonists ranged between 41.0 to 56.0% respectively.

The plant growth promoting rhizobacteria (PGPR) viz., *Azotobacter* sp., *Bacillus cereus*, *B. megaterium* and *B. subtilis* produce biologically active compounds (antibiotics and toxic substances) that have antifungal activity, besides bioactive compounds, including plant growth regulators (El-Mohamedy & Ahmad, 2009; Sharma *et al.*, 2009; Baset *et al.*, 2010; Almeida *et al.*, 2011; Ismail *et al.*, 2011; Abdel Monaim *et al.*, 2012 and Asma *et al.*, 2013).

Two bio-control agents were evaluated individually and in combinations and in integration with carbendazim seed treatment and hexaconazole seed soaking for management of dry root rot of chickpea. Seed and soil application with bioagents was effective for the maximum germination with less root rot incidence as against seed treatment with carbendazim alone and seed soaking in hexaconazole alone integration of seed and soil application of biocontrol agents resulted in higher germination and reduced mortality due to disease as against check. The application of biocontrol agents individually either soil application or seed treatment also showed good germination with reduced mortality due to dry root rot as compared to treatments imposed with fungicides (Manjunatha *et al.*, 2011).

Seed treatment with carbendazim @ 2g/kg of seed+ seed treatment with *T. viride* @ 4 g/kg of seed + soil application of FYM fortified with *T. viride* recorded (T₇) was the best treatment to decrease the per cent mortality of root rot caused by *R. bataticola*. It also recorded highest growth parameters like shoot length and root length and also highest yield (Nagamani *et al.*, 2011).

The efficacies of eight fungicides against *R. bataticola* were revealed, depending upon their inhibitory effect on the growth of the fungus. Among eight fungicides, mancozeb, carbendazim companion, copper-oxy-chloride and benomyl completely inhibited the growth of the fungus compared to control (Khan *et al.*, 2012).

Ten *Trichoderma* spp. were isolated from chickpea rhizosphere and root endophytic region by using serial dilution technique and purified by single hyphal tip method. Out of the ten isolates tested against *R. bataticola*, *Trichoderma* isolate-7 showed highest inhibition percentage (83.33). In compatibility tests with commonly used fungicides, *Trichoderma* isolate-7 showed highest compatibility with validamycin (72.22%) followed by copper oxychloride (66.66%) (Veena *et al.*, 2013).

Integrated management study of root rot disease of chickpea under field condition showed that soil application of *T. harzianum* @ 5 kg in 500 kg neem cake/ha in furrow 5 days prior to sowing resulted in higher seed germination (74.90%). In case of disease incidence and yield of crop, soil application of *T. harzianum* @ 5 kg in 500 kg FYM/ha in furrow 5 days prior to sowing has lowest disease incidence (11.81%) and gave highest yield (1533 kg/ha) (Dhingani and Solanky, 2016).

Material and Methods

III. MATERIAL AND METHODS

All the investigations were conducted under laboratory and glasshouse during 2014-15 and 2015-16 at Department of Plant Pathology, University of Agricultural Sciences, Raichur. Raichur which is situated in North eastern dry zone (zone-2) of Karnataka state at 16° 12' N longitude, 77° 21' E latitude and at an altitude of 389.37 m above mean sea level. The details of the materials used and the methodology followed are described under this chapter.

3.1 General laboratory procedure

3.1.1 Glasswares and culture media sterilization

Laboratory experiments were conducted using Borosil, Qualigens and Technico glasswares. The glasswares were kept in the cleaning solution containing potassium dichromate ($K_2Cr_2O_7$), concentrated sulphuric acid (H_2SO_4) at 60 g and 60 ml respectively in one litre of water. All glassware's, solid and liquid culture media were sterilized by autoclaving at 1.1 kg pressure sq^{-1} cm (121.6 °C) for 20 minutes. Soil used for experiments was sterilized for two hours at 1.33 kg pressure/sq. cm in an autoclave. All cultural studies were conducted in aseptic condition under laminar air flow. The blades, scissor, fungal inoculation needle and forceps were sterilized by autoclaving, surface sterilization by 70 per cent alcohol and using flame.

3.1.2 Potato dextrose agar

The following nutrient media were used for isolation, culturing and maintenance of the *R. bataticola*. The composition of the PDA is as follows.

Agar-Agar	: 20.00 g.
Dextrose ($C_6H_{12}O_6$)	: 20.00 g.
Potato	: 20.00 g.
Distilled water (Volume to make up)	: 1000.00 ml

Two hundred grams of peeled potatoes were cut into small pieces and boiled in distilled water and then extract was collected by filtering through muslin cloth. Dextrose and agar (20.0 g each) were dissolved in the potato extract and the final volume was made

up to 1000 ml with distilled water and sterilized as described earlier and preserved for further use.

3.1.3 Potato dextrose broth

Potato dextrose broth was prepared as same as that of potato dextrose agar except the addition of solidifying agent agar agar. Two hundred gram of peeled potatoes were cut into small bits, boiled in distilled water and then extract was collected by filtering through muslin cloth. Dextrose 20 g was dissolved in the potato extract and the final volume was made up to 1000 ml with distilled water and sterilized as described earlier and preserved for further use.

3.1.4 Preparation of PDA slants

PDA slants were prepared by transferring 3 ml of the medium to culture tubes. The tubes were plugged with non-absorbant cotton and sterilized in an autoclave. After sterilization, the tubes were removed from the autoclave when they were still in hot condition (*i.e.* approximately 40 °C) and kept in a slanting position for the medium to solidify. After solidification, the slants were kept in refrigerator for further use.

3.1.5 Plating of medium

The sterilized medium was melted and distributed in petriplates (9 mm diameter) at the rate of 20 ml per plate aseptically in a laminar air flow chamber and allowed to solidify. The plates containing the medium were used for culturing and maintenance of fungus in the laboratory.

3.2 Isolation and purification of *R. bataticola* isolates

Chickpea plants showing typical dry root rot symptoms were washed under running tap water and blot dried. Infected roots were cut into pieces of 5-6 mm size and were surface sterilized by dipping in 0.8 per cent sodium hypochlorite for 2 min. After thorough washing in sterile distilled water, the pieces were then transferred by using forceps on to sterilized potato dextrose agar (PDA) medium in Petri dishes and incubated at 25 ± 2 °C to obtain mycelial growth. After 48h of incubation, hyphal tips of the growing mycelium were marked on the underside of the petri dish with a glass marker by viewing through a light microscope. The hyphal tips from margins of resulting colonies were cut with the help of sterilized 2 mm cork borer and transferred to petri dish

containing PDA. The cultures were purified by single sclerotial isolation and transferred to PDA slants. On the basis of morphological characters of mycelium and sclerotia, the isolates were identified as *R. bataticola*. The pure cultures of all the 60 isolates grown on PDA slants were stored at $4 \pm 1^\circ\text{C}$ for further studies.

3.2.1 Proving the pathogenicity

3.2.2 Mass multiplication of *R. bataticola*

In order to get mass multiplication of inoculum, sorghum grain media was used. About 200 g sorghum grains were soaked in 2 per cent sucrose solution for 16 hr, these sorghum grains were filled in $\frac{1}{4}^{\text{th}}$ of 1000 ml conical flasks and autoclaved at 121°C for 45 min. Each flask was seeded with a mycelial disc (1 cm) from a 10 days old culture of *R. bataticola* grown on PDA under aseptic conditions and incubated at $28 \pm 2^\circ\text{C}$ for 20 days. The flasks were shaken on alternate days to get uniform growth. The giant culture so obtained was used for preparing sick pots in glass house.

3.2.3 Raising of seedlings

Twenty days old inoculum developed on sorghum grain medium was used to conduct pathogenicity test. The inoculum 50 g/kg of soil was mixed in sterilized soil. Plants for untreated control were kept without the inoculation of the inoculum. After 10 days of incubation, chickpea seedlings of the susceptible genotype Annigeri-1 were sown in the pots (30 cm diameter). Observations on root rot infection were recorded. Reinoculation was made from such symptoms produced on plant tissue to compare with that of original isolate for conformity.

3.3 To survey for incidence of dry root rot of chickpea in North Karnataka

A roving survey was conducted to record the occurrence and distribution of dry root rot of chickpea in 8 districts of the major chickpea growing areas in North Karnataka viz., Bidar, Kalaburagi, Yadagiri, Raichur, Koppal, Ballari, Bijapur and Dharwad during *rabi*, 2015-16 and 2016-17. In each district, three taluks were selected, in each taluk minimum two villages were selected, in each village minimum three fields were selected and in each field, five plots of 10 m^2 were selected and the incidence of disease was recorded after counting the diseased and healthy plants. While surveying, data on variety grown, soil type, plant protection measures taken, **number of infected plants and total**

number of plants observed, **were recorded**. Chickpea plants showing the typical dry root rot symptoms were collected from surveyed areas, packed in labeled paper bags and brought to the laboratory for isolation of the pathogen. Based on observations the disease incidence was calculated by following formula.

$$\text{Dry root rot incidence (\%)} = \frac{\text{Number of plants infected}}{\text{Total number of plants}} \times 100$$

Meantime, the co-ordinates of surveyed fields were also recorded with the help of GPS and details of co-ordinates are listed in Table-1 and Table-2 and **the isolates were designated as Rb-1 to Rb-60 were listed in table-3**.

3.4 Studies on cultural, morphological and molecular variability of pathogen

3.4.1 Cultural and morphological characterization of *R. bataticola*

All the 60 isolates of *R. bataticola* were characterized for morphological and cultural characteristics. The mycelial discs of 5 mm diameter were cut from the edge of a three days old culture and transferred aseptically to 90 mm Petri dish containing 15 ml PDA. These plates were incubated at 27 ± 2 °C with 12 h photoperiod. Each treatment was replicated thrice. The colonies of isolates were characterized for growth rate at 72 h after incubation. Seven days old cultures were used to record radial growth, colony morphology, colony pigmentation, width of **hyphae** and aerial mycelium. At the twentieth day after incubation, production of sclerotia, number of sclerotia/microscopic field (10x), sclerotial size and shape were recorded by using Q-capture image analyzer.

3.4.1.1 Cultural variability of *R. bataticola*

Cultural characters such as radial growth, colony morphology, colony pigmentation and aerial mycelium were studied by growing *R. bataticola* isolates on PDA at 27 ± 2 °C.

3.4.1.1.1 Radial growth

Fifteen ml of PDA medium was poured in sterilized Petriplates and allowed to solidify. Mycelial disc of 5 mm diameter was cut from the margin of the three day old culture of *R. bataticola* for each isolate and placed at center of the Petriplate under aseptic

Table 1. Co-ordinates of villages selected for survey of dry root rot of chickpea during rabi, 2015-16

District	Taluks	Villages	Longitude	Latitude
Bidar	Bidar	Bidar	17.897105	77.526881
		Andura	17.913207	77.405731
		Manhalli	17.771061	77.507000
		Gadgi	17.953411	77.545862
	Bhalki	Halahalli	18.029307	77.340514
		Madgatti	18.005446	77.150970
		Godhiperga	18.127828	77.329884
	Humanabad	Humanabad	17.761088	77.142910
		Sanahallikhed	17.855169	77.269616
		Chitguppa	17.687892	77.210164
		Bachnal	17.622143	77.009979
Kalaburgi	Kalaburgi	Kalaburgi	17.361080	76.814770
		Kadagancha	17.441848	76.662136
		Itga Khandal	17.231929	76.845382
		Sannur	17.404638	76.680398
		Kamalapur	17.571672	76.981425
	Aland	Aland	17.594733	76.553024
		Bhusnur	17.455346	76.544972
		Tadola	17.691027	76.537503
		Khajuri	17.680116	76.564148
	Jewargi	Raddewadgi	17.029994	76.808640
		Jewargi	17.020578	76.781226
		Andola	16.960342	76.865528
		Ijeri	16.916100	76.697632
		Nelogi	17.049348	76.620339

Table 1. Contd....

District	Taluks	Villages	Longitude	Latitude
Yadagiri	Yadagiri	Yadagiri	16.725221	77.106478
		Mailapur	16.740168	77.244887
		Munderga	16.785419	77.191905
		Gurmitkal	16.854045	77.390804
	Shahapur	Bheemarayangudi	16.732428	76.799144
		Vibhutihalli	16.680355	76.853580
		Gogi	16.726805	76.757560
		Madriki	16.795874	76.786894
	Shorapur	Malanoor	16.448429	76.453206
		Devapur	16.438709	76.731338
		Tintani	16.399160	76.663228
		Kakkeri	16.373908	76.606067
		Manjalapur	16.418384	76.533654
		Aldahal	16.473441	76.712408
Raichur	Raichur	Raichur	16.199280	77.336521
		Kallur	16.149537	77.239300
		Kalmala	16.184329	77.221864
		Kasbe camp	16.1630194	77.2548388
		Sirawar	16.164593	77.063903
	Lingasugur	Bhupur	16.115301	76.462964
		Honnahalli	16.213370	76.559755
		Hatti	16.175674	76.605108
	Devadurga	Masarakal	16.368906	77.039925
		Jalahalli	16.368626	76.768741
		Kollur (M)	16.500680	76.918887
		Huvinahadgi	16.480466	76.922042

Table 1. Contd....

District	Taluks	Villages	Longitude	Latitude
Koppal	Koppal	Hyati	15.256968	76.179048
		Halagere	15.084096	76.574033
	Kushtagi	Kyadiguppa	15.904767	76.154074
		Matur	15.888317	76.213779
	Yelbarga	Bewoor	15.562820	76.185240
		Kuknoor	15.481605	76.007042
Ballari	Ballari	Kolagallu	15.195614	76.878232
		Belagallu	15.128429	76.832356
	Siraguppa	Siraguppa	15.617373	76.908835
		Halekota	15.553795	76.903061
	Kudligi	Kudligi	14.906958	76.389448
		Hansi	14.956646	76.297143
Vijayapura	Vijayapura	Vijayapura	16.802953	75.663548
		Arakeri	16.931608	75.660793
		Jalageri	16.926457	75.631757
	Muddebihal	Kuntoji	16.374913	76.165753
		Gangur	16.239120	76.082410
	Sindagi	Sindagi	16.929132	76.239687
		Golageri	16.801919	76.363606
		Bableshtar	17.01958	76.253395
Dharwad	Dharwad	Dharwad	15.496338	74.980106
		Yettingudda	15.487509	74.995014
		Kallapur	15.581384	75.031031
	Nargund	Nargund	15.721025	75.389690
		Tadhal	15.669251	75.456266
	Navalgund	Yamanur	15.538961	75.297595
		Bhadrapur	15.386460	75.387481
		Javoor	15.619934	75.234168

Table 2. Co-ordinates of villages selected for survey of dry root rot of chickpea during *rabi*, 2016-17

District	Taluks	Villages	Longitude	Latitude
Bidar	Bidar	Bidar	17.898848	77.527213
		Andura	17.910527	77.402570
		Astoor	17.933991	77.557930
		Manhalli	17.762704	77.505922
		Gadgi	17.943497	77.546724
	Bhalki	Halahalli	17.886974	77.350925
		Madgatti	17.998072	77.150559
		Nidoda	18.140833	77.317351
		Godihipperga	18.037002	77.284958
		Halbarga	17.991792	77.337351
	Humanabad	Nimbura	17.77571	77.23956
		Sannahalliked	17.674737	77.070224
		Chitguppa	17.690959	77.220274
		Humanabad	17.766160	77.136736
Kalaburgi	Kalaburgi	Kalaburagi	117.361016	76.815173
		Kadagancha	17.445095	76.654380
		Nilkod	17.482071	76.830975
		Itga Khandal	17.228785	76.820469
		Sannur	17.296936	76.904852
	Aland	Aland	17.593719	76.574661
		Bhusnur	17.457590	76.544086
		Tadola	17.595280	76.547649
		Khajuri	17.691410	76.563869
	Jewargi	Raddewadgi	17.036648	76.798990
		Jewargi	17.024069	76.776921
		Andola	16.945849	76.838435
		Nelogi	17.048970	76.613197

Table 2. Contd....

District	Taluks	Villages	Longitude	Latitude
Yadagiri	Yadagiri	Yadagiri	16.736225	77.153476
		Mailapur	16.737274	77.240700
		Saidapur	16.556863	77.247064
		Munderga	16.788272	77.186410
	Shahapur	Bheemarayangudi	16.723267	76.806538
		Vibhutihalli	16.686494	76.845954
		Hattigudar	16.596514	76.877137
		Biranur	16.542008	76.87685
	Shorapur	Malanoor	16.467321	76.393373
		Devapur	16.434149	76.716162
		Hunasagi	16.487229	76.519042
		Tintani	16.376292	76.656132
		Kakkeri	16.377880	76.611179
		Manjalapur	16.427346	76.564281
		Aldahal	16.500822	76.614396
Raichur	Raichur	Raichur	16.195914	77.322161
		Gabbur	16.309939	77.151692
		Kalmala	16.199629	77.215199
		Yargera	16.274539	77.376985
		Kasbe camp	16.155243	77.251602
	Lingasugur	Karadkal	16.151586	76.565077
		Honnahalli	16.211007	76.548341
		Hatti	16.177939	76.668947
	Devadurga	Masarakal	16.368600	77.024031
		Jalahalli	16.362033	76.781033
		Ganadhal	16.299689	76.782525
		Huvinahadgi	16.479897	76.922332

Table 2. Contd....

District	Taluks	Villages	Longitude	Latitude
Koppal	Koppal	Hyati	15.258942	76.186680
		Halagere	15.378362	76.077404
	Kushtagi	Kyadiguppa	15.894825	76.154915
		Matur	15.881914	76.217894
	Yelbarga	Bewoor	15.571000	76.180646
		Bandi	15.306581	76.170313
Ballari	Ballari	Kolagallu	15.192084	76.866249
		Belagallu	15.128338	76.829552
	Siraguppa	Siraguppa	15.900576	76.247735
		Halekota	15.563748	76.891611
	Kudligi	Kudligi	14.904919	76.393205
		Amalapur	14.795700	76.437164
Vijayapura	Vijayapura	Vijayapura	16.765885	75.744378
		Toravi	16.781632	75.621745
		Jalageri	16.937539	75.614927
	Muddebihal	Kuntoji	16.358251	76.180538
		Gangur	16.261005	76.088106
	Sindagi	Sindagi	16.926966	76.236421
		Yankanchi	16.883339	76.331653
		Madbal	16.812732	76.285759
Dharwad	Dharwad	Dharwad	15.490488	74.981003
		Yettingudda	15.555546	75.040422
		Kallapur	15.581486	75.031261
	Nargund	Nargund	15.724136	75.377341
		Banahatti	15.681722	75.411653
	Navalgund	Yamanur	15.532322	75.316156
		Bhadrapur	15.384732	75.393169
		Javoor	15.497238	75.397837

Table 3. Source and designation of the isolates of *R. bataticola* in chickpea

Sl. No.	Place of collection			Name of the isolate
	District	Taluk	Village	
1.	Bidar	Bidar	Bidar	Rb-1
			Andura	Rb-2
		Bhalki	Madgatti	Rb-3
		Humanabad	Humanabad	Rb-4
			Sannahalliked	Rb-5
			Bachnal	Rb-6
2.	Kalaburgi	Kalaburgi	Kalaburgi	Rb-7
			Khadagancha	Rb-8
			Kamalapur	Rb-9
			Sannur	Rb-10
		Aland	Khajuri	Rb-11
			Aland	Rb-12
			Bhusnur	Rb-13
		Jewargi	Jewargi	Rb-14
			Raddewadgi	Rb-15
			Ijeri	Rb-16
3.	Yadgiri	Yadgiri	Yadagiri	Rb-17
			Gurumitkal	Rb-18
		Shahapur	Bheemarayanagudi	Rb-19
			Gogi	Rb-20
			Vibhutihalli	Rb-21
		Shorapur	Kakkeri	Rb-22
			Malnoor	Rb-23
			Hunasagi	Rb-24
4.	Raichur	Raichur	Raichur	Rb-25
			Yergera	Rb-26
			Kallur	Rb-27
			Kalmala	Rb-28
		Lingasugur	Huvinahadgi	Rb-29
			Karadkal	Rb-30
			Bhupur	Rb-31
		Devadurga	Masarkal	Rb-32
			Jalahalli	Rb-33
			Hatti	Rb-34

Table 3. Contd....

Sl. No.	Place of collection			Name of the isolate
	District	Taluk	Village	
5.	Koppal	Koppal	Hyati	Rb-35
			Halagere	Rb-36
		Kushtagi	Matur	Rb-37
			Kyadiguppa	Rb-38
		Yelburga	Hanampur	Rb-39
			Bewoor	Rb-40
			Kuknur	Rb-41
6.	Ballari	Ballari	Kolagallu	Rb-42
			Belagallu	Rb-43
		Siraguppa	Siraguppa	Rb-44
			Halekota	Rb-45
		Kudligi	Kudligi	Rb-46
			Amalapur	Rb-47
7.	Vijayapura	Vijayapura	Vijayapura	Rb-48
			Toravi	Rb-49
		Muddebihal	Gangur	Rb-50
			Kuntoji	Rb-51
		Sindagi	Sindagi	Rb-52
			Bableshwar	Rb-53
			Madabal	Rb-54
			Golageri	Rb-55
8.	Dharwad	Dharwad	Dharwad	Rb-56
			Kallapur	Rb-57
		Navalgund	Yamanur	Rb-58
			Javoor	Rb-59
			Bhadrapura	Rb-60

conditions. The plates were incubated for seventh days in an incubator at 27 ± 2 °C and the radial growth for each isolate was measured in mm by using Q-capture image analyzer.

3.4.1.1.2 Colony morphology

Observation on colony morphology was recorded at seventh day after inoculation. Colony growth of each of isolate was recorded as fluffy, appressed and velvety.

3.4.1.1.3 Colony pigmentation

Observation on colony pigmentation was recorded at seventh day after inoculation. Colony pigmentation of each of isolate was recorded as black with grey aerial mycelia, black, grey and white pista colour.

3.4.1.1.4 Aerial mycelium

Observation on aerial mycelium was recorded at seventh day after inoculation. Presence and absence of aerial mycelium growth of each of isolate was recorded.

3.4.1.2 Morphological variability of *R. bataticola*

Morphological characters **which** include width of the hyphae, microsclerotial production, number of days to produce microsclerotia, number of microsclerotia produced, shape and size of microsclerotial **were** studied.

3.4.1.2.1 Width of the hyphae

Five mm mycelial disc of each isolate was placed in the center of Petriplate containing PDA medium under aseptic conditions. The seventh day old cultures were measured for width of the hyphae under Q-capture image analyzer.

3.4.1.2.2. Microsclerotial production

Five mm mycelial disc of each isolate was placed in the center of Petriplate containing PDA medium under aseptic conditions. The plates were incubated for 20 days and then observed for the presence or absence of microsclerotial bodies.

3.4.1.2.3 Number of days to produce microsclerotia

Five mm mycelial disc of each isolate was placed in the center of Petriplate containing PDA medium under aseptic conditions. The plates were incubated in an

incubator at 27 ± 2 °C. Number of days taken by an each isolate to produce microsclerotia were recorded.

3.4.1.2.4 Number of microsclerotia produced

Fully grown culture was observed under the Q-capture image analyzer and then number of microsclerotia was counted per microscopic field (10x). Three replications were maintained for each isolate.

3.4.1.2.5 Shape of the microsclerotia

Fully grown cultures were observed in the Q-capture image analyzer for shape of the microsclerotia and recorded as round, irregular and ovoid shapes.

3.4.1.2.6 Size of microsclerotia

Individual microsclerotial bodies were focused in Q-capture image analyzer and size of the sclerotia was measured.

3.4.2 Identification and molecular variability of *R. bataticola* isolates

Twenty isolates of *R. bataticola* representing different geographic regions were selected and used for detection using ITS 1 and ITS 4 fungal universal primers and molecular variability by using Random Amplified Polymorphic DNA (RAPD) primers. Standard protocols were used for the isolation of DNA and RAPD analysis according to [Liu *et al.* \(2000\)](#). *R. bataticola* isolates were selected based on location, a total 20 isolates were selected from eight districts to study the molecular diversity. List of isolates selected for molecular diversity study are listed hereunder (Table 5).

3.4.2.1 Fungal cultures (*R. bataticola*)

Mycelial discs of respective isolates of *R. bataticola* from the periphery of an actively growing 7 days old culture were grown on 30 ml potato dextrose broth contained in 100 ml flask for 5 days at 27 ± 2 °C. Resulting fungal mat was placed on the Whatman No. 1 filter paper in funnel to drain excess of medium for 30 min and kept in the deep freeze for overnight before homogenizing it. Later, fungal mat was used for isolation of total genomic DNA.

3.4.2.2 DNA extraction

Total DNA of *R. bataticola* was extracted using Cetyl-Trimethylammonium Bromide (CTAB) method from young vegetative mycelium using the procedure given by Murray and Thompson (1980) with slight modifications. Cultures were grown at 28 °C in an incubator without shaking in 100 ml conical flasks containing 30 ml of potato dextrose broth (PDB). Mycelium was harvested after 5 days, preceding sclerotium formation. The mycelium was powdered in liquid nitrogen. Fungal mat of 0.5 g grounded to fine powder in liquid nitrogen and transferred to sterile eppendorf tube (Tarsons Products Pvt. Ltd, Kolkata, India). To this, one ml of extraction buffer (1 M Tris-5.0 ml; 5 M NaCl-14.0 ml; 0.5 M EDTA- 2.0 ml; 0.1 per cent Mercaptoethanol-100.0 µl; 2 per cent CTAB-1.0g; 1 per cent PVP-0.5 g) was added and incubated for 1 h in water bath at 65 °C.

Then the tubes were centrifuged at 10,000 g for 10 minutes at room temperature. The supernatant was transferred into other tubes. To the supernatant, equal ml of chloroform and isoamyl alcohol (24:1) and RNase (1 µl/100 µl) was added and incubated at room temperature for 10-20 minutes. The tubes were centrifuged at 10,000 g for 10 minutes, separated the supernatant and added 0.6 vol of ice cold isopropanol + 0.1 vol of sodium acetate and incubated at -20 °C for overnight. Next day, the tubes were centrifuged at 13,000 g for 20 min at 4 °C, the supernatant was discarded and the pellet was washed with 70 per cent ethanol and centrifuged at 13,000 g for 2 min at 4 °C. Again the supernatant discarded, the pellet air dried and dissolved in 100 µl of sterile distilled water or TE. The DNA samples were stored at -20 °C for further studies.

3.4.2.3 Agarose gel electrophoresis

Agarose gel of 1.5 per cent was prepared by 1.5 grams of agarose was added to a conical flask containing 250 ml of 1 x TAE buffer. The agarose was melted by heating the solution in an oven and the solution was stirred to ensure even in mixing and complete dissolution of agarose. The solution was allowed to cool to 60°C. After cooling, 5µl of ethidium bromide solution was added and the resulting mixture was poured into the gel casting tray for solidification. Before the gel solidified, an acrylic comb of desired well number was placed on the agarose solution to form wells for loading the samples. Each well was loaded with 15µl of amplified product and 1µl orange dye. The gel was run at 70V for 20min. After completing the electrophoresis run, DNA bands on the gel was visualized under UV light and photographed.

3.4.2.4 Measurement of DNA concentration

The quality and quantity of DNA was analysed by running 2 µl of each sample mixed with 1 µl of 10x loading dye on 1.5 per cent agarose gel. The gel was visualized on UV transilluminator and photographs were taken using a gel documentation system. The quality and quantity of DNA was also measured by spectrophotometer (Nanodrop ND 1000) to get desired concentration of 15-20 ng/µl.

3.4.3 PCR amplification for ITS region

PCR (Mastercycler, Hamburg, Germany) amplifications of the Internal Transcribed Spacers (ITS) of rDNA was performed by using universal primers ITS-1 (5'CCTGTGCACCTGTGAGACAG-3') as forward primer and ITS-4 (5'-TGTCCAAGTCAATGGACTAT-3') as reverse primer (Camporota *et al.*, 2000). Amplification was carried out in 0.2 ml eppendorf tubes with 25 µl reaction mixture.

PCR mixers (25 µl) per reaction

i.	10x PCR buffer	2.5 µl
ii.	MgCl ₂ (25 mM)	4 µl
iii.	Forward primer (5 pmol/ µl)	1 µl
iv.	Reverse primer (5 pmol/ µl)	1 µl
v.	dNTP mix (5 mM)	1 µl
vi.	Taq DNA polymerase (1.5 U)	0.5 µl
vii.	Nuclease free water	12.0 µl
viii.	Template DNA	3 µl

The PCR amplifications were conducted in a thermal cycler with an initial denaturation of 5 min at 94 °C followed by 35 cycles (each) of denaturation of 1 min at 94 °C, annealing for 1 min at 56 °C, extension for 1 min at 72 °C and a final extension of 5 min at 72 °C. Amplified products were subjected to 1.5 per cent agarose gel using TBE (electrophoresis buffer, 40 mM Tris 2mM EDTA, Ph 8.0) containing 3 µl of ethidium bromide. The size of the PCR product was estimated by comparison with known DNA marker of 1kb DNA ladder. The banding profiles of ITS-PCR products were documented in gel documentation system.

3.4.3.1 Separation of amplified products by agarose gel electrophoresis

Agarose gel electrophoresis was performed to resolve the amplified product using 1.5 per cent agarose in 1x TBE (Tris Borate EDTA) buffer, 0.5 µg/ml of ethidium bromide and loading buffer. Four µl of the loading dye was added to 20 µl of PCR product loaded to the agarose gel. Electrophoresis was carried at 65 V for 1.5 hr. the gel was observed under UV light and documentation unit.

3.4.3.2 Sequencing of ITS region

The ITS region was sequenced from eight isolates belonging to different locations of eight districts (each isolate from each district) to confirm organism and to know the variability present in them.

3.4.3.3 Sequencing and in silico analysis

The PCR product was sequenced using forward and reverse primers at Medauxin Biotech. Ltd., Bengaluru. Homology search was done using BLAST algorithm available at the <http://www.ncbi.nlm.nih.gov>., multiple alignments for homology search were performed using the clusteral algorithm software and the phylogenetic tree was constructed.

3.4.4 Molecular diversity of *R. bataticola* isolates by RAPD

RAPD was carried out for molecular diversity of the 20 isolates as per the protocol described by Williams *et al.* (1990). A set of four RAPD primers were selected to study the polymorphism among 20 selected isolates of *R. bataticola*. The primers were synthesized by Xcelaris genomic, Xcelaris Lab Ltd., Ahmedabad, India. The primers details are mentioned below.

Table 4. List of RAPD primers used for PCR amplification

Sl. No.	RAPD marker	Sequences (5'-3')
1	OPC-06	GAACGGACTC
2	OPG-15	ACTGGGACTC
3	OPP-14	CCAGCCGAAC
4	OPU-07	CCTGCTCATC

Table 5. List of isolates of *R. bataticola* selected for molecular variability studies

Sl. No.	Location	Isolates
1	Bidar	Rb-1
2	Madgatti	Rb-3
3	Sannur	Rb-10
4	Bhusnur	Rb-13
5	Jewargi	Rb-14
6	Yadagiri	Rb-17
7	Malnoor	Rb-23
8	Bheemarayanagudi	Rb-19
9	Raichur	Rb-25
10	Huvinahadgi	Rb-34
11	Kallur	Rb-27
12	Halagere	Rb-36
13	Kyadiguppa	Rb-38
14	Kuknur	Rb-41
15	Belagallu	Rb-43
16	Siraguppa	Rb-44
17	Vijayapura	Rb-48
18	Golageri	Rb-55
19	Dharwad	Rb-56
20	Bhadrapura	Rb-60

3.4.4.1. RAPD profiles through Polymerised Chain Reaction (PCR)

PCR mixers (25 µl) per reaction

i.	10x PCR buffer	2.5 µl
ii.	MgCl ₂ (25 mM)	2 µl
iii.	Primer (5 pmol/ µl)	1 µl
iv.	dNTP mix (5 mM)	1 µl
v.	Taq DNA polymerase (1.5 U)	0.5 µl
vi.	Nuclease free water	16.0 µl
vii.	Template DNA	2 µl

PCR amplification was performed in a thermal cycler with the temperature profiles of 95°C for 5 min of initial denaturation, followed by 35 cycles of denaturation at 95°C for 60 sec, with constant annealing at 39°C for 60 sec and extension at 72°C for 60 sec with final extension at 72°C for 10 min. After completion of the polymerase chain reaction, the amplified PCR products were subjected to 1.5 per cent agarose gel electrophoresis with 1x TBE as ethidium bromide staining. The DNA banding profiles were documented in the gel documentation system and compared with 1 kb DNA ladder.

3.4.4.2 Scoring and data analysis of the amplified fragments

The amplified profiles for all the primers were compared with each other and bands of DNA fragment were scored as '1' for presence and '0' for absence, generating '0' and '1' matrix. Faint bands were not scored. Binary matrices were analyzed by NTSYS-PC (version 2.0; Exeter Biological Software, Setauket, NY) to calculate Jaccard's similarity coefficient for each pair wise comparison. Jaccard's coefficients were clustered to generate dendrogram using the SAHN clustering program selecting the unweighted pair group method with arithmetic average (UPGMA) algorithm in NTSYS PC (Rohlf, 1998).

3.5 Effect of climate change variables (temperature, moisture and carbon dioxide) on dry root rot disease

3.5.1 Effect of temperature on pathogen

Effect of temperature on pathogen virulence was studied by using three isolates (Rb-11, Rb-17 and Rb-29) of dry root rot pathogen, *R. bataticola*. The isolates were

selected based on the disease incidence in field condition and virulence nature of quick growth in the PDA media. The isolates were inoculated in PDA media and kept for inoculation in different temperature regimes 15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C and 45 °C, observed for the radial growth of mycelia.

3.5.2 Effect of temperature on disease

3.5.2.1 Raising of chickpea seedlings

Chickpea seedlings were raised in small plastic containers which were filled with pre-autoclaved sand, chickpea seeds are sown in those containers with 2 cm deep. Moisture was maintained by watering for daily basis, after ten days seedlings are ready for the further use.

Effect of temperature was carried out by using three isolates (Rb-13, Rb-17 and Rb-34) of dry root rot pathogen, *R. bataticola*. Ten days old seedlings was inoculated with *R. bataticola* and grown under *in vitro* conditions using paper towel technique. Inoculated seedlings were placed in folded, moist blotting paper with the shoots left outside and then incubated at different temperature regimes 15°C, 20°C, 25°C, 30°C, 35°C, 40°C, 45°C and 50°C with a 12 h photoperiod. The experiment was conducted in Completely Randomized Block Design (CRBD) with three replications (each replication consisted of 5 plants). Total 15 plants per treatment were scored for disease severity. Disease severity was recorded at seven days after inoculation on a 1 - 9 rating scale (Sharma and Pande, 2013).

Table 6. Disease rating scale of 1 - 9 for dry root rot of chickpea.

Rating	Observation
1	No infection on roots
>1 and <3	Very few small lesions (black discoloration) on roots
>3 and <5	Lesions (black discoloration) on roots clear but less; new roots free from infection
>5 and <7	Lesions (black discoloration) on roots more; many new roots generally free from lesions
>7 and 9	Roots infected and completely discoloured (black)



a. Raising of seedlings in sand tray



b. Washing of roots with water



c. Inoculation with *R. bataticola* pathogen



d. Rapping with moist blotting paper and incubation

Plate 1. Procedure followed for paper towel technique

3.5.3 Effect of soil moisture

Pot experiment was conducted on dry root disease in glasshouse conditions. To obtain large amount of inoculum, the fungus was multiplied on sorghum grain medium (Preparation 3.1.7.2). The giant culture so obtained was mixed with autoclaved black and red soils separately @ 50 g/kg which was used to fill pots having 4 kg capacity. The soil in pots was incubated for 4 days to allow the pathogen to multiply in soil. After 4 days, seeds of highly susceptible variety Annigeri-1 were sown in each pot at 2-3 cm depth.

Effect of seven soil moisture regimes, *i.e.* 40, 50, 60, 70, 80, 90 and 100 per cent was studied with two types of soils *viz.*, black and red soils on the development of disease. Each treatment was replicated four times and each replication consisted of three pots (five plants/pot).

Deionized water was used for maintaining the soil moisture content (SMC) in each treatment. The SMC was determined using the gravimetric method on an oven-dry basis. The method includes saturation of soil sample followed by removal of available soil moisture by oven drying (100 °C – 110 °C) until the weight remains constant. After removing from oven, **samples were cooled** slowly to room temperature and weighed again. The difference in weight was amount of moisture in the soil. The available SMC in the soil was calculated by the following formula.

$$\text{SMC (\%)} = \frac{\text{Saturated soil weight} - \text{oven dry soil weight}}{\text{Oven dry soil weight}} \times 100$$

The seven levels of SMC (40, 50, 60, 70, 80, 90 and 100 %) were adjusted by maintaining the constant weight by regular weighing and replacing the moisture deficit in each pot by watering. Observations were recorded on growth parameters, disease severity at 45 DAS days after sowing.

3.5.4 Effect of carbon dioxide (CO₂)

3.5.4.1 Open top chamber (OTC)

Open top chambers are widely used to study the effects of elevated CO₂ and other atmospheric gases on vegetation. They are plastic enclosures, with an open top, constructed

of an aluminum frame covered by panels of polyvinyl chloride plastic film. Air is pulled into the bottom of the chamber, enriched with CO₂, and then blown through the open top of the chamber. The study was conducted in the OTPs maintained by Department of Entomology, AC, Raichur under five sets of treatments.

T₁: Elevated CO₂ @ 550 ± 25 ppm alone

T₂: Elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature

T₃: Ambient CO₂ @ 390 ± 25 ppm with 2 °C rise in temperature

T₄: Reference Open Top Chamber

T₅: Open plot

Under the open top chamber, susceptible chickpea variety (A-1) was sown in the sick pots (preparation 3.5.2.1). Each pot was sown with five plants and for each treatment three replications were maintained. Observations on growth parameters and disease severity were recorded at 75 DAS.

3.6 Identification of dry root rot resistant sources of chickpea

3.6.1 Preliminary disease screening

A field experiment was conducted to identify the sources of resistance to dry root rot of chickpea. Totally hundred chickpea germplasm enteries (IIPR, Kanpur) were screened at Agricultural Research Station, Kalaburgi, during two seasons of *rabi*-2015-16 and *rabi*-2016-17 under sick plot was maintained by ARS, Kalaburgi.

Germplasm enteries were sown during second week of October with spacing of 60 x 30 cm spacing and with rows of 5 m length. The crop was raised as per the recommended package of practices except the plant protection measures against dry root rot. Observations were recorded at 30, 60 and 90 DAS on dry root rot and yield parameters. The observations on number of disease plants and total number of plants in each germplasm line were recorded. Later, disease incidence (%) was calculated using the following formula and genotypes were categorized for their reaction.

$$\text{Disease incidence (\%)} = \frac{\text{No. of diseased seedlings}}{\text{Total no. of seedlings}} \times 100$$

Table 7. List of genotypes used for screening against *R. bataticola*

Sl. No.	Germplasm entry	Sl. No.	Germplasm entry	Sl. No.	Germplasm entry
1.	EC 219999	34.	EC498765	67	IC83510
2.	EC 220078	35	EC498767	68	IC83511
3.	EC 267240	36	EC498773	69	IC83512
4.	EC 267484	37	EC498779	70	IC83513
5.	EC 267504	38	EC498780	71	IC83514
6.	EC 441805	39	EC498812	72	IC83515
7.	EC 489870	40	EC498818	73	IC83516
8.	EC 489870	41	EC498823	74	IC83517
9.	EC 489907	42	EC498825	75	IC83722
10.	EC 528345	43	EC498827	76	IC83723
11.	EC 555002	44	IC83358	77	IC83725
12.	IC 555217	45	IC83359	78	IC83728
13.	IC 83344	46	IC83361	79	IC83729
14.	IC 83370	47	IC83362	80	IC83730
15.	IC 83409	48	IC83363	81	IC83731
16.	IC 83526	49	IC83353	82	IC83732
17.	IC 83538	50	IC83354	83	IC83733
18.	IC83603	51	IC83355	84	IC83734
19.	IC 83775	52	IC83356	85	IC83735
20.	IC83797	53	IC83357	86	IC83736
21.	IC83798	54	IC83487	87	IC83739
22.	IC83803	55	IC83489	88	IC83740
23.	IC83816	56	IC83490	89	IC83741
24.	IC83965	57	IC83493	90	IC83742
25.	IC83996	58	IC83494	91	IC83742
26.	IC83997	59	IC83497	92	IC83743
27.	IC84008	60	IC83500	93	IC83744
28.	IC84011	61	IC83501	94	IC83745
29.	IC95135	62	IC83502	95	IC83746
30.	IC116287	63	IC83504	96	IC83747
31.	IC116304	64	IC83506	97	IC83748
32.	IC116325	65	IC83508	98	IC83749
33.	IC116343	66	IC83509	99	IC83750
				100	IC83751

Table 8. Categorization of germplasm lines based on per cent mortality (Khan *et al.*, 2013)

Sl. No.	Per cent mortality	Reaction
1	0-10 % mortality	Resistance
2	10.1- 20 % mortality	Moderately resistance
3	20.1- 30 % mortality	Moderately susceptible
4	30.1 – 50 % mortality	Susceptible
5	More than 50 % mortality	Highly susceptible

3.6.3 Advanced screening of promising entries of chickpea against dry root rot by phenotyping technique

Nineteen germplasm lines which were found resistant /moderately resistant were further screened for their performance using phenotyping technique (Bera *et al.*, 2016).

3.6.3.1 Preparation of sick block for the phenotyping technique

Field suitable for chickpea crop with proper drainage facility was selected. Nineteen germplasm lines were sown in block with 2 meter width bed with spacing of 0.5 meter between beds; 30 cm between lines and 10 cm between plants.

3.6.3.2 Mulching of block with chickpea stubbles

Each block (2 m length x 2 m width) was mulched with sun dried chickpea stubbles chopped into small size (2-3 cm). After mulching with chickpea stubbles uniformly, blocks were watered to field capacity at 24 hours interval for five days to ensure maximum soil moisture and softening of chickpea fodder. Care was taken to ensure uniform distribution of chickpea fodder in each block; stubbles are not floated into one place in the block during watering and moist to its maximum capacity. Observation on initial plant population of each genotype was recorded prior to mulching on 30 days after sowing.

3.6.3.3 Inoculation of *R. bataticola* to the advanced screening block

Blocks were watered sufficiently immediate before inoculation *R. bataticola* grown on sorghum grain were mixed with sterilized sand (1:1 ratio) just before

Table 9. List of promising entries selected for phenotyping technique for dry root rot of chickpea

Sl. No.	Germplasm entry
1	IC83736
2	IC 83526
3	IC 83538
4	IC83798
5	IC95135
6	EC498827
7	IC83358
8	IC83357
9	IC83487
10	IC83500
11	IC83501
12	IC83506
13	IC83514
14	IC 83736
15	IC83603
16	IC84011
17	EC498818
18	IC83732
19	IC83742

inoculation to ensure easy and uniform delivery of inoculum. Each block (3 m length x 2 m width) was inoculated with *R. bataticola* multiplied on one kg of sorghum grain to ensure sufficient and equal inoculum load to each block as well as plant. Inoculation was done in the evening hours to avoid exposure of inoculum to the sun and loss of moisture from sorghum grain.

3.6.3.4 Maintaining optimum moisture and temperature in the advanced screening block

After inoculation a rectangular structure made with wooden pole was placed around the block. Further, each rectangular structure was covered with 5.2 mm green shade net over the block to avoid high temperature (more than 40 °C during experiment). The height (2.5 ft.) of the rectangular structure used in our experiment ensured sufficient space above the top of the 40 days old chickpea crop. Maximum care was taken to ensure temperature 30-37 °C inside the net covered structure for fullest growth and extended mycelial stage of fungus to maximum period. We observed increase in temperature (from 34 °C to 40 °C) inside the chambers between 11 am to 2 pm during phenotyping. However, temperature (air temperature around 35 °C) inside the block was maintained by opening of green shade net in corners for 2-3 hours.

Simulated conditions were maintained for well growth of mycelia of *R. bataticola*. Observations on disease incidence (%) were taken at 75 DAS. Based on disease scoring, the most promising genotypes were identified.

3.7 Integrated management of dry root rot of chickpea

An experiment was laid out as per randomized block design (RBD) during *rabi*-2015-16 and *rabi*-2016-17 at experimental block of Department of Plant Pathology UAS, Raichur. The susceptible variety Annigeri-1 was grown with nine treatments which were replicated thrice with plot size of 5 X 3 m². Bioagents like *T. harzianum* and *P. fluorescens* were collected from Department of Plant Pathology UAS, Raichur. Recommended agronomical practices were followed to conduct the experiment. The artificial inoculation was done by using the fungus *R. bataticola*, mass multiplied on autoclaved sand sorghum medium. Pre cultured fungal mycelia (and sclerotia) was then placed aseptically in the medium, after ten days of fully grown culture was inoculated to soil. The observations like per cent disease, grain yield were recorded.

Table 10. Treatments for integrated disease management

Sl. No.	Treatment details
T ₁	Seed treatment with <i>T. harzianum</i> @ 5 g/kg seed followed by soil application of enriched <i>T. harzianum</i> 2.5 kg/250 kg FYM during sowing
T ₂	Seed treatment with <i>P. fluorescens</i> @ 5 g/kg seed followed by soil application of enriched <i>T. harzianum</i> 2.5 kg/250 kg FYM during sowing
T ₃	Seed treatment with mancozeb 50% + carbendazim 25% WS (Sprint) @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/l water to infected and surrounding plants
T ₄	Seed treatment with <i>T. harzianum</i> @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/l water to infected and surrounding plants
T ₅	Seed treatment with <i>P. fluorescens</i> @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/l water to infected and surrounding plants
T ₆	Seed treatment with <i>T. harzianum</i> @ 5 g/kg seed + <i>P. fluorescens</i> @ 5 g/kg seed followed by soil application of enriched <i>T. harzianum</i> + <i>P. fluorescens</i> 2.5 kg/250 kg FYM during sowing
T ₇	Seed treatment with <i>T. harzianum</i> @ 5 g/kg seed + <i>P. fluorescens</i> @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/lit water to infected and surrounding plants
T ₈	Seed treatment with mancozeb 50% + carbendazim 25% WS (Sprint) @ 3.5 g/kg
T ₉	Untreated control

a) Disease incidence (%)

Based on observations the disease incidence was calculated by following formula.

$$\text{Dry root rot incidence (\%)} = \frac{\text{Number of plants infected}}{\text{Total number of plants}} \times 100$$

b) Seed yield (q/ha)

At the time of harvest, chickpea plants from all the treatments and replications were collected separately. After threshing and winnowing, the net plot yield was recorded and later converted to ha and expressed as q per ha.

c) Test weight (g)

The test weight was also recorded by taking 100 seeds randomly from each treatment and expressed as gram. The data were subjected to statistical analysis by using standard statistical procedures.

3.7.1 Estimation of Benefit Cost (B:C) ratio

The economic analysis of the experiment was done by taking into consideration of market prices prevailing during 2016-17 for the produce and cost of treatment. B:C ratio was calculated by gross returns and total cost of cultivation.

$$\text{Benefit : Cost} = \frac{\text{Gross returns}}{\text{Total cost of cultivation}}$$

Experimental Results

IV. EXPERIMENTAL RESULTS

The experiments were conducted on various aspects of dry root rot of chickpea which includes the study on incidence of dry root rot in chickpea growing areas of North Karnataka and to collect the isolates of *Rhizoctonia bataticola* from different locations. Isolates of *R. bataticola* were characterized based on the cultural, morphological and molecular features by different methods. Experiments were taken to know of effects climate change variables like temperature, moisture and carbon dioxide on dry root rot disease. The studies on identification of resistance sources against dry root rot disease were taken up. Integrated management of the dry root rot by using different treatment components under field condition was undertaken. The results are presented in this chapter.

4.1 Symptoms of the disease

The first symptom was yellowing and partial or complete wilting of the stem or one or more branches. In advanced stage of the disease, affected plants are became straw coloured, but in some cases the lower leaves and stems showed brown discolouration. The tap root appeared black, rotten and devoid of most of the lateral and fine roots. The dead root became quite brittle and shows shredding of bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil when plants are uprooted, these symptoms are most commonly observed in chickpea in field condition. In severe cases there was no formation of the seeds (Plate 2).

4.2 Pathogen

The *R. bataticola* pathogen produced black, brown to grey coloured mycelium that become darker with age. The young hyphae were thin, hyaline, septate and dichotomously branched and later produce typical black sclerotia. The characteristic features of *R. bataticola* were right angle branching of the mycelium and constriction of the branch near the point of origin. The sclerotia formed were black, smooth, varying from spherical through oblong to irregular shapes.

4.3 Isolation, identification and pathogenicity

Standard tissue isolation was followed to isolate *R. bataticola* culture from diseased sample of infected root with typical symptom of shredding of bark, collected



Initiation of yellowing of leaves



Drying and drooping of leaves



**Completely dried plants
under glasshouse condition**



Roots infected and completely discoloured



Infection under field condition



Plate 2. Manifestation of symptoms of dry root rot of chickpea

from chickpea field. The fungus isolated from infected root was confirmed as *R. bataticola* based on their cultural, morphological and formation of sclerotial bodies.

Pathogenecity test was conducted in sick pot by growing susceptible variety (A-1) of chickpea. The first symptom of yellowing of one or more branches started on 20 day after sowing later, drooping of branches was observed on 32th to 35th day after sowing. Drying and shredding of leaves were observed near the soil surface and wilting of plants took place on 45 to 50 day after sowing. Uprooting of the affected plants was easy and root portion shown black rotted as well as shredding of roots. Symptoms due to dry root rot in the pots inoculated with *R. bataticola* culture were similar to dry root rot affected plants in the field. Reisolation of the fungus from root region of the plants was made and pathogenic cultures obtained were compared with original culture of *R. bataticola*. The culture was found similar with regard to all morphological characters on PDA and thus pathogen was identified as *R. bataticola* (Plate 3, 4, 5 and 6).

4.4 To survey for incidence of dry root rot of chickpea in North Karnataka

A roving survey was conducted during *rabi*, 2015-16 and 2016-17 in eight chickpea growing districts of North Karnataka. The per cent dry root rot incidence was recorded at 55–80 days old crop. The results obtained during roving survey are presented in Table 11, 12 and 13. The results indicated that the incidence was noticed in all the fields wherever the survey was conducted during both the years.

Results indicated that Bidar district recorded the mean dry root rot incidence of 23.73 per cent (Table 13). During *rabi* 2015-16, highest dry root rot (34.20 %) was recorded in Madagatti village of Bhalki taluk followed by Humnabad village (32.11%) and Gadgi (30.32%) village of Bidar taluk and lowest incidence of 16.12% was recorded in Manhalli village of Bidar taluk (Table 11). During *rabi* 2016-17, the highest incidence of dry root rot (32.15%) was observed in Gadgi village of Bidar taluk followed by Humnabad village (31.00%) of Humnabad taluk and Andura village of Bidar taluk (30.44%) and lowest dry root rot of 10.08% was recorded in Halbarga village of Bhalki taluk and Nidoda village (11.14%) of Bhalki taluk (Table 12).

Survey conducted in Kalaburgi district indicated that mean incidence of dry root rot was 32.93% for the two years (Table 13). During *rabi* 2015-16, highest incidence of dry root rot (45.55%) was observed in Bhusnur village of Aland taluk followed by Sannur

village (42.33%) of Kalaburgi taluk and Aland village of Aland taluk (40.00%). While, Ijeri village of Jewargi taluk recorded least dry root rot of 24.56%. (Table 11). *Rabi* 2016-17, results also indicated that Sannur village of Kalaburgi taluk recorded highest dry root rot (43.00%) followed by Jewargi (41.00%) and Andola village of Jewargi taluk (38.00%) and least incidence of 18.00% in Raddewadgi village of Jewargi taluk (Table 12).

Survey conducted in three taluks of Yadagiri district during *rabi* 2015-16, revealed that highest dry root rot was found in Bheemarayanagudi village (11.10%) of Shahapur taluk followed by Malanoor (7.00%) and Yadagiri (5.60%) and least dry root rot was recorded in Aldhal village (0.50%) of Shorapur taluk (Table 11). During *rabi* 2016-17, highest dry root rot was found in Yadagiri village (08.55%) of Yadagiri taluk followed by Gurmitkal (6.58%) and Bheemarayanagudi (6.11%) and least dry root rot was recorded in Mailapur village and saidapur (1.00%) of Yadagiri taluk (Table 12). The mean incidence in Yadagiri district for dry root rot was 3.35% (Table 13).

Among twelve villages in Raichur district, Huvinahadgi village of Devadurga taluk recorded highest dry root rot (45.23%) followed by Jalahalli (36.44%) village of Devadurga taluk and Kollur (35.56%), the least dry root rot (18.35%) was recorded in Bhupur village of Lingasugur during *rabi* 2015-16, (Table 11). Whereas in *rabi* 2016-17, Huvinahadgi village of Devadurga taluk recorded maximum dry root rot (43.20%) followed by Gabbur village of Raichur taluk (33.70%) and least (16.00%) was observed in Hatti village of Lingasugur taluk (Table 12). The district mean incidence of dry root rot was 29.87% for two years (Table 13)

The mean data over two years indicated that Koppal district recorded 13.91% dry root rot (Table 13). The highest disease incidence (23.00%) was found in Halagere village of Koppal taluk followed by Kuknoor village of Yelbarga taluk (17.00%) and there was least incidence of dry root rot in the fields of Bewoor village (08.00%) of Yelbarga taluk during *rabi*, 2015-16 (Table 11). During *rabi*, 2016-17, Halagere village of Koppal taluk recorded maximum dry root rot (22.08%) followed by Hanamapur village of Yelbarga taluk (16.15%) and lowest (5.18%) was in Bewoor village of Yelbarga taluk (Table 12).

The average incidence dry root rot in Ballari district was 13.39% (Table 13). Maximum incidence was observed in Belagallu village (24.59%) of Ballari taluk followed by Kolagallu village of Ballari taluk (21.53%) and Hansi village of Kudligi taluk 13.00%.

Least incidence was in Siraguppa village of Siraguppa taluk (04.81%) during *rabi*, 2015-16 (Table 11). During *rabi*, 2016-17, Belagallu village of Ballari taluk recorded dry root rot incidence of 25.00% followed by Kolagallu village of Ballari taluk (17.90%) and Amalapur village of Kudligi taluk (11.00%) and the lowest (06.00%) was found in Kudligi village of Kudligi taluk (Table 12).

In Vijayapura district survey revealed that highest dry root rot (45.00%) was found in Bableshtar village of Sindagi taluk followed by Kuntoji village (34.35%) of Muddebihal taluk and Golageri village (32.85%) of Sindagi taluk and lowest was in Vijayapura village (17.39%) of Vijayapura taluk during *rabi*, 2015-16 (Table 11). Whereas survey conducted during *rabi* 2016-17, highest dry root rot of 30.05% was recorded in Sindagi village of Sindagi taluk followed by Yankanchi village (28.00%) of Sindagi taluk and least was in Jalageri village (09.00%) of Vijayapura taluk (Table 12). However, an average incidence for the district for two years survey is 24.84% (Table 13).

Dharwad district recorded the mean incidence of dry root rot were 8.39 per cent (Table 13). During *rabi*, 2015-16, survey was conducted in eight villages of Dharwad district, among them highest dry root rot (12.35%) was found in Bhadrapur village followed by Tadhal village of Nargund taluk (11.21%) and least dry root rot (02.00%) was recorded in Dharwad village of Dharwad taluk (Table 11). During *rabi* 2016-17, highest dry root rot (13.00%) was recorded in Banahatti village of Nargund taluk followed by Nargund village of Nargund taluk (11.53%) and least incidence (03.00%) was in Yamanur village of Navalgund taluk (Table 12).

The data on mean incidence (Table 13) with respect to taluk over two seasons indicated that the highest mean incidence of dry root rot was recorded in Devadurga taluk (35.70%) of Raichur district followed of Kalaburgi taluk (33.81%) and Jewargi taluk (33.40%) of Kalaburgi district. The least incidence of dry root rot (2.88%) was recorded in Shorapur taluk of Yadagiri district.

Kalaburagi showed highest mean incidence of dry root rot (32.93%) followed by Raichur (29.87%) and Vijayapura for dry root rot (24.84%) over two seasons. However, the lowest mean incidence over two seasons was observed in Yadagiri (3.35%) followed by Dharwad (8.39%) and Ballari districts (13.39%). Data also indicated that the overall mean incidence was in the North Karnataka state for dry root rot was 35.72% (Table 13).

Table 11. Incidence of dry root rot of chickpea in major chickpea growing districts of North Karnataka during *rabi*, 2015-16

[illegible]

Table 11. Contd....

[illegible]

Table 11. Contd....

[illegible]

Table 12. Incidence of dry root rot of chickpea in major chickpea growing districts of North Karnataka during *rabi*, 2016-17

[illegible]

Table 12. Contd....

[illegible]

Table 12. Contd....[illegible]

Table 13. Talukwise incidence of dry root rot of chickpea in parts of North Karnataka during *rabi*, 2015-16 and 2016-17

Sl. No.	District	Taluku	Dry root rot (%)		Mean
			2015-16	2016-17	
1	Bidar	Bidar	25.35	24.71	23.03
		Bhalki	25.73	18.31	22.04
		Humanabad	26.50	25.76	26.13
Mean			25.87	22.73	23.73
2	Kalaburagi	Kalaburagi	34.22	33.41	33.81
		Aland	38.43	24.73	31.58
		Jewargi	34.56	32.25	33.40
Mean			35.73	30.13	32.93
3	Yadagiri	Yadagiri	2.85	3.66	3.25
		Shahapur	4.57	3.32	3.945
		Shorapur	2.40	3.36	2.88
Mean			3.27	3.44	3.35
4	Raichur	Raichur	31.69	29.87	30.78
		Lingasugur	21.98	24.3	23.14
		Devadurga	37.53	33.88	35.70
Mean			30.40	29.35	29.87
5	Koppal	Koppal	17.50	17.59	17.54
		Kustagi	12.61	12.62	12.61
		Yelaburga	12.50	10.66	11.58
Mean			14.20	13.62	13.91
6	Ballari	Ballari	23.06	21.45	22.25
		Siraguppa	7.90	8.49	8.19
		Kudligi	11.00	8.50	9.75
Mean			13.98	12.81	13.39
7	Vijayapura	Vijayapura	21.44	10.85	16.14
		Muddebihal	30.30	18.5	24.4
		Sindagi	35.95	26.05	31.00
Mean			29.23	18.46	24.84
8	Dharawad	Dharawad	5.07	4.83	4.95
		Nargund	10.72	12.26	11.49
		Navalagund	9.83	7.66	8.74
Mean			8.54	8.25	8.39
Overall mean incidence			38.72	32.73	35.72



Plate 3. Severity of dry root rot of chickpea noticed in farmer field during survey



A. Water agar



B. Potato dextrose agar



C. Potato dextrose broth



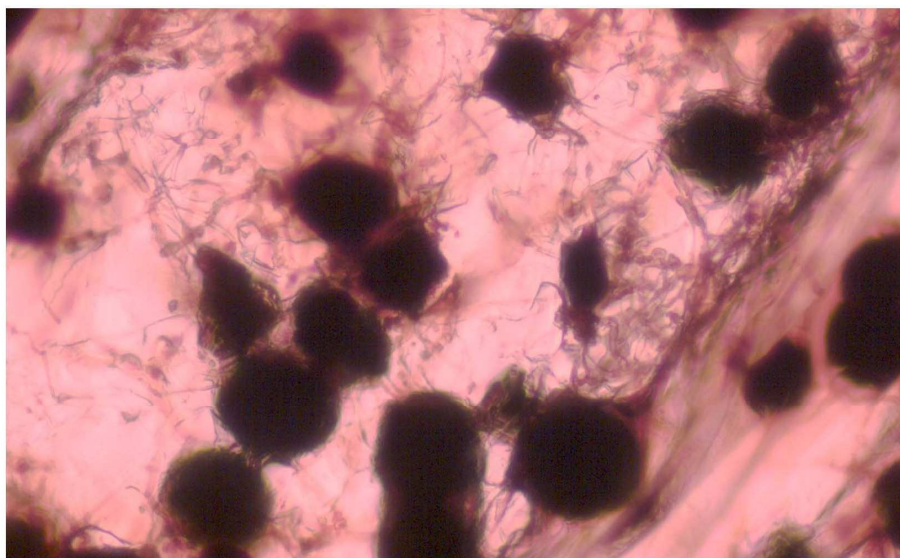
D. Sorghum media



Right and acute angle branching of mycelia



Monilioid cell formation



Dark microsclerotial bodies

Plate 5. Mycelial characters of *R. bataticola* causing dry root rot of chickpea



a) Initiation of yellowing of leaves



b) Drying and drooping of infected plants



c) Roots infected and completely discolored

Plates 6. Proving of pathogenicity test

4.5 Studies on cultural, morphological and molecular variability of pathogen

4.5.1 Collection of *R. bataticola* isolates infecting chickpea

Infected roots of chickpea were collected from different geographical regions of different districts of North Karnataka during the survey. Isolation of as many as sixty isolates of *R. bataticola* infecting chickpea was carried out by tissue isolation. The identity of all the isolates was confirmed based on the cultural and morphological observations (Plate 7a and 7b).

4.5.2 Cultural variability of *R. bataticola*

The cultural characters such as radial growth, colony morphology, production of sclerotia mycelial growth, colony pigmentation and arial mycelium of sixty isolates were studied on PDA as described in material and methods. (Table 14, 14a, 14b, 14c and 14d).

4.5.2.1 Radial growth

The results (Table 14 and 14a) on radial growth of *R. bataticola* ranged from 23.90 to 90 mm. Out of sixty isolates, fifty five isolates (Rb- 1, Rb- 2, Rb- 3, Rb- 4, Rb- 5, Rb- 6, Rb- 7, Rb- 8, Rb- 10, Rb- 11, Rb- 12, Rb- 13, Rb- 14, Rb- 15, Rb- 16, Rb- 17, Rb- 18, Rb- 19, Rb- 20, Rb- 21, Rb- 22, Rb- 23, Rb- 24, Rb- 25, Rb- 26, Rb- 27, Rb- 28, Rb- 29, Rb- 30, Rb- 31, Rb- 32, Rb- 33, Rb- 34, Rb- 35, Rb- 36, Rb- 37, Rb- 38, Rb- 39, Rb- 40, Rb- 42, Rb- 43, Rb- 44, Rb- 45, Rb- 46, Rb- 47, Rb- 48, Rb- 49, Rb- 50, Rb- 51, Rb- 52, Rb- 53, Rb- 54, Rb- 56, Rb- 57, Rb- 59) were fast growing (75-90 mm), two isolates (Rb- 55, Rb- 58) were moderate (60-75mm) and only three isolates (Rb- 9, Rb- 41, Rb- 60) were found slow growing (45-60mm) based on radial growth (Plate 7a, 7b).

4.5.2.2 Colony morphology

The results indicated that 17 isolates (Rb- 5, Rb- 8, Rb- 10, Rb- 11, Rb- 23, Rb- 26, Rb- 27, Rb- 34, Rb- 38, Rb- 39, Rb- 41, Rb- 48, Rb- 49, Rb- 50, Rb- 51, Rb- 54, Rb- 53) recorded Appressed growth, 13 isolates (Rb- 1, Rb- 2, Rb- 3, Rb- 4, Rb- 9, Rb- 17, Rb- 18, Rb- 28, Rb- 43, Rb- 46, Rb- 47, Rb- 55, Rb- 52) were shown velvety and rest of the 30 isolates (Rb- 6, Rb- 7, Rb- 12, Rb- 13, Rb- 14, Rb- 15, Rb- 16, Rb-19, Rb- 20, Rb- 21, Rb- 22, Rb- 24, Rb- 25, Rb- 29, Rb- 30, Rb- 31, Rb- 32, Rb- 33, Rb- 35, Rb- 36, Rb- 37, Rb- 40, Rb- 42, Rb- 44, Rb- 45, Rb- 56, Rb- 57, Rb- 58, Rb- 59, Rb- 60) showed fluffy growth pattern (Table 14b and Plate 8).

Table 14. Cultural variability in isolates of *R. bataticola* affecting chickpea

Sl. No.	Isolates	Cultural characters			
		Radial growth (mm)	Colony morphology	Colony pigmentatin	Aerial mycelium
1	Rb- 1	90.00	Velvetty	Black with grey aerial mycelium	Present
2	Rb- 2	90.00	Velvetty	Black with grey mycelium	Absent
3	Rb- 3	90.00	Velvetty	Black with grey aerial mycelium	Absent
4	Rb- 4	90.00	Velvetty	Grey	Absent
5	Rb- 5	88.50	Appressed	White pista	Absent
6	Rb- 6	90.00	Fluffy	White pista	Present
7	Rb- 7	90.00	Fluffy	Grey	Absent
8	Rb- 8	89.00	Appressed	Grey	Absent
9	Rb- 9	52.5	Velvetty	Grey	Absent
10	Rb- 10	90.00	Appressed	Grey	Absent
11	Rb- 11	89.00	Appressed	Grey	Absent
12	Rb- 12	90.00	Fluffy	Black	Absent
13	Rb- 13	90.00	Fluffy	Black	Absent
14	Rb- 14	90.00	Fluffy	Grey	Absent
15	Rb- 15	88.90	Fluffy	Black	Present
16	Rb- 16	89.00	Fluffy	Grey	Present
17	Rb- 17	88.00	Velvetty	Grey	Absent
18	Rb- 18	89.00	Velvetty	Black	Absent
19	Rb- 19	90.00	Fluffy	Black	Present
20	Rb- 20	90.00	Fluffy	Black with grey aerial mycelium	Present
21	Rb- 21	90.00	Fluffy	Grey	Present
22	Rb- 22	90.00	Fluffy	White pista	Present
23	Rb- 23	87.50	Appressed	Grey	Absent
24	Rb- 24	90.00	Fluffy	Grey	Present
25	Rb- 25	88.00	Fluffy	Black	Present
26	Rb- 26	89.00	Appressed	Black	Absent
27	Rb- 27	89.00	Appressed	Black	Absent
28	Rb- 28	87.00	Velvetty	Black with grey aerial mycelium	Present
29	Rb- 29	90.00	Fluffy	Black	Present
30	Rb- 30	89.00	Fluffy	Black with grey aerial mycelium	Present
31	Rb- 31	90.00	Fluffy	Black with grey	Present

				aerial mycelium	
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Table 14. Contd...

Sl. No.	Isolates	Cultural characters			
		Radial growth (mm)	Colony morphology	Colony pigmentatin	Aerial mycelium
32	Rb- 32	90.00	Fluffy	Black with grey aerial mycelium	Present
33	Rb- 33	86.00	Fluffy	Grey	Present
34	Rb- 34	89.00	Appressed	Black	Absent
35	Rb- 35	90.00	Fluffy	Black	Absent
36	Rb- 36	89.00	Fluffy	Grey	Present
37	Rb- 37	89.70	Fluffy	Grey	Present
38	Rb- 38	89.00	Appressed	Grey	Absent
39	Rb- 39	90.00	Appressed	Black	Absent
39	Rb- 40	90.00	Fluffy	Grey	Present
39	Rb- 41	45.3	Appressed	Grey	Absent
42	Rb- 42	90.00	Fluffy	Black with grey aerial mycelium	Present
43	Rb- 43	90.00	Velvetty	Black with grey aerial mycelium	Present
44	Rb- 44	89.00	Fluffy	Grey	Present
45	Rb- 45	90.00	Fluffy	Black with grey aerial mycelium	Present
46	Rb- 46	88.5	Velvetty	Grey	Present
47	Rb- 47	90.00	Velvetty	Black with grey aerial mycelium	Present
48	Rb- 48	89.00	Appressed	Black	Absent
49	Rb- 49	90.00	Appressed	Black with grey aerial mycelium	Present
50	Rb- 50	90.00	Appressed	Grey	Absent
51	Rb- 51	90.00	Appressed	Black	Absent
52	Rb- 52	90.00	Velvetty	Grey	Absent
53	Rb- 53	89.00	Appressed	Grey	Absent
54	Rb- 54	80.00	Appressed	White pista	Absent
55	Rb- 55	72.00	Velvetty	White pista	Absent
56	Rb- 56	89.00	Fluffy	Black with grey aerial mycelium	Present
57	Rb- 57	90.00	Fluffy	Black with grey aerial mycelium	Present
58	Rb- 58	73.90	Fluffy	Grey	Absent
59	Rb- 59	94.50	Fluffy	Grey	Absent

60	Rb- 60	23.90	Fluffy	Black	Absent
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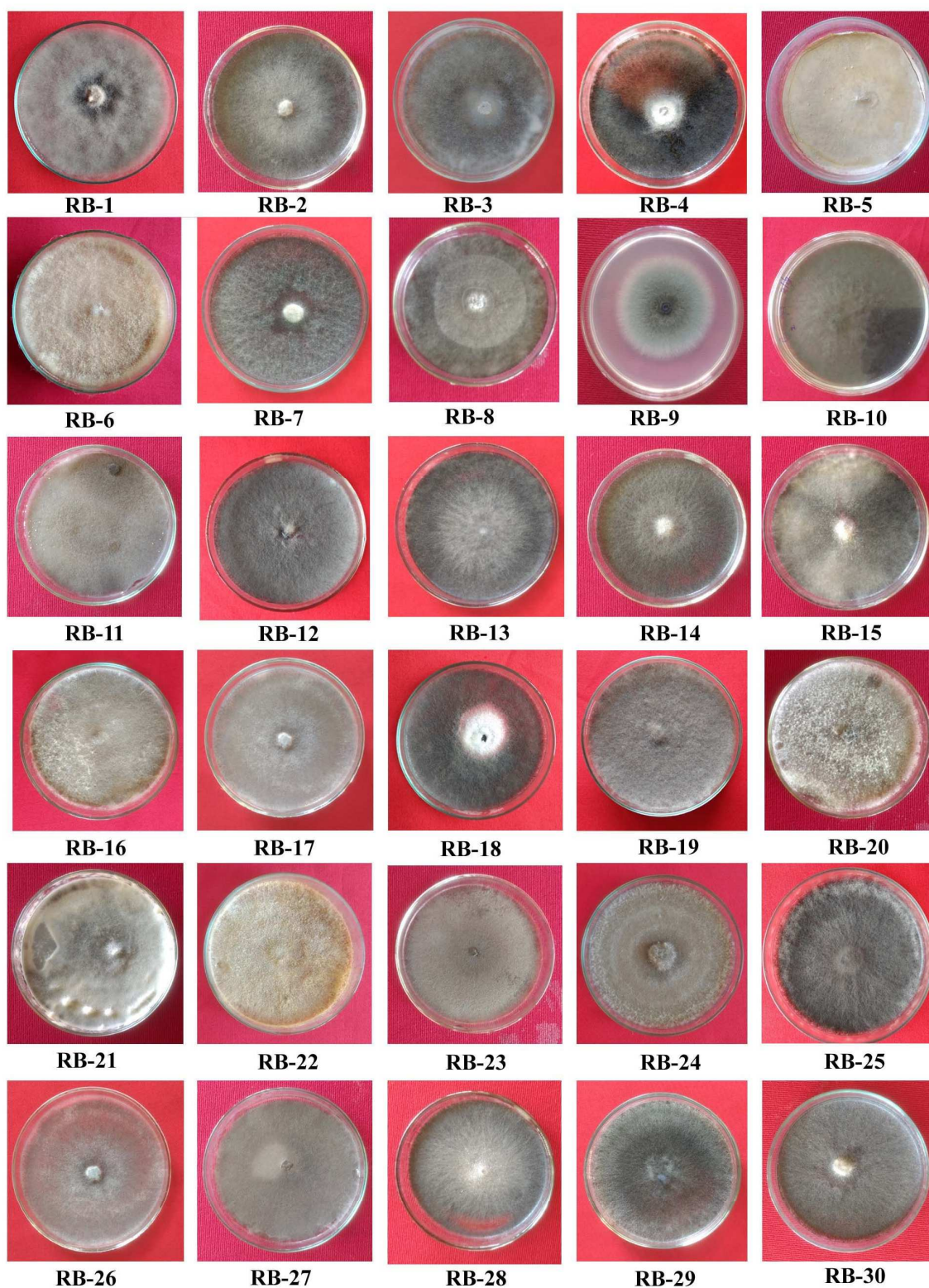


Plate 7a: Diverse growth pattern in isolates of *R. bataticola* isolated from chickpea on PDA

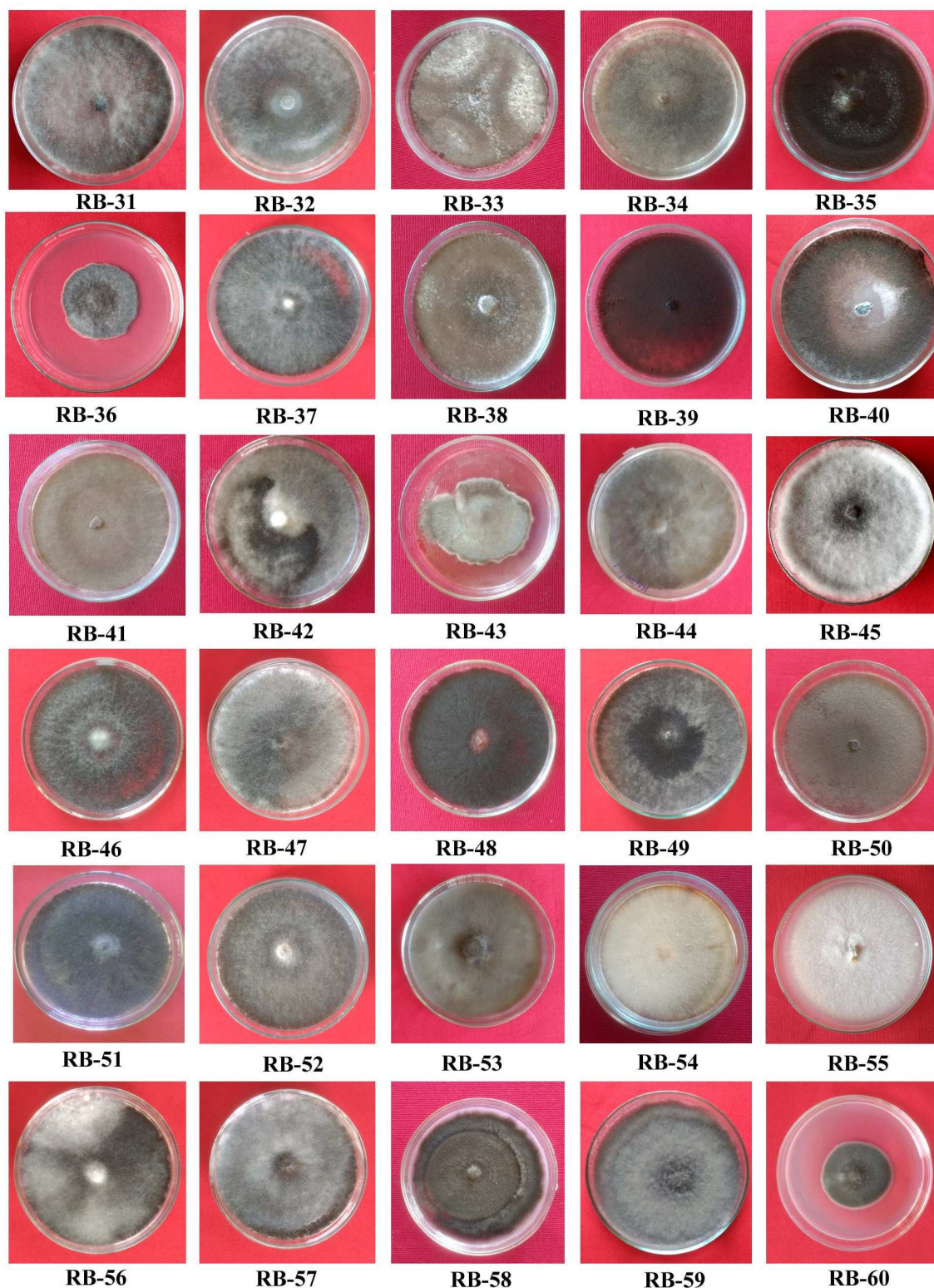


Plate 7b: Diverse growth pattern in isolates of *R. bataticola* isolated from chickpea on PDA

Table 14a. Grouping of *R. bataticola* isolates based on radial mycelial growth

Sl. No.	Isolate	Mycelial growth rate (mm)	Number of isolates
1	Rb- 1, Rb- 2, Rb- 3, Rb- 4, Rb- 5, Rb- 6, Rb- 7, Rb- 8, Rb- 10, Rb- 11, Rb- 12, Rb- 13, Rb- 14, Rb- 15, Rb- 16, Rb- 17 ,Rb- 18, Rb- 19, Rb- 20, Rb- 21, Rb- 22, Rb- 23, Rb- 24, Rb- 25, Rb- 26, Rb- 27, Rb- 28, Rb- 29, Rb- 30, Rb- 31, Rb- 32, Rb- 33, Rb- 34, Rb- 35, Rb- 36, Rb- 37, Rb- 38, Rb- 39, Rb- 40, Rb- 42, Rb- 43, Rb- 44, Rb- 45, Rb- 46, Rb- 47, Rb- 48, Rb- 49, Rb- 50, Rb- 51, Rb- 52, Rb- 53, Rb- 54, Rb- 56, Rb- 57, Rb- 59	Fast (75-90 mm)	55
2	Rb- 55, Rb- 58	Moderate (60-75 mm)	2
3	Rb- 9, Rb- 41, Rb- 60	Slow (45-60 mm)	3

Table 14b. Grouping of isolates of *R. bataticola* based on colony morphology

Sl. No.	Isolate	Colony morphology	Number of isolates
1	Rb- 5, Rb- 8, Rb- 10, Rb- 11, Rb- 23, Rb- 26, Rb- 27, Rb- 34, Rb- 38, Rb- 39, Rb- 41, Rb- 48, Rb- 49, Rb- 50 ,Rb- 51, Rb- 54, Rb- 53	Appressed	17
2	Rb- 1, Rb- 2, Rb- 3, Rb- 4, Rb- 9, Rb- 17, Rb- 18, Rb- 28, Rb- 43, Rb- 46, Rb- 47, Rb- 55, Rb- 52	Velvety	13
3	Rb- 6, Rb- 7, Rb- 12, Rb- 13, Rb- 14, Rb- 15, Rb- 16, Rb-19, Rb- 20, Rb- 21, Rb- 22, Rb- 24, Rb- 25, Rb- 29, Rb- 30, Rb- 31, Rb- 32, Rb- 33, Rb- 35, Rb- 36, Rb- 37, Rb- 40, Rb- 42, Rb- 44, Rb- 45, Rb- 56, Rb- 57, Rb- 58, Rb- 59, Rb- 60	Fluffy	30

Table 14c. Grouping of isolates of *R. bataticola* based on colony pigmentation

Sl. No.	Isolate	Colony pigmentation	Number of isolates
1	Rb- 1, Rb- 2, Rb- 3, Rb- 20, Rb- 28, Rb- 30, Rb- 31, Rb- 32, Rb- 42, Rb- 43, Rb- 45, Rb- 47, Rb- 49, Rb- 56, Rb- 57	Black with grey aerial mycelium	15
2	Rb- 12, Rb- 13, Rb- 15, Rb- 18, Rb- 19, Rb- 25, Rb- 26, Rb- 27, Rb- 29, Rb- 34, Rb- 35, Rb- 39, Rb- 48, Rb- 51, Rb- 60	Black	15
3	Rb- 4, Rb- 7, Rb- 8, Rb- 9, Rb- 10, Rb- 11, Rb- 14, Rb- 16, Rb- 17, Rb- 21, Rb- 23, Rb- 24, Rb- 33, Rb- 36, Rb- 37, Rb- 38, Rb- 40, Rb- 41, Rb- 44, Rb- 46, Rb- 50, Rb- 52, Rb- 53, Rb- 59, Rb- 58	Grey	25
4	Rb- 5, Rb- 6, Rb- 22, Rb- 55, Rb- 54	White pista	5

Table 14d. Grouping of isolates of *R. bataticola* based on aerial mycelia

Sl. No.	Isolate	Aerial mycelia	Number of isolates
1	Rb- 27, Rb- 28, Rb- 29, Rb- 30, Rb- 24, Rb- 25, Rb- 21, Rb- 22, Rb- 19, Rb- 20, Rb- 15, Rb- 16, Rb- 6, Rb- 1, Rb- 31, Rb- 32, Rb-33, Rb- 36, Rb- 37, Rb- 40, Rb- 43, Rb- 44, Rb- 45, Rb- 46, Rb- 47, Rb- 49, Rb- 56, Rb- 57	Present	28
2	Rb- 2, Rb- 3, Rb- 4, Rb- 5, Rb- 7, Rb- 8, Rb- 9, Rb- 10, Rb- 11, Rb- 12, Rb- 13, Rb- 14, Rb- 17, Rb- 18, Rb- 23, Rb- 26, Rb- 27, Rb- 34, Rb- 35, Rb- 38, Rb- 39, Rb- 41, Rb- 48, Rb- 50, Rb- 51, Rb- 52, Rb- 53, Rb- 54, Rb- 55, Rb- 58, Rb- 59, Rb- 60	Absent	32



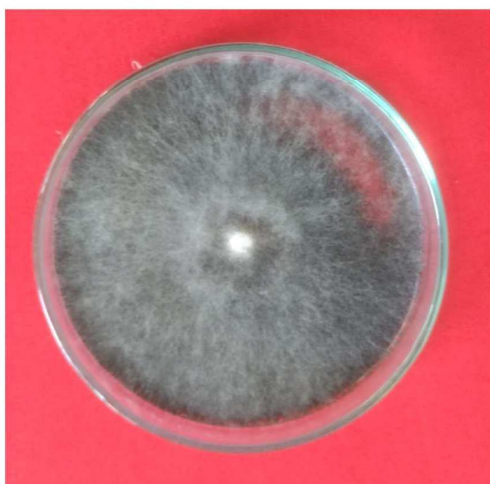
Aerial mycelia present



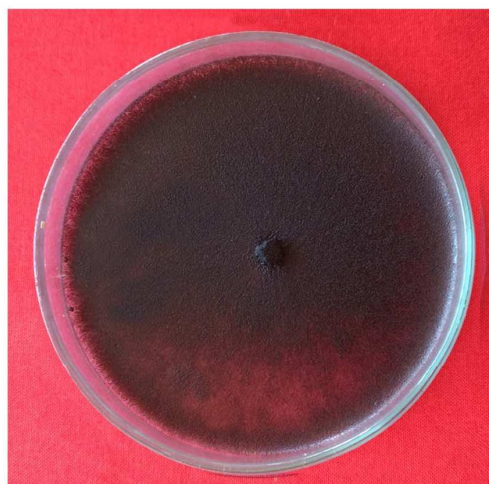
Aerial mycelia absent



Velvety



Fluffy



Appressed

Plate 8. Cultural variability based on aerial mycelia and colony morphology of isolates of *R. bataticola*



Black



Grey



Black with grey aerial mycelium



White pista

Plate 9. Cultural variability based on colony pigmentation of isolates of *R. bataticola*

4.5.2.3 Colony pigmentation

Based on visual observation on colony colour, the isolate were divided into four groups, black with grey aerial mycelia, black, grey and white pista colour. Black with grey aerial mycelial colony was found in fifteen isolates (Rb- 1, Rb- 2, Rb- 3, Rb- 20, Rb- 28, Rb- 30, Rb- 31, Rb- 32, Rb- 42, Rb- 43, Rb- 45, Rb- 47, Rb- 49, Rb- 56, Rb- 57), black pigmentation was in fifteen isolates (Rb- 12, Rb- 13, Rb- 15, Rb- 18, Rb- 19, Rb- 25, Rb- 26, Rb- 27, Rb- 29, Rb- 34, Rb- 35, Rb- 39, Rb- 48, Rb- 51, Rb- 60), grey pigmentation of the colony shown by twenty five isolates (Rb- 4, Rb- 7, Rb- 8, Rb- 9, Rb- 10, Rb- 11, Rb- 14, Rb- 16, Rb- 17, Rb- 21, Rb- 23, Rb- 24, Rb- 33, Rb- 36, Rb- 37, Rb- 38, Rb- 40, Rb- 41, Rb- 44, Rb- 46, Rb- 50, Rb- 52, Rb- 53, Rb- 59, Rb- 58) and peculiar character of white pista coloured colony was found in five isolates (Rb- 5, Rb- 6, Rb- 22, Rb- 55, Rb- 54) (Table 14c and Plate 9).

4.5.2.4 Aerial mycelium

Observations on aerial mycelium was recorded at fifteenth day after inoculation. The results indicated that 28 isolates (Rb- 27, Rb- 28, Rb- 29, Rb- 30, Rb- 24, Rb- 25, Rb- 21, Rb- 22, Rb- 19, Rb- 20, Rb- 15, Rb- 16, Rb- 6, Rb- 1, Rb- 31, Rb- 32, Rb- 33, Rb- 36, Rb- 37, Rb- 40, Rb- 43, Rb- 44, Rb- 45, Rb- 46, Rb- 47, Rb- 49, Rb- 56, Rb- 57) showed the production of arial mycelia and in rest of the 32 isolates (Rb- 2, Rb- 3, Rb- 4, Rb- 5, Rb- 7, Rb- 8, Rb- 9, Rb- 10, Rb- 11, Rb- 12, Rb- 13, Rb- 14, Rb- 17, Rb- 18, Rb- 23, Rb- 26, Rb- 27, Rb- 34, Rb- 35, Rb- 38, Rb- 39, Rb- 41, Rb- 48, Rb- 50, Rb- 51, Rb- 52, Rb- 53, Rb- 54, Rb- 55, Rb- 58, Rb- 59, Rb- 60) aerial mycelia was absent (Table 14d).

4.5.3 Morphological variability of *R. bataticola*

The parameters for morphological variability such as width of the hyphae, microsclerotial production, number of days to produce microsclerotia, number of microsclerotia produced and shape and size of microsclerotia were recorded in this laboratory study. These features were considered to assess the existence of variation in the pathogen and results are presented in Table 15, 15a, 15b, 15c and 15d.

4.5.3.1 Width of the hyphae

There was much variation in the width of the hyphae, among the sixty isolates highest width was found in the isolates Rb-58 (191.23 μ m) followed by Rb-18

(171.24 μm) and Rb-57 (168.00 μm). Least width of the hyphae was recorded in isolates Rb-14 (86.56 μm) and Rb-51 (87.49 μm) (Table 15).

4.5.3.2 Microsclerotial production

A laboratory study on production of microsclerotial bodies, out of sixty isolates fifty one isolates were found to produce microsclerotia, but in nine isolates (Rb-5, Rb-6, Rb-16, Rb-21, Rb-22, Rb-24, Rb-43, Rb-54, Rb-55) there was absence of microsclerotia (Table 15a and Plate 10).

4.5.3.3 Number of days to produce microsclerotia

All the isolates showed variation in respect of number of days to produce microsclerotia (Table 15b). Grouping of *R. bataticola* isolates was made on the basis of number of days taken to production of microsclerotial bodies and sixty isolates were categorized into three groups. Eighteen isolates (Rb-1, Rb-2, Rb-4, Rb-7, Rb-9, Rb-10, Rb-13, Rb-25, Rb-26, Rb-27, Rb-30, Rb-35, Rb-39, Rb-42, Rb-49, Rb-50, Rb-52, Rb-53, Rb-57) were started to produce within 2-3 days, twenty six isolates (Rb-3, Rb-8, Rb-11, Rb-12, Rb-14, Rb-15, Rb-18, Rb-19, Rb-20, Rb-23, Rb-28, Rb-29, Rb-31, Rb-32, Rb-33, Rb-36, Rb-37, Rb-38, Rb-40, Rb-44, Rb-45, Rb-46, Rb-51, Rb-60, Rb-59, Rb-58) were taken 4-5 days, rest of the six isolates (Rb-17, Rb-34, Rb-41, Rb-47, Rb-48, Rb-56) started to produce microsclerotial bodies in 6-7 days.

4.5.3.4 Number of microsclerotia produced

The number of microsclerotial bodies produced by isolates of *R. bataticola* per microscopic field (10x) also varied and were categorized into three groups (Table 15c). Among sixty isolates, twenty one isolates (Rb-2, Rb-19, Rb-20, Rb-23, Rb-26, Rb-31, Rb-32, Rb-33, Rb-34, Rb-36, Rb-40, Rb-41, Rb-42, Rb-46, Rb-47, Rb-48, Rb-52, Rb-56, Rb-58, Rb-59, Rb-60) produced 0-25 sclerotial bodies per microscopic field (10x), twenty six isolates (Rb-1, Rb-3, Rb-4, Rb-7, Rb-8, Rb-9, Rb-10, Rb-11, Rb-12, Rb-14, Rb-15, Rb-17, Rb-18, Rb-25, Rb-27, Rb-30, Rb-35, Rb-37, Rb-38, Rb-39, Rb-44, Rb-45, Rb-49, Rb-50, Rb-51, Rb-57) produced 26-50 sclerotial bodies per microscopic field (10x), four isolates (Rb-13, Rb-28, Rb-29, Rb-53) produced 51-100 sclerotial bodies per microscopic field (10x) (Plate 10).

Table 15. Morphological variability in isolates of *R. bataticola* infecting chickpea

Sl. No.	Isolates	Width of mycelium (µm)	Production of microsclerotia	Number of days to produce microsclerotia (DAI)	Morphology of microsclerotia		
					(No./microscopic field, 10x)	Shape	Size (mm) (40x)
1	Rb-1	120.57	+	3	41.24	Round	2.75×2.00
2	Rb-2	152.50	+	3	13.33	Irregular	1.59×1.00
3	Rb-3	136.59	+	5	28.33	Irregular	1.91×1.34
4	Rb-4	124.50	+	3	35.33	Round	2.75×2.50
5	Rb-5	113.70	-	-	-	-	-
6	Rb-6	99.23	-	-	-	-	-
7	Rb-7	103.30	+	3	38.00	Round	2.54×1.17
8	Rb-8	105.82	+	4	35.00	Irregular	1.40×1.36
9	Rb-9	163.26	+	3	36.60	Round	3.10×2.71
10	Rb-10	135.20	+	3	27.66	Ovoid	1.33×1.24
11	Rb-11	127.60	+	4	48.00	Ovoid	1.55×1.47
12	Rb-12	97.18	+	5	45.00	Ovoid	2.20×1.33
13	Rb-13	134.79	+	3	54.00	Irregular	3.05×1.49
14	Rb-14	86.556	+	4	30.66	Round	1.76×1.66
15	Rb-15	93.98	+	4	31.00	Round	1.64×1.27
16	Rb-16	120.65	-	-	-	-	-
17	Rb-17	121.90	+	6	26.00	Ovoid	1.77×1.50
18	Rb-18	171.24	+	4	27.66	Round	2.06×1.29
19	Rb-19	122.54	+	4	20.00	Irregular	1.22×1.43
20	Rb-20	100.32	+	4	16.00	Round	2.13×1.91
21	Rb-21	130.45	-	-	-	-	-
22	Rb-22	123.00	-	-	-	-	-
23	Rb-23	131.66	+	5	24.33	Irregular	1.75×1.15
24	Rb-24	143.80	-	-	-	-	-
25	Rb-25	119.53	+	33	25.66	Round	2.01×1.88
26	Rb-26	112.56	+	3	22.33	Ovoid	1.09×1.00
27	Rb-27	117.89	+	3	31.33	Ovoid	1.31×0.94
28	Rb-28	94.35	+	4	50.33	Irregular	1.75×1.50
29	Rb-29	130.50	+	5	61.33	Ovoid	1.29×1.20
30	Rb-30	165.64	+	3	30.00	Round	1.53×1.37

Table 15. Contd....

Sl. No.	Isolate s	Width of mycelium (µm)	Production of microsclerotia	Number of days to produce microsclerotia (DAI)	Morphology of microsclerotia		
					(No./microscopic field, 10x)	Shape	Size (mm) (40X)
31	Rb-31	123.80	+	4	17.03	Round	1.54×1.47
32	Rb-32	133.42	+	5	20.05	Ovoid	1.83×1.54
33	Rb-33	138.26	+	5	17.66	Irregular	3.20×2.72
34	Rb-34	150.23	+	6	07.66	Irregular	3.05×1.47
35	Rb-35	145.80	+	3	35.00	Irregular	1.75×1.45
36	Rb-36	136.78	+	4	20.50	Ovoid	1.40×1.36
37	Rb-37	134.19	+	4	34.66	Ovoid	1.55×1.47
38	Rb-38	106.70	+	4	25.66	Round	1.67×1.56
39	Rb-39	93.19	+	3	31.00	Round	1.64×1.47
40	Rb-40	101.46	+	5	15.00	Ovoid	1.70×1.33
41	Rb-41	163.60	+	6	05.00	Irregular	1.52×1.13
42	Rb-42	141.24	+	3	17.66	Round	1.86×1.19
43	Rb-43	115.60	-	-	-	-	-
44	Rb-44	122.12	+	4	35.00	Irregular	2.80×1.90
45	Rb-45	156.22	+	5	32.45	Irregular	1.25×1.30
46	Rb-46	94.50	+	5	18.33	Ovoid	2.60×2.29
47	Rb-47	101.36	+	6	14.33	Irregular	0.86×0.19
48	Rb-48	120.39	+	6	18.24	Ovoid	1.83×1.69
49	Rb-49	120.53	+	3	35.66	Irregular	2.21×1.88
50	Rb-50	92.50	+	3	42.33	Irregular	2.09×2.00
51	Rb-51	87.49	+	4	31.33	Round	2.31×2.74
52	Rb-52	170.35	+	3	23.33	Round	2.65×2.30
53	Rb-53	140.00	+	3	61.33	Irregular	1.99×1.80
54	Rb-54	153.08	-	-	-	-	-
55	Rb-55	129.25	-	-	-	-	-
56	Rb-56	143.86	+	5	10.00	Irregular	1.56×2.09-
57	Rb-57	168.00	+	3	35.66	Round	3.20×2.71
58	Rb-58	191.23	+	4	17.64	Round	1.33×1.14
59	Rb-59	104.90	+	55	25.00	Round	1.54×1.47
60	Rb-60	116.34	+	5	10.00	Round	2.26×2.14

Table 15a. Grouping of isolates of *R. bataticola* based on production of microsclerotia

Sl. No.	Isolate	Microsclerotia	Number of isolates
1	Rb-1, Rb-2, Rb-3, Rb-4, Rb-7, Rb-8, Rb-9, Rb-10, Rb-11, Rb-12, Rb-13, Rb-14, Rb-15, Rb-17, Rb-18, Rb-19, Rb-20, Rb-23, Rb-25, Rb-26, Rb-27, Rb-28, Rb-29, Rb-30, Rb-31, Rb-32, Rb-33, Rb-34, Rb-35, Rb-36, Rb-37, Rb-38, Rb-39, Rb-40, Rb-41, Rb-42, Rb-44, Rb-45, Rb-46, Rb-47, Rb-48	Present	51
2	Rb-5, Rb-6, Rb-16, Rb-21, Rb-22, Rb-24, Rb-43, Rb-54, Rb-55	Absent	09

Table 15b. Grouping of *R. bataticola* isolates based on number of days to produce microsclerotia

Sl. No.	Isolate	Number of days to produce microsclerotia (DAI)	Number of isolates
1	Rb-1, Rb-2, Rb-4, Rb-7, Rb-9, Rb-10, Rb-13, Rb-25, Rb-26, Rb-27, Rb-30, Rb-35, Rb-39, Rb-42, Rb-49, Rb-50, Rb-52, Rb-53, Rb-57	2-3	18
2	Rb-3, Rb-8, Rb-11, Rb-12, Rb-14, Rb-15, Rb-18, Rb-19, Rb-20, Rb-23, Rb-28, Rb-29, Rb-31, Rb-32, Rb-33, Rb-36, Rb-37, Rb-38, Rb-40, Rb-44, Rb-45, Rb-46, Rb-51, Rb-60, Rb-59, Rb-58	4-5	26
3	Rb-17, Rb-34, Rb-41, Rb-47, Rb-48, Rb-56	6 – 7	6

Table 15c. Grouping of *R. bataticola* isolates based on production of number of microsclerotia

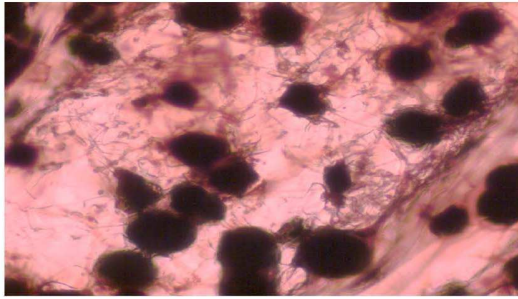
Sl. No.	Isolate	No./microscopic field, (10x)	Number of isolates
1	Rb-2, Rb-19, Rb-20, Rb-23, Rb-26, Rb-31, Rb-32, Rb-33, Rb-34, Rb-36, Rb-40, Rb-41, Rb-42, Rb-46, Rb-47, Rb-48, Rb-52, Rb-56, Rb-58, Rb-59, Rb-60	0-25	21
2	Rb-1, Rb-3, Rb-4, Rb-7, Rb-8, Rb-9, Rb-10, Rb-11, Rb-12, Rb-14, Rb-15, Rb-17, Rb-18, Rb-25, Rb-27, Rb-30, Rb-35, Rb-37, Rb-38, Rb-39, Rb-44, Rb-45, Rb-49, Rb-50, Rb-51, Rb-57	26-50	26
3	Rb-13, Rb-28, Rb-29, Rb-53	51-100	4

Table 15d. Grouping of isolates of *R. bataticola* based on shape of microsclerotia

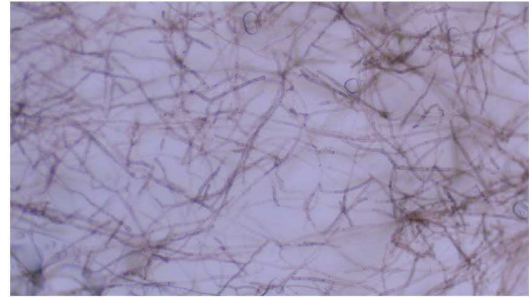
Sl. No.	Isolate	Shape	Number of isolates
1	Rb-1, Rb-4, Rb-7, Rb-9, Rb-14, Rb-15, Rb-18, Rb-20, Rb-25, Rb-30, Rb-31, Rb-38, Rb-39, Rb-42, Rb-51, Rb-52, Rb-57, Rb-58, Rb-59, Rb-60	Round	20
2	Rb-10, Rb-11, Rb-12, Rb-17, Rb-26, Rb-27, Rb-29, Rb-32, Rb-36, Rb-37, Rb-40, Rb-48, Rb-46	Ovoid	13
3	Rb-2, Rb-3, Rb-8, Rb-13, Rb-19, Rb-23, Rb-28, Rb-33, Rb-34, Rb-35, Rb-41, Rb-44, Rb-45, Rb-47, Rb-49, Rb-50, Rb-53, Rb-56	Irregular	18

Table 16. Grouping of different *R. bataticola* isolates based on cultural and morphological characters

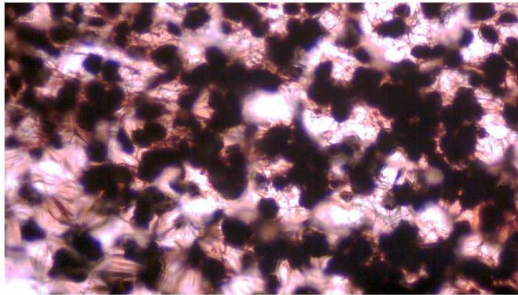
Group	G-I	G-II	G-III	G-IV		
Shape of microsclerotia	Round	Ovoid	Irregular	-		
No. of microsclerotia	0-25	26-50	51-100	-		
Days to initiation of microsclerotia	1-2	3-4	5-6	-		
Production of microsclerotia	Present	Present	Present	Absent		
Radial growth	Fast	Moderate	Slow	Fast		Moderate
Colony morphology	Appressed	Velvety	Fluffy	Fluffy	Appressed	Velvety
	Rb-1, Rb-2, Rb-3, Rb-4, Rb-7, Rb-8, Rb-9, Rb-10, Rb-11, Rb-12, Rb-13, Rb-14, Rb-15, Rb-17, Rb-18, Rb-19, Rb-20, Rb-23, Rb-25, Rb-26, Rb-27, Rb-28, Rb-29, Rb-30, Rb-31, Rb-32, Rb-33, Rb-34, Rb-35, Rb-36, Rb-37, Rb-38, Rb-39, Rb-40, Rb-41, Rb-42, Rb-44, Rb-45, Rb-46, Rb-47, Rb-48, Rb-49, Rb-50, Rb-51, Rb-53, Rb-56, Rb-57, Rb-58, Rb-59, Rb-60	Rb-1, Rb-2, Rb-3, Rb-4, Rb-7, Rb-8, Rb-9, Rb-10, Rb-11, Rb-12, Rb-14, Rb-15, Rb-17, Rb-18, Rb-23, Rb-25, Rb-26, Rb-27, Rb-28, Rb-29, Rb-30, Rb-31, Rb-32, Rb-33, Rb-35, Rb-36, Rb-37, Rb-38, Rb-39, Rb-40, Rb-44, Rb-45, Rb-46, Rb-47, Rb-48, Rb-49, Rb-50, Rb-51, Rb-52, Rb-56, Rb-57, Rb-58, Rb-59, Rb-60	Rb-2, Rb-3, Rb-7, Rb-8, Rb-9, Rb-12, Rb-13, Rb-14, Rb-15, Rb-19, Rb-20, Rb-23, Rb-25, Rb-28, Rb-29, Rb-30, Rb-31, Rb-32, Rb-33, Rb-34, Rb-35, Rb-36, Rb-37, Rb-40, Rb-41, Rb-42, Rb-44, Rb-45, Rb-47, Rb-49, Rb-50, Rb-53, Rb-56, Rb-57, Rb-58, Rb-59, Rb-60	Rb-6, Rb-16, Rb-21, Rb-22, Rb-24, Rb-43	Rb-5, Rb-54	Rb-55



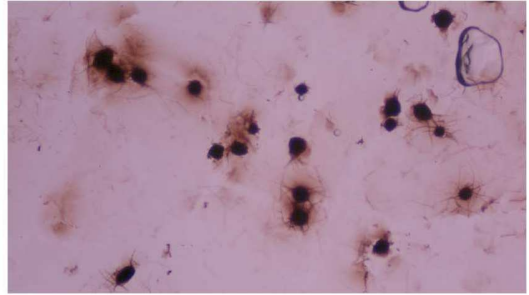
Microsclerotia present



Microsclerotia absent



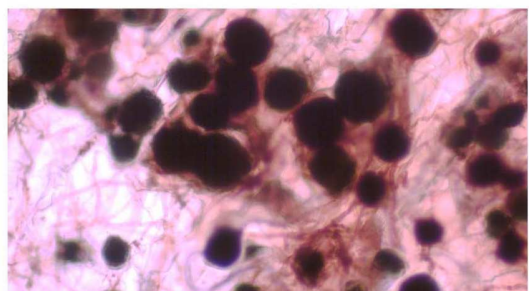
More number of microsclerotia



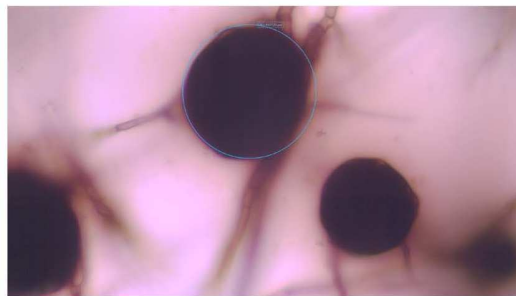
Less number of microsclerotia



Ovoid



Irregular



Round

Plate10. Morphological variability based on presence/absence of microsclerotia, number of microsclerotia and shape of microsclerotia in isolates of *R. bataticola* infecting chickpea

4.5.3.5 Shape of the microsclerotia

The isolates were categorized into round, ovoid and irregular groups based on shape of microsclerotia (Table 15d). Round shaped microsclerotia were observed in twenty isolates (Rb-1, Rb-4, Rb-7, Rb-9, Rb-14, Rb-15, Rb-18, Rb-20, Rb-25, Rb-30, Rb-31, Rb-38, Rb-39, Rb-42, Rb-51, Rb-52, Rb-57, Rb-58, Rb-59, Rb-60) while ovoid shaped microsclerotia were observed in thirteen isolates (Rb-10, Rb-11, Rb-12, Rb-17, Rb-26, Rb-27, Rb-29, Rb-32, Rb-36, Rb-37, Rb-40, Rb-48, Rb-46). Irregular shape of microsclerotia was observed in eighteen isolates (Rb-2, Rb-3, Rb-8, Rb-13, Rb-19, Rb-23, Rb-28, Rb-33, Rb-34, Rb-35, Rb-41, Rb-44, Rb-45, Rb-47, Rb-49, Rb-50, Rb-53, Rb-56) (Plate 10).

4.5.3.6 Size of microsclerotia

Among the sixty isolates, there was much variation in size of the microsclerotia was observed. The maximum size was recorded in isolates Rb-33 (3.20 x 2.71 mm) followed by Rb-34 (3.05×1.47 mm) and Rb-1 (2.75×2.00 mm). Least size of the microsclerotia was recorded in isolates Rb-27 (1.31×0.94 mm) and Rb-26 (1.09 × 1.00 mm) (Table 15 and Plate 10).

4.5.3.7 Grouping of *R. bataticola* isolates based on cultural and morphological characters

Diversity in cultural characters such as colony radial growth, colony morphology, production of sclerotia mycelial growth and colony pigmentation and morphological characters such as width of the hyphae, microsclerotial production, number of days to produce microsclerotia, number of microsclerotia produced and shape and size of microsclerotia was closely observed in sixty isolates of *R. bataticola*. Based on the cultural and morphological characters isolates were categorized into four groups designated as G-I, G-II, G-III, G-IV (Table 16).

4.5.4 Molecular variability of *R. bataticola* isolates

4.5.4.1 Identification of *R. bataticola* isolates by ITS

Twenty isolates ITS-1 and ITS-4 primers were used for PCR amplification of ITS region. Both the primers produced amplified product size of 500-650 bp in all the twenty isolates indicating that all the isolates belong to genus *Rhizoctonia bataticola*.

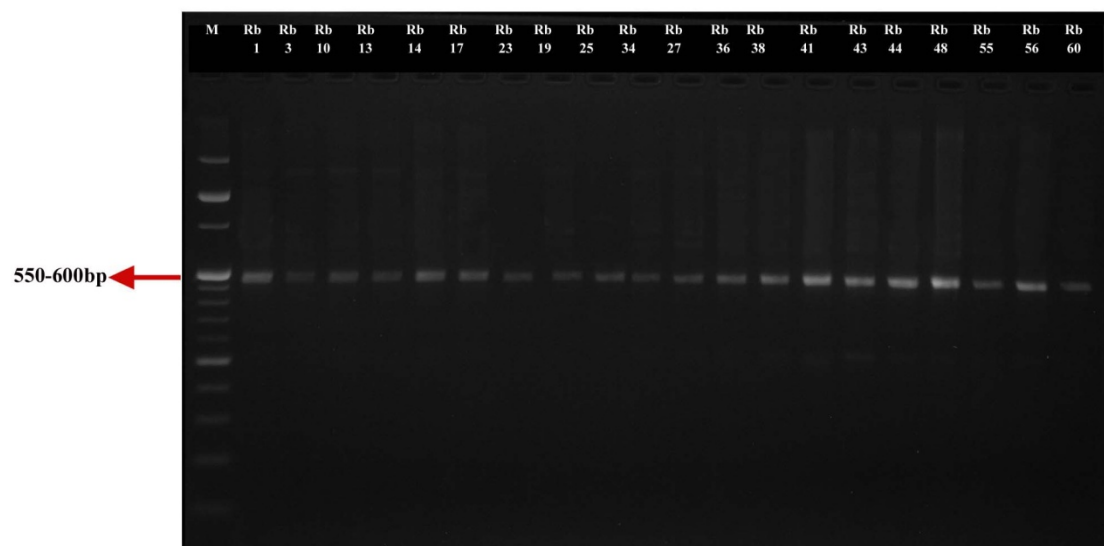


Plate 11. Amplification product of Internal Transcribed Spacer (ITS) with ITS-1 and ITS-4 primers

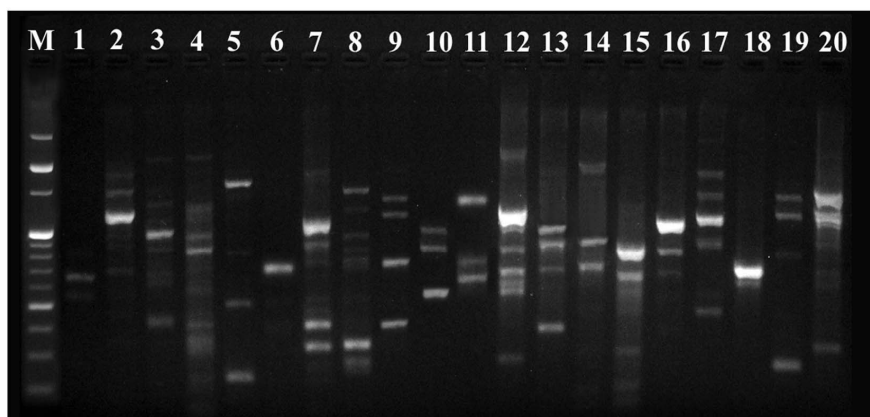
Further eight isolates *R. bataticola* are selected from eight districts (each isolate from one district) and nucleotide sequencing was done for ITS region of 18S rRNA. The BLAST data results revealed that the *R. bataticola* species matched with the reference strains of NCBI results and identified as *Rhizoctonia bataticola* (Table 17 and Plate 11). Out of eight isolates, all the isolates were distinct from one another in the cluster but identical to each other in homology and divergence with different nodes. The sequences are deposited in NCBI GeneBank, Maryland, USA along with location of the isolates.

4.5.4.2 Molecular diversity of *R. bataticola* isolates by RAPD

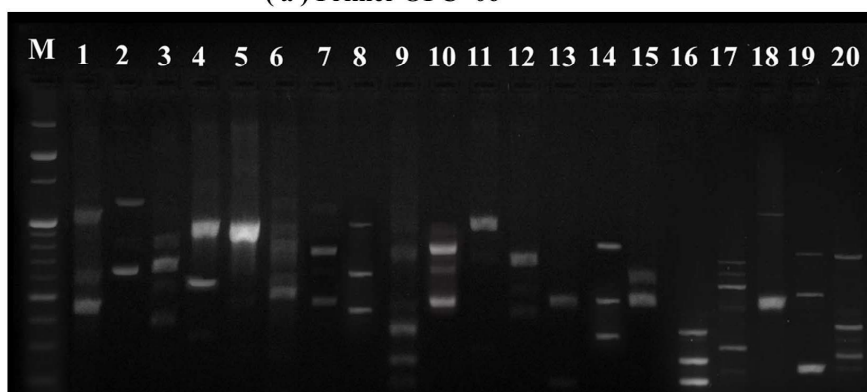
In twenty isolates, four random primers viz., OPC-06, OPG-15, OPP-14 and OPU-07 generated reproducible polymorphism. Amplified products with all the primers have showed polymorphic and distinguishable banding pattern indicating the genetic diversity among all the isolates of *R. bataticola*. A total of 389 reproducible and scorable polymorphic bands ranging approximately as low as 150 bp to as high as 1000 bp was generated with four primers among the twenty isolates studied (Plate 12).

The genetic dissimilarity estimates for twenty isolates were employed to generate dendrogram for tree construction by Unweighted Neighbourhood-joining method using the program DARwin 6.0 software. Based on the results obtained, all the twenty isolates were grouped into two main clusters indicating there is genetic diversity in the isolates of *R. bataticola*. Cluster I contained twelve isolates, main cluster divided into two sub clusters, sub cluster I had seven isolates (Rb-1, Rb-3, Rb-14, Rb-27, Rb-25, Rb-60 and Rb-41) and sub cluster II had five isolates (Rb-36, Rb-48, Rb-10, Rb-13 and Rb-34). Main cluster II has two sub clusters, sub cluster I consisted of two isolates (Rb-55 and Rb-43) and in sub cluster II six isolates (Rb-19, Rb-23, Rb-17, Rb-38, Rb-44 and Rb-56) were grouped.

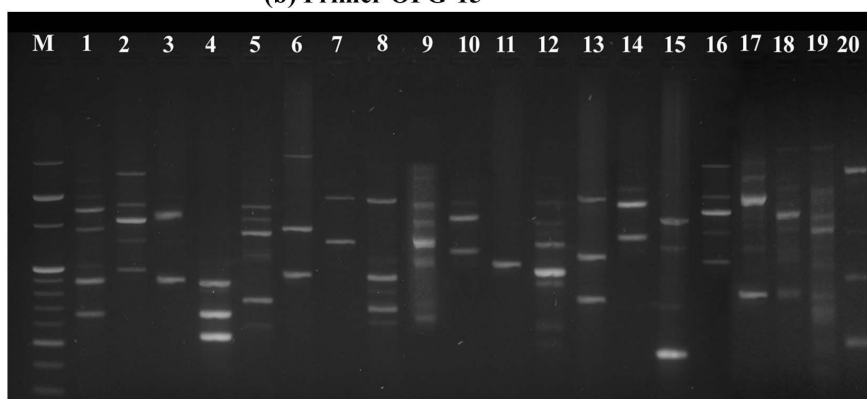
Based on simple matching co-efficient, a genetic similarity matrix was constructed to assess the genetic relatedness among the isolates of *R. bataticola*. Genetic similarity based on RAPD analysis of 20 isolates of *R. bataticola* is given in Table 18. The similarity coefficient ranged from 14 to 92 per cent. The maximum genetic similarity of 92 was between Rb-1 and Rb-3 whereas least genetic similarity (14 per cent) was observed between Rb-17 and Rb-27. The suitability of RAPD used was to detect the variations among the isolates of *R. bataticola*.



(a) Primer OPC -06



(b) Primer OPG-15



(c) Primer OPP- 14



(d) Primer OPU-07

Plate 12. DNA profiling of *R. bataticola* isolates banding on RAPD pattern

Table 17. Identity of *Rhizoctonia bataticola* of chickpea sequencing and BLAST results

Sl. No.	Isolate designation	Identified as	Per cent homology	Ref.accession number	Sequences
1.	<i>R. bataticola</i> (Rb-1)	<i>Rhizoctonia bataticola</i>	99%	HQ649832.1	GACGGGTCTACTGATCCGAGGTACCTTGAGAAAGTTCAGAAGGTTCTGTCGGGCGGGCGACGCCTTAC GCTCCGAAGCGAGGTGTATTCTACTACGCTTGAGGCAAGACGCCACCGCCGAGGTCTTTGAGGCGCGC CCGCAAAGGACGGTGCCCAATACCAAGCAGAGCTTGAGGGTTGAAATGACGCTCGAACAGGCATGCC CCCCGAATACCAAGGGGCGCAATGTGCGTTCAAAGATTGATGATTCACTGAATTCTGCAATTCACA TTACTTATCGCATTTTCGCTGCGTTCTTCATCGATGAAGAACGCAGCGAAATGCGATAAGTAATGTGAAT TGCAGAATTCAGTGAATCATCGAATCTTTGAACGCACATTGCGCCCCTTGGTATTCCGGGGGGCATGCC TGTTTCGAGCGTCATTTCAACCCTCAAGCTCTGCTTGGTATTGGGCACCGTCCTTTGCGGGCGCGCCTCA AAGACCTCGGCGGTGGCGTCTTGCCCTCAAGCGTAGTAGAATACACCTCGCTTCGGAGCGTAAGGCGTC GCCCCGCGGACGAACCTTCTGAACCTTCTCAAGGTGACCTCGGATCAGTAGACCCGTC
2.	<i>R. bataticola</i> (Rb-13)	<i>Rhizoctonia bataticola</i>	99%	KX270355.1	AGCCGCCCCCGATTGTTGGGGGTGGCTAGTGCCCGCCAGAGGACTATCAAACCTCCAGTCAGTAAACGT TGCAGTCTGAAAAAATATTAATAAACTAAACCTTCAACAACGGATCTCTTGGTTCTGGCATCGATG AAGAACGCAGCGAAATGCGATAAGTAATGTGAATTGCAGAATTCAGTGAATCATCGAATCTTTGAACG CACATTGCGCCCCTTGGTATTCCGGGGGGCATGCCTGTTTCGAGCGTCATTTCAACCCTCAAGCTCTGCT TGGTATTGGGCACCGTCCTTTGCGGGCGCGCCTCAAAGACCTCGGCGGTGGCGTCTTGCCCTCAAGCGTA GTAGAATACACCTCGCTTCGGAGCGTAAGGCGTCGCCCCCGGACGAACCTTCTGAACCTTTTCTCAAG GTTGACCTCGGATCAGGTAGGGATAC
3.	<i>R. bataticola</i> (Rb-17)	<i>Rhizoctonia bataticola</i>	99%	MG001962.1	TTTGGGGGTGGCTAGTGCCCGCCAGAGGACTATCAAACCTCCAGTCAGTAAACGTTGCAGTCTGAAAA AAATATTAATAAACTAAACCTTCAACAACGGATCTCTGTTCTGGCATCGATGAAGAACGCAGCG AAATGCGATAAGTAATGTGAACAGCAGAATTCAGTGTATCATAGAATCCTTGAACGCACATTGCGCTC CTTGGTATTCCGGGGGGCAAGCCTGTTTCGAGCGTCATTTCAACCCTCAAGCTCTGCTTGGCATTGGGG ACCGCCCTTGGCGGGGCGCGCGCAAATACCTCTGCGGAGGCGTCTTGCCCTCGCCGTTGAAAA
4.	<i>R. bataticola</i> (Rb-25)	<i>Rhizoctonia bataticola</i>	99%	KX447538.1	TACCTACCTCTGTTGCTTTGGCGGGCGCGGTCTTCCGCGGCCGCCCCCGATTGTTGGGGGTGGCTAG TGCCCGCCAGAGGACTATCAAACCTCCAGTCAGTAAACGTTGCAGTCTGAAAAAATATTAATAAACT AAAACTTTCAACAACGGATCTCTTGGTTCTGGCATCGATGAAGAACGCAGCGAAATGCGATAAGTAAT GTGAATTGCAGAATTCAGTGAATCATCGAATCTTTGAACGCACATTGCGCCCCTTGGTATTCCGGGGGG CATGCCTGTTTCGAGCGTCATTTCAACCCTCAAGCTCTGCTTGGTATTGGGCACCGTCCTTTGCGGGCGC GCCTCAAAGACCTCGGCGGTGGCGTCTTGCCCTCAAGCGTAGTAGAATACACCTCGCTTCGGAGCGTAA GGCGTCGCCCCGCGGACGAACCTTCTGAACCTTTCTCAAGGTTGACCTCGGATCAGGTAGGGATACCC GCTGAACCTTAAGCATATCAATAAGGCGGAGGAAC

Table 17. Contd....

Sl. No.	Isolate designation	Identified as	Per cent homology	Ref.accession number	Sequences
5.	<i>R. bataticola</i> (Rb-36)	<i>Rhizoctonia bataticola</i>	99%	HQ392772.1	GAACGTGCGGAGGGATCATTAGCGAGTTTACAACCTCCCAAACCCCTGTGAACATACCAATTGTTGCC TCGGCGGATCAGGGCCGCCCGGATTTGGGGGGTGGCTAGTGCCCGCCAGAGGACTATCAAACCTC CAGTCAGTAAACGTTGCAGTCTGAAAAAATATTAATAAACTAAACCTTCAACAACGGATCTCTT GGTTCTGGCATCGATGAAGAACGCAGCGAAATGCGATAAGTAATGAGAATTGAGAATTCAGTGAA TCATAGAATCTTTGAACGCACATTGCGCTCCTTGGTATTCCGGGGGGCAAGCCTGTTGAGCGTCAT TTCAACCCTCAAGCTCTGCTTGGTATTGGGCACCGTCCTTTGCGGGGCGCGCCTCAAAGACCTCGGCG GTGGCGTCTTGCATCAAGCGTAGGAAAATACACCTCCCTTCGCAGCGTACCGCC
6.	<i>R. bataticola</i> (Rb-44)	<i>Rhizoctonia bataticola</i>	99%	HQ392794.1	TGTCTTCCGCGGCCCGCCCCGATTTTGGGGGTGGCTAGTGCCCGCCAGAGGACTATCAAACCTCCA GTCAGTAAACGTTGCAGTCTGAAAAAATATTAATAAACTAAACCTTCAACAACGGATCTCTTGG TTCTGGCATCGATGAAGAACGCAGCGAAATGCGATAAGTAATGTGAATTGAGAATTCAGTGAATC ATCGAATCTTTGAACGCACATTGCGCCCCCTTGGTATTCCGGGGGGCATGCCTGTTGAGCGTCATTT CAACCCCTCAAGCTCTGCTTGGTATTGGGCACCGTCCTTTGCGGGCGCGCCTCAAAGACCTCGGCGGT GGCGTCTTGCTCAAGCGTAGTAGAATACACCTCGCTTCGGAGCGTAAGGCGTCGCCCGCCGACG AACCTTCTGAACCTTTCTCAAGGTTGACCTCGGATCAGGTAGGGATACCCGCTGAACCTTAAGCATAT CAATAAGCGGAGGAAATTACCGAGTTGATTCCGGCTCC
7.	<i>R. bataticola</i> (Rb-48)	<i>Rhizoctonia bataticola</i>	99%	HQ392778.1	GTCCCGATCCTCCACCCCTTTGTATACCTACCTCTGTTGCTTTGGCGGGCCGCGGTCTCCGCGGCCG CCCCCCGATTTTGGGGGTGGCTAGTGCCCGCCAGAGGACTATCAAACCTCCAGTCAGTAAACGTTG CAGTCTGAAAAAATATTAATAAACTAAACCTTCAACAACGGATCTCTTGGTTCTGGCATCGATG AAGAACGCAGCGAAATGCGATAAGTAATGTGAATTGCAGAATTCAGTGAATCATCGAATCTTTGAA CGCACATTGCGCCCCCTTGGTATTCCGGGGGGCATGCCTGTTGAGCGTCATTTCAACCCCTCAAGCTC TGCTTGGTATTGGGCACCGTCCTTTGCGGGCGCGCCTCAAAGACCTCGGCGGTGGCGTCTTGCTCA AGCGTAGTAGAATAACACCTCGCTTCGGAGCGTAAGGCGTCGCCCGCCGACGAACCTTCTGAACCTT TCTCAAGGTTGACCTCGGATCAGGTAGGGATACCCGCTGAACCTTAAGCATATCAATAAGGCGGAGG
8.	<i>R. bataticola</i> (Rb-56)	<i>Rhizoctonia bataticola</i>	99%	KT862032.1	TCCGCGGCCGCCCCCGATTTGGGGGGTGGCTAGTGCCCGCCAGAGGACTATCAAACCTCCAGTCA GTAAACGTTGCAGTCTGAAAAAATATTAATAAACTAAACCTTCAACAACGGATCTCTTGGTTCT GGCATCGATGAAGAACGCAGCGAAATGCGATAAGTAATGTGAATTGCAGAATTCAGTGAATCATCG AATCTTTGAACGCACATTGCGCCCCCTTGGTATTCCGGGGGGCATGCCTGTTGAGCGTCATTTCAAC CCTCAAGCTCTGCTTGGTATTGGGCACCGTCCTTTGCGGGCGCGCCTCAAAGACCTCGGCGGTGGCG TCTTGCTCAAGCGTAGTAGAATACACCTCGCTTCGGAGCGTAAGGCGTCGCCCGCCGACGAACCT TCTGAACCTTTCTCAAGGTTGACCTCGGATCAGGTAGGGATACCCGCTGAACCTTAAGCATATCAATA AGCGGAGGAACCTTACCGAGTTGATTCCGGCTCCGTCCCGATCCTCCACCTTTGTA

Table 18. Similarity coefficients of *R. bataticola* isolates based on RAPD analysis

	Rb-1	Rb- 2	Rb-3	Rb-4	Rb-5	Rb-6	Rb-7	Rb-8	Rb-9	Rb-10	Rb-11	Rb-12	Rb-13	Rb-14	Rb-15	Rb-16	Rb-17	Rb-18	Rb-19	Rb-20
Rb-1	1.0																			
Rb-2	0.92	1.0																		
Rb-3	0.65	0.40	1.0																	
Rb-4	0.54	0.70	0.62	1.0																
Rb-5	0.61	0.68	0.38	0.70	1.0															
Rb-6	0.47	0.67	0.57	0.46	0.45	1.0														
Rb-7	0.62	0.60	0.49	0.49	0.56	0.48	1.0													
Rb-8	0.65	0.68	0.36	0.57	0.70	0.43	0.53	1.0												
Rb-9	0.68	0.58	0.46	0.46	0.67	0.53	0.34	0.43	1.0											
Rb-10	0.47	0.45	0.63	0.66	0.55	0.53	0.34	0.50	0.36	1.0										
Rb-11	0.57	0.54	0.39	0.55	0.76	0.19	0.35	0.45	0.65	0.38	1.0									
Rb-12	0.50	0.34	0.29	0.29	0.35	0.42	0.5	0.43	0.33	0.5	0.14	1.0								
Rb-13	0.55	0.41	0.23	0.42	0.54	0.37	0.45	0.41	0.44	0.29	0.32	0.38	1.0							
Rb-14	0.55	0.33	0.47	0.41	0.42	0.18	0.49	0.76	0.46	0.46	0.39	0.42	0.50	1.0						
Rb-15	0.54	0.45	0.72	0.60	0.61	0.53	0.48	0.50	0.36	0.61	0.65	0.48	0.49	0.46	1.0					
Rb-16	0.73	0.46	0.70	0.65	0.54	0.52	0.40	0.57	0.52	0.76	0.37	0.54	0.14	0.61	0.60	1.0				
Rb-17	0.40	0.57	0.45	0.55	0.48	0.44	0.57	0.18	0.44	0.78	0.75	0.62	0.61	0.62	0.36	0.43	1.0			
Rb-18	0.47	0.39	0.54	0.40	0.71	0.70	0.45	0.43	0.53	0.45	0.38	0.41	0.71	0.72	0.53	0.36	0.52	1.0		
Rb-19	0.53	0.52	0.36	0.45	0.42	0.44	0.82	0.64	0.34	0.60	0.18	0.55	0.54	0.45	0.36	0.50	0.71	0.7	1.0	
Rb-20	0.60	0.63	0.53	0.39	0.54	0.68	0.61	0.47	0.44	0.36	0.37	0.61	0.48	0.61	0.52	0.50	0.50	0.60	0.58	1.0

4.6 Effect of climate change variables (temperature, moisture and carbon dioxide) on dry root rot disease

4.6.1 Effect of temperature

4.6.1.1 Effect of temperature on pathogen

Temperature is known as one of the crucial factor for the growth of pathogen. Present studies were conducted to know the optimum, minimum and maximum temperature requirements for the growth of pathogen *R. bataticola*. Three isolates (Rb 13, Rb 17 and Rb 34) were grown at eight different temperatures viz., 15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C, 45 °C and 50 °C and the data on colony growth was recorded at 48 h and 96 h after inoculation (Table 19 and Plate 13).

At 48 h after inoculation, among the three different isolates tested, the maximum colony growth was observed in Rb 34 (4.5, 31, 47, 52, 68 and 15 mm) at 15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C, respectively and least growth was observed in Rb 17 (1.40, 29, 32, 40, 50 and 10 mm) at 15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C respectively. No growth was observed in all the three isolates at 45 °C and 50 °C. However, there was significant difference between all the three isolates with respect to temperature requirement for their growth (Table 19 and Plate 13).

Among the eight different temperatures tested, the maximum colony growth was observed at 35 °C (65, 50 and 68 mm) followed by 30 °C (50, 40 and 52 mm) and 25 °C (43, 32 and 47 mm) and the least growth was observed at 15 °C (2.5, 1.4 and 4.5 mm) at 48 h after inoculation. No growth was observed at 45 °C and 50 °C temperature in all the three isolates (Rb 13, Rb 17 and Rb 34). Among different temperatures, there was significant difference between all eight temperatures except 45 °C and 50 °C (Table 19 and Plate 13).

At 96 h after inoculation, the maximum colony growth was observed in Rb 34 (18.15, 61.00, 75.50, 86.50, 90.00 and 18.00 mm) at 15, 20, 25, 30, 35, 40 °C, respectively and this is followed by isolate Rb 13 (12.55, 50.00, 64.00, 85.00, 90.00 and 17.00 mm) and least growth was observed in Rb 17 (10.20, 34.20, 37.00, 47.70, 80.00 and 15.00 mm) at 15, 20, 25, 30, 35 and 40 °C, respectively. There was no growth in all the three isolates at 45 and 50 °C. There was significant difference between all three isolates with respect to growth at different temperature (Table 19).

Table 19. Effects of temperature regimes on radial growth of *R. bataticola* at 48 and 96 hours after inoculation

*Radial growth of <i>R. bataticola</i> at 48 and 96 HAI (mm)								
Temperature regimes (°C)	Rb 13		Rb 17		Rb 34		Mean	
	48 HAI	96 HAI	48 HAI	96 HAI	48 HAI	96 HAI	48 HAI	96 HAI
15	2.50	12.55	1.40	10.20	4.50	18.15	2.80	13.66
20	30.00	50.00	29.00	34.20	31.00	61.00	30.00	48.40
25	43.00	64.00	32.00	37.00	47.00	75.50	40.66	58.80
30	50.00	85.00	40.00	47.70	52.00	86.50	47.33	73.06
35	65.00	90.00	50.00	80.00	68.00	90.00	61.00	86.66
40	12.00	17.00	10.00	15.00	15.00	18.00	12.33	16.66
45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mean	25.31	39.81	20.31	28.01	27.18	43.64	24.26	37.15

*Mean of three replications HAI- Hours after inoculation

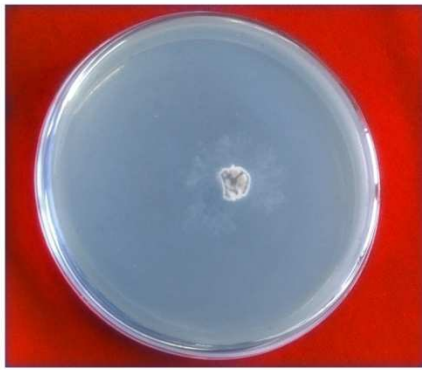
Factors	S.Em±	CD @ 1%
Temperature (T)	0.30	0.95
Isolate (I)	0.32	0.98
T x I	0.34	1.11

Table 20. Effects of temperature regimes on dry root rot disease caused by *R. bataticola* isolates on Annigeri-1 variety

Temperature (°C)	*Disease severity (1-9 scale)		
	Rb 13	Rb 17	Rb 34
15	1.3	1.0	1.7
20	3.0	2.9	3.1
25	7.2	6.7	7.4
30	8.0	7.2	8.2
35	9.0	8.5	9.0
40	3.00	2.5	3.0
45	0.00	0.00	0.00
50	0.00	0.00	0.00
Mean	3.93	3.60	4.05

*Mean of three replications

Factors	S.Em±	CD @ 1%
Temperature (T)	0.20	0.65
Isolate (I)	0.22	0.72
T x I	0.16	0.69



15 °C



20 °C



25 °C



30 °C



35 °C



40 °C



45 °C



50 °C

Plate 13. Effect of temperature on radial growth of pathogen *R. bataticola*



15 °C



20 °C



25 °C



30 °C



35 °C



40 °C



45 °C



50 °C

Plate 14. Effect of temperature regimes on dry root rot disease of chickpea plants

Eight different temperatures were tested for the growth of pathogen and among them, the maximum colony growth of the fungal isolates was found at 35 °C (90.0, 80.0 and 90.0 mm). The next best temperature was 30 °C (85.0, 47.70 and 86.50 mm) and 25 °C (64.00, 37.00 and 75.50 mm), There was very meager growth at 15 °C (12.55, 10.20 and 18.15 mm). No growth was observed at 45 °C and 50 °C in all the three isolate (Table 19 and Plate 13).

4.6.1.2 Effect of temperature on disease

The maximum disease severity rating was recorded in isolate Rb 34 with 1.7, 3.1, 7.4, 8.2, 9.0 and 3.0, followed by isolate Rb 13 (1.3, 3.0, 7.2, 8.0, 9.0 and 3.0) at 15, 20, 25, 30, 35 and 40 °C, respectively. Least disease severity rating was recorded in Rb 17 (1.0, 2.9, 6.7, 7.2, 8.5 and 2.5) at 15, 20, 25, 30, 35 and 40 °C respectively. In all the three isolates at 45 °C and 50 °C, no disease severity was recorded (Table 20 and Plate 14).

Among the eight different temperatures tested, maximum growth of the disease was found at 35 °C (9.0 8.5 and 9.0). The next best temperature for disease was 30 °C (8.0, 7.2 and 8.2) and 25 °C (7.20, 6.70 and 7.40), there was very less disease at 15 °C (1.3, 1.0 and 1.7). However, there was no disease at 45 °C and 50 °C (Table 20).

4.6.2 Effect of soil moisture

4.6.2.1 Effect of soil moisture on growth parameters

The effect of dry root rot at different moisture levels was studied on root length, shoot length and total biomass (Table 21, 22 and Plate 15).

4.6.2.1.1 Root length

The roots are the beginning of the vascular system pipeline that moves water and minerals from the soil up to the leaves. The *R. bataticola* is the dry root rot causing pathogen so the experiment was conducted to know the effect of soil moisture on growth parameters of chickpea as influenced by *R. bataticola*.

The highest mean root length of three isolates (Rb 13, Rb 17 and Rb 34) was observed in red soil (3.67 cm) compared to the black soil (3.60 cm), but no significant difference was observed (Table 21 and 22). Among the three isolates, maximum mean root length in both the soils was recorded in Rb 17 (3.74 cm and 3.90 cm) followed by Rb 13 (3.59 cm and 3.60 cm) and least was observed in Rb 34 (3.43 cm and 3.54 cm).

It was also observed that there was significant difference in the root length at different moisture regimes, among the seven soil moisture regimes tested against the dry root rot and its effect on plant root length, the maximum mean root length was recorded in both black soil and red soil at 100% (4.97 cm and 5.29 cm) soil moisture followed by 90% (4.45 cm and 4.80 cm), 80% (4.16 cm and 4.60 cm) and 70% (3.50 cm and 3.46 cm) and least was observed in 40% (2.14 cm and 1.98 cm) soil moisture regime (Table 21 and 22).

4.6.2.1.2 Shoot length

The shoots are the vegetative parts of the plant system which helps in the photosynthesis and production of food. The experiment was conducted to know the effect of *R. bataticola* with soil different moisture regimes on the shoot growth of chickpea.

The highest mean shoot length of three isolates (Rb 13, Rb 17 and Rb 34) was observed in red soil (19.38 cm) compared to the black soil (19.29 cm). Among the three isolates, maximum mean shoot length was recorded in both the soils in Rb 17 (19.96 cm and 20.26 cm) followed by Rb 13 (19.13 cm and 19.26 cm) and least was observed in Rb 34 (18.79 cm and 18.63 cm) (Table 21 and 22).

The results also indicated that there was significant difference in the shoot length in different moisture regimes, the maximum mean shoot length was recorded in both black soil and red soil at 100% (29.56 cm and 29.13 cm) followed by 90% (29.48 cm and 28.85 cm), 80% (25.21 cm and 27.14 cm) and 70% (20.76 cm and 21.66 cm). However, least shoot length was observed in 40% (8.45 cm and 7.30 cm) (Table 21 and 22).

4.6.2.1.3 Total biomass

The highest mean total biomass of three isolates (Rb 13, Rb 17 and Rb 34) was observed in red soil (1.59 g) compared to the black soil (1.42 g). Among the three isolates, maximum mean total biomass in both the soils was in Rb 17 (1.89 g and 1.68 g) followed by Rb 13 (1.36 g and 1.61 g) and next was observed in Rb 34 (1.35 g and 1.52 g) (Table 21 and 22).

Among the seven soil moisture regimes tested against the dry root rot and its effect on plant total biomass, the maximum mean total biomass was recorded in both soils at 100% (3.82 g and 3.81 g) soil moisture followed by 90% (3.45 g and 3.35 g),

Table 21. Effect of soil moisture levels on growth parameters of chickpea in black soil inoculated with *R. bataticola*

Moisture (%)	Growth parameters											
	Root length (cm)				Shoot length (cm)				Total biomass (g)			
	Rb 13	Rb 17	Rb 34	Mean	Rb 13	Rb 17	Rb 34	Mean	Rb 13	Rb 17	Rb 34	Mean
40	2.05	2.10	2.00	2.14	8.20	8.67	8.50	8.45	0.37	0.40	0.38	0.38
50	2.76	2.80	2.70	2.75	10.00	11.00	10.07	10.35	0.45	0.55	0.37	0.45
60	3.10	3.45	3.00	3.18	11.23	11.50	11.00	11.24	0.48	0.65	0.35	0.49
70	3.57	3.55	3.38	3.50	20.10	23.00	19.19	20.76	0.67	0.70	0.61	0.66
80	4.56	4.80	4.00	4.16	25.04	26.00	24.59	25.21	1.10	1.54	1.33	1.32
90	4.16	4.20	4.15	4.45	29.54	29.70	29.20	29.48	3.40	3.27	3.51	3.45
100	4.88	5.23	4.80	4.97	29.81	29.88	29.00	29.56	3.77	3.90	3.80	3.82
Mean	3.59	3.74	3.43	3.60	19.13	19.96	18.79	19.29	1.36	1.89	1.35	1.42

* Mean of three replications

Factors	S.Em±	CD @ 1%
Moisture (M)	0.29	0.93
Isolates (I)	0.32	1.01
M x I	0.41	1.31

Table. 22. Effect of soil moisture levels on growth parameters of chickpea in red soil inoculated with *R. bataticola*

Moisture (%)	Growth parameters											
	Root length (cm)				Shoot length (cm)				Total biomass (g)			
	Rb 13	Rb 17	Rb 34	Mean	Rb 13	Rb 17	Rb 34	Mean	Rb 13	Rb 17	Rb 34	Mean
40	1.84	2.00	2.1	1.98	7.50	7.65	7.00	7.30	0.35	0.50	0.25	0.36
50	2.64	2.67	2.63	2.60	9.53	10.89	9.55	10.00	0.66	0.85	0.70	0.73
60	2.60	3.40	3.00	3.00	11.46	11.66	11.38	11.54	0.60	0.64	0.57	0.60
70	3.33	3.60	3.30	3.46	20.76	26.23	18.00	21.66	0.98	1.00	0.83	0.93
80	4.50	4.75	4.56	4.60	27.20	27.24	27.00	27.14	1.39	1.43	1.35	1.39
90	5.00	5.06	4.35	4.80	29.07	29.00	28.50	28.85	3.54	3.45	3.10	3.35
100	5.24	5.85	4.8	5.29	29.33	29.20	29.00	29.13	3.88	3.90	3.90	3.81
Mean	3.60	3.90	3.54	3.67	19.26	20.26	18.63	19.38	1.61	1.68	1.52	1.59

Factors	S.Em±	CD @1%
Moisture (M)	0.21	0.70
Isolates (I)	0.26	0.82
M x I	0.32	1.06

Table 23. Effect of soil moisture levels on disease severity of dry root rot caused by *R. bataticola* isolates in black soil and red soil

Moisture (%)	Disease severity (1-9 scale)							
	Black soil				Red soil			
	Rb 13	Rb 17	Rb 34	Highest scale	Rb 13	Rb 17	Rb 34	Highest scale
40	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
50	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
60	8.90	8.35	8.91	8.91	9.0	8.45	8.95	8.95
70	7.80	6.60	8.03	8.03	8.00	7.82	8.33	8.33
80	5.00	4.87	5.62	5.62	5.25	5.22	5.73	5.73
90	4.55	4.32	4.86	4.86	5.08	4.25	5.40	5.40
100	3.10	2.80	3.30	3.30	3.16	3.00	3.56	3.56

*Mean of three replications

Factors	S.Em±	CD @ 1%
Soil type (S)	0.20	0.81
Isolate (I)	0.23	0.93
Moisture (M)	0.20	0.42
S x I x M	0.35	1.17



Plate 15. Effect of soil moisture levels on disease severity of dry root rot of chickpea

80% (1.32 g and 1.39 g) and 70% (0.66 and 0.93 g) and least was observed in 40% (0.38 g and 0.36 g), 50% (0.45 g and 0.73g) and 60% (0.49 g and 0.60 g). There was significant difference in the total biomass at different moisture regimes (Table 21 and 22).

4.6.2.2 Effect of soil moisture on disease incidence

Soil moisture plays a vital role in the development of plants. It also influences the growth of the soil microorganisms. In order to determine the effect of the soil moisture on development of dry root rot disease in chickpea, highly susceptible variety Annigeri-1 was grown in sick pots with black and red soils maintained at different moisture levels by gravimetric method (Table 23 and Plate 13).

The results indicated that there was significant difference between moisture levels but not with respect to soil types. In black soil the highest disease severity rating was 9.0, 9.0, 8.91, 8.03, 5.62, 4.86 and 3.30 while in red soil, it was 9.0, 9.0, 8.95, 8.33, 5.73, 5.40 and 3.56 at 40, 50, 60, 70, 80, 90 and 100 per cent soil moisture, respectively. The disease severity decreased as the soil moisture increased in both the types of soils (Table 23).

Among the seven levels of moisture content, the disease severity decreased slowly with respect to the increase in soil moisture. At 40% and 50% soil moisture, there was early development of disease symptoms. Yellowing of leaves was started at ten days after sowing and at twenty days after sowing the plants completely dried and root system was completely black. The plants recorded the maximum disease severity scale of 9.0 in both the soils at 40% and 50% soil moisture. While in 60% (8.91 and 8.95) and 70% (8.03 and 8.33) soil moisture, the dry root symptoms such as yellowing and upward turning of leaflets was started after 20 days after sowing and recorded the moderate disease severity in both black and red soils. The lesser disease severities were recorded at 100% (3.30 and 3.56), 90% (4.86 and 5.40) and 80% (5.62 and 5.73) soil moisture levels in black soil and red soil, respectively (Table 23).

4.6.3 Effect of carbon dioxide combined with temperature

Effect of carbon dioxide and temperature on growth and development of chickpea as well as dry root rot disease was studied under Open Top Chambers. The results obtained on phenology of crop and disease severity was presented here.

4.6.3.1 Effect of carbon dioxide combined with temperature on growth parameters

Growth parameters viz., shoot length, root length, number of branches and total biomass of chickpea due to combined effect of disease and carbon dioxide concentrations were observed under the Open Top Chambers (Table 24 and Plate 16).

4.6.3.1.1 Shoot length

Elevated carbon dioxide combined with temperature on phenology of chickpea crop revealed that the maximum shoot length of 31.66 cm was observed in open plot (T₅) followed by 27.66 cm in ambient CO₂ @ 390 ± 25 ppm (T₄) so, these two treatments significantly differed in respect of shoot length 12.91 cm in elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂) followed by in elevated CO₂ @ 550 ± 25 ppm alone (T₁) recorded shoot length of 20.33 cm and 24.58 cm in ambient CO₂ @ 390 ± 25 ppm with 2 °C rise in temperature (T₃) (Table 24). The results indicated that the carbon dioxide alone and coupled with increased temperature has detrimental impact on shoot length of chickpea plants.

4.6.3.1.2 Root length

There was a significant difference between root length of plants in the different treatments were observed. The more root length of 10.41 cm was observed in open plot (T₅) followed by 7.45 cm in ambient CO₂ @ 390 ± 25 ppm (T₄). The least growth of 3.66 cm was observed in elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂) followed by elevated CO₂ @ 550 ± 25 ppm alone (T₁) recorded root length of 5.30 cm and 6.61cm in ambient CO₂ @ 390 ± 25 ppm with 2°C rise in temperature (T₃) (Table 24). The elevated carbon dioxide combined with increased temperature of 2 °C has increased the severity of disease and reduced the root length of plants.

4.6.3.1.3 Number of branches

The results indicated that there was a significant difference between number of branches of plants in the different sets of treatments. The maximum number of branches of 17.26 was observed in open plot (T₅) followed by 13.33 in ambient CO₂ @ 390 ± 25 ppm (T₄) and 12.55 in ambient CO₂ @ 390 ± 25 ppm with 2°C rise in temperature (T₃). Reduction in the number of branches were observed in elevated CO₂ @ 550 ± 25 ppm with 2°C rise in temperature (T₂) as 8.30 followed by elevated CO₂ @ 550 ± 25 ppm alone (T₁) recorded 10.29 number of branches (Table 24).

Table 24. Effect of elevated carbon dioxide and temperature on disease severity of dry root rot of chickpea caused by *R. bataticola*

Sl. No.	Treatments	Mean shoot length (cm)	Mean root length (cm)	No. of branches	Total biomass (g)	*Disease severity
T ₁	Elevated CO ₂ @ 550 ± 25 ppm alone	20.33	5.30	10.29	0.54	8.10
T ₂	Elevated CO ₂ @ 550 ± 25 ppm with 2 °C rise in temperature	12.91	3.66	8.30	0.36	9.00
T ₃	Ambient CO ₂ @ 390 ± 25 ppm with 2 °C rise in temperature	24.58	6.61	12.55	0.85	5.83
T ₄	Ambient CO ₂ @ 390 ± 25 ppm	27.66	7.45	13.33	1.07	5.50
T ₅	Open plot	31.66	10.41	17.26	1.43	5.10
	S.Em±	0.41	0.25	0.19	0.12	0.11
	CD at 1%	1.34	0.71	0.61	0.39	0.36

*Mean of four replications



T₁ - Elevated CO₂ @ 550 ± 25 ppm alone



T₂ - Elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature



T₃-Ambient CO₂ @ 390 ± 25 ppm with 2 °C rise in temperature



T₄-Ambient CO₂ @ 390 ± 25 ppm



T₅- Open plot

Plate 16. Effect of elevated carbon dioxide and temperature on disease severity of dry root rot of chickpea

4.6.3.1.4 Total biomass

Effect of carbon dioxide combined with temperature on total biomass of the chickpea plants was reduced in elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂) as 0.36 g followed by elevated CO₂ @ 550 ± 25 ppm alone (T₁) recorded 0.54 g total biomass. The maximum total biomass was observed 1.43 g in open plot (T₅) followed by 1.07 g in ambient CO₂ @ 390 ± 25 ppm (T₄) and 0.85 in ambient CO₂ @ 390 ± 25 ppm with 2°C rise in temperature (T₃) (Table 24).

4.4.3.2 Effect of carbon dioxide on disease severity

Effect of different treatments of carbon dioxide and temperature were studied on host mediated effect on dry root rot. The results revealed that at higher concentration of carbon dioxide alone and in combination with increased temperature of 2 °C has aggravated the disease. Among the five treatments, elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂) showed higher disease severity scale of 9.00 with early infection showing drying of leaves and wilting of entire plant and roots were completely rotten. This was followed by elevated CO₂ @ 550 ± 25 ppm alone (T₁) showed disease rating scale of 8.10, here also same symptoms but expression of symptoms were slow compared to the (T₂). The least disease severity was observed in open plot (T₅) with 5.10 disease severity. Ambient CO₂ @ 390 ± 25 ppm (T₄) showed disease severity of 5.50 and ambient CO₂ @ 390 ± 25 ppm with 2°C rise in temperature (T₃) with 5.83 (Table 24 and Plate 16).

4.7 Identification of dry root rot resistant sources of chickpea

The studies on screening of one hundred germplasm lines of chickpea was done to identify the sources of resistance against dry root rot of chickpea (Table 25, 26 and 27).

4. 7.1 Preliminary screening of chickpea germplasm lines against dry root rot of chickpea under field condition

One hundred germplasm lines of chickpea were screened against dry root rot disease under field condition with artificial inoculation during *rabi*, 2015-16 and *rabi*, 2016-17. Observations on per cent disease incidence was recorded at 30, 60 and 90 DAS. The pooled data are presented in Table 30 and 31 (Plate 17).

Out of one hundred germplasm lines screened, 14 germplasm lines (IC83736, IC 83526, IC 83538, IC83798, IC95135, EC498827, IC83358, IC83357, IC83487,

IC83500, IC83501, IC83506, IC83514, IC 555217) showed resistant reaction to dry root rot with disease reaction of 0-10 per cent. Five germplasm lines (IC83603, IC84011, EC498818, IC83732, IC83742) showed moderately resistant reaction (10.1-20 per cent), three germplasm lines (EC 267504, EC498825, IC83356) moderately susceptible (20.1-30 per cent), twelve germplasm lines (EC 220078, IC 83409, IC 83775, IC83803, IC83816, EC498767, EC498773, IC83722, IC83723, IC83735, IC83743, IC83749) susceptible reaction (30.1-50 per cent). However, sixty six germplasm lines (EC 219999, EC 267240, EC 267484, EC 441805, EC 489870, EC 489870, EC 489907, EC 555002, IC 83344, IC83797, IC83965, IC83996, IC84008, IC116325, IC116343, EC498765, EC498779, EC498780, EC498812, EC498823, IC83359, IC83361, IC83362, IC83363, IC83353, IC83354, IC83355, IC83489, IC83490, IC83493, IC83494, IC83497, IC83502, IC83504, IC83508, IC83510, IC8351, IC83512, IC83513, IC83515, IC83516, IC83517, IC83725, IC83728, IC83729, IC83730, IC83731, IC83733, IC83734, IC83739, IC83740, IC83741, IC83742, IC83750, IC83751, IC83744, IC83745, IC83746, IC83747, IC83748, EC 528345, IC 83370, IC83997, IC116287, IC116304, IC83509) showed highly susceptible reaction of more than 50 per cent dry root rot incidence (Table 25, 26 and Plate 13).

4.7.2 Advanced screening of promising lines of chickpea against dry root rot by phenotyping technique

Nineteen promising (Resistant and moderately resistant) germplasm lines (Table 27) selected in preliminary screening were subjected to screening under phenotyping technique for confirmation of their disease reaction recorded under the field condition. The results indicated that only four genotypes (IC83798, IC95135, IC83506 and IC83500) showed resistant reaction and five germplasm lines viz., IC83736, IC 83526, IC83358, IC83501 and IC83514 were moderately resistant and rest of the lines were moderately susceptible/susceptible (Plate 18).

4.8 Integrated management of dry root rot of chickpea

Field experiments on integrated management of dry root rot of chickpea were conducted during *rabi*, 2015-16 and *rabi*, 2016-17. The results are presented in Table 28 and 29 (Plate 19).

4.8.1 Disease incidence of dry root rot (%)

4.8.1.1 Disease data of *rabi*, 2015-16

Table 25. Reaction of chickpea germplasm lines against dry root rot under field conditions during *rabi*, 2015-16 and *rabi*, 2016-17

Sl. No.	Germplasm lines	Incidence (%)				Reaction
		<i>Rabi</i> , 2015-16	<i>Rabi</i> , 2016-17	Mean	Maximum	
1.	EC 219999	100	89.00	94.50	100	HS
2.	EC 220078	47.00	42.95	44.97	47.00	S
3.	EC 267240	100	100	100	100	HS
4.	EC 267484	100	100	100	100	HS
5.	EC 267504	23.83	15.06	19.44	23.83	MS
6.	EC 441805	90.00	100	95	100	HS
7.	EC 489870	88.00	96.11	92.05	96.11	HS
8.	EC 489870	100	100	100	100	HS
9.	EC 489907	100	100	100	100	HS
10.	EC 528345	64.67	100	82.33	100	HS
	CHECK	100	100	100	100	HS
11.	EC 555002	87.5	100	93.75	100	HS
12.	IC 555217	06.50	02.20	4.35	6.50	R
13.	IC 83344	77.00	79.00	78.00	79.00	HS
14.	IC 83370	49.18	53.15	51.16	53.15	HS
15.	IC 83409	38.00	45.41	41.70	45.41	S
16.	IC 83526	08.20	6.00	7.10	8.20	R
17.	IC 83538	10.00	10.00	10.00	10.00	R
18.	IC83603	14.81	15.24	15.02	15.24	MR
19.	IC 83775	32.50	50.00	41.25	50.0	S
20.	IC83797	87.0	100	93.50	100	HS
	CHECK	100	100	100	100	HS
21.	IC83798	3.00	0.0	1.50	3.00	R
22.	IC83803	43.33	39.08	41.20	43.33	S
23.	IC83816	43.12	48.10	45.61	48.10	S
24.	IC83965	100	100	100	100	HS
25.	IC83996	100	100	100	100	HS
26.	IC83997	48.84	89.00	68.92	89.00	HS
27.	IC84008	100	100	100	100	HS

Table 25. Contd.....

Sl. No.	Germplasm lines	Incidence (%)				Reaction
		<i>Rabi</i> , 2015-16	<i>Rabi</i> , 2016-17	Mean	Maximum	
28.	IC84011	18.70	17.5	18.10	18.70	MR
29.	IC95135	05.60	2.60	4.10	5.60	R
30.	IC116287	46.98	81.00	63.99	81.00	HS
	CHECK	100	100	100	100	HS
31.	IC116304	49.22	65.60	57.41	65.60	HS
32.	IC116325	79.00	100	89.50	100	HS
33.	IC116343	100	100	100	100	HS
34.	EC498765	81.40	87.53	84.46	87.53	HS
35.	EC498767	34.40	40.96	37.68	40.96	S
36.	EC498773	31.66	34.12	32.89	34.12	S
37.	EC498779	100	100	100	100	HS
38.	EC498780	100	100	100	100	HS
39.	EC498812	100	100	100	100	HS
40.	EC498818	10.90	14.90	12.90	14.90	MR
	CHECK	100	100	100	100	HS
41.	EC498823	100	100	100	100	HS
42.	EC498825	24.85	29.71	27.28	29.71	MS
43.	EC498827	8.70	3.80	6.25	8.70	R
44.	IC83358	4.25	1.45	2.85	4.25	R
45.	IC83359	100	100	100	100	HS
46.	IC83361	90.56	100	95.28	100	HS
47.	IC83362	88.00	100	94.00	100	HS
48.	IC83363	100	100	100	100	HS
49.	IC83353	100	100	100	100	HS
50.	IC83354	95.0	100	97.50	100	HS
	CHECK	100	100	100	100	HS
51.	IC83355	100	100	100	100	HS
52.	IC83356	28.41	30.00	29.20	30.00	MS
53.	IC83357	5.50	1.30	3.40	5.50	R
54.	IC83487	9.34	9.15	9.24	9.34	R

Table 25. Contd.....

Sl. No.	Germplasm lines	Incidence (%)				Reaction
		<i>Rabi</i> , 2015-16	<i>Rabi</i> , 2016-17	Mean	Maximum	
55	IC83489	100	100	100	100	HS
56	IC83490	89.00	100	94.50	100	HS
57	IC83493	90.00	100	95.00	100	HS
58	IC83494	100	100	100	100	HS
59	IC83497	100	100	100	100	HS
60	IC83500	02.50	1.25	1.87	2.50	R
	CHECK	100	100	100	100	HS
61	IC83501	06.70	3.57	5.13	6.70	R
62	IC83502	100	100	100	100	HS
63	IC83504	76.00	100	88.00	100	HS
64	IC83506	6.00	6.78	6.39	6.78	R
65	IC83508	78.00	90.00	84.00	90.00	HS
66	IC83509	49.21	80.20	64.70	80.20	HS
67	IC83510	100	100	100	100	HS
68	IC83511	100	100	100	100	HS
69	IC83512	100	100	100	100	HS
70	IC83513	98.20	100	99.10	100	HS
	CHECK	100	100	100	100	HS
71	IC83514	07.80	5.45	6.62	7.80	R
72	IC83515	100	100	100	100	HS
73	IC83516	100	100	100	100	HS
74	IC83517	100	100	100	100	HS
75	IC83722	34.35	34.35	34.35	34.35	S
76	IC83723	36.56	40.73	38.64	40.73	S
77	IC83725	100	100	100	100	HS
78	IC83728	99.0	100	99.50	100	HS
79	IC83729	100	100	100	100	HS
80	IC83730	100	100	100	100	HS

Table 25. Cond.....

Sl. No.	Germplasm lines	Incidence (%)				Reaction
		<i>Rabi</i> , 2015-16	<i>Rabi</i> , 2016-17	Mean	Maximum	
	CHECK	100	100	100	100	HS
81	IC83731	100	81.20	90.60	100	HS
82	IC83732	15.40	19.83	17.61	19.83	MR
83	IC83733	100	100	100	100	HS
84	IC83734	100	100	100	100	HS
85	IC83735	49.19	41.16	45.17	49.19	S
86	IC83736	9.50	7.45	8.47	9.50	R
87	IC83739	75.00	82.00	78.50	82.00	HS
88	IC83740	86.00	92.00	89.00	92.00	HS
89	IC83741	100	100	100	100	HS
90	IC83742	100	100	100	100	HS
	CHECK	100	100	100	100	HS
91	IC83742	16.00	16.45	16.22	16.45	MR
92	IC83743	32.70	32.26	32.47	32.27	S
93	IC83744	100	100	100	100	HS
94	IC83745	100	100	100	100	HS
95	IC83746	90.00	100	95	100	HS
96	IC83747	93.50	100	96.75	100	HS
97	IC83748	99.00	90.13	94.56	99.00	HS
98	IC83749	44.05	44.21	44.11	44.21	S
99	IC83750	100	70.40	85.20	100	HS
100	IC83751	100	100	100	100	HS
	CHECK	100	100	100	100	HS

Table 26. Disease reaction of chickpea germplasm entries against dry root rot under field conditions during *rabi*, 2015-16 and *rabi*, 2016-17

Scale	Reaction	Germplasm lines	Entries
0-10%	Resistance (R)	IC83736, IC 83526, IC 83538, IC83798, IC95135, EC498827, IC83358, IC83357, IC83487, IC83500, IC83501, IC83506 IC83514, IC 555217	14
10.1- 20%	Moderately resistance (MR)	IC83603, IC84011, EC498818, IC83732, IC83742	5
20.1- 30%	Moderately susceptible (MS)	EC 267504, EC498825, IC83356	3
30.1 – 50%	Susceptible (S)	EC 220078, IC 83409, IC 83775, IC83803, IC83816, EC498767, EC498773, IC83722, IC83723, IC83735, IC83743, IC83749	12
More than 50 %	Highly susceptible (HS)	EC 219999, EC 267240, EC 267484, EC 441805, EC 489870, EC 489870, EC 489907, EC 555002, IC 83344, IC83797, IC83965, IC83996, IC84008, IC116325, IC116343, EC498765, EC498779, EC498780, EC498812, EC498823, IC83359, IC83361, IC83362, IC83363, IC83353, IC83354, IC83355, IC83489, IC83490, IC83493, IC83494, IC83497, IC83502, IC83504, IC83508, IC83510, IC8351, IC83512, IC83513, IC83515, IC83516, IC83517, IC83725, IC83728, IC83729, IC83730, IC83731, IC83733, IC83734, IC83739, IC83740, IC83741, IC83742, IC83750, IC83751, IC83744, IC83745, IC83746, IC83747, IC83748, EC 528345, IC83370, IC83997, IC116287, IC116304, IC83509	66

Table 27. Reaction of chickpea germplasm lines against dry root rot under phynotypic technique

Sl. No.	Germplasm lines	Disease incidence (%)	Reaction
1	IC83736	16.00	MR
2	IC 83526	17.5	MR
3	IC 83538	36.09	S
4	IC83798	2.8	R
5	IC95135	4.9	R
6	EC498827	29.00	MS
7	IC83358	15.40	MR
8	IC83357	26.00	MS
9	IC83487	38.11	S
10	IC83500	2.87	R
11	IC83501	16.00	MR
12	IC83506	5.6	R
13	IC83514	18.33	MR
14	IC 83736	29.00	MS
15	IC83603	25.00	MS
16	IC84011	33.06	S
17	EC498818	21.50	MS
18	IC83732	35.00	S
19	IC83742	38.55	S



Overview of screening plot

Plate 17. Screening of chickpea germplasm entries against dry root rot of chickpea under field condition



Plate 18. Advanced screening of promising enteries of chickpea against dry root rot by phenotyping technique

Results also indicated that all the treatments were significantly superior over control. Among the treatments employed, the treatment containing seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants (T₃) recorded significantly lowest incidence of dry root rot (7.70%) when compared to all other treatments except (T₇) seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants (9.57%). However, untreated control treatment recorded the highest incidence of dry root rot (33.45%) (Table 28 and Plate 13).

4.8.1.2 Disease data of *rabi*, 2016-17

The experiment was repeated during *rabi*, 2016 with same set of treatments. The results obtained during *rabi*, 2016-17 followed similar trend of results as observed during *rabi*, 2015-16. All the treatments were significantly superior to untreated control. Among them, seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants (T₃) recorded significantly lowest dry root rot incidence of 8.25% and next best treatment was (T₇) seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescence* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants (10.30%) and (T₄) seed treatment with *T. harzianum* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants and (T₆) seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* + *P. fluorescens* 2.5 kg/250 kg FYM during sowing were on par with each other by recording dry root rot incidences of 20.59% and 20.47%, respectively. The highest incidence of 36.25% was recorded in untreated control treatment (Table 28).

4.8.1.3 Pooled disease data (*rabi*, 2015-16 and *rabi*, 2016-17)

The results on pooled data indicated that the treatment combination seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants (T₃) was significantly superior in recording least disease incidences of dry root rot

(7.97%) from rest of the treatment combinations (Table 28). The next best treatments were seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants (T₇) (9.93%) and seed treatment with *T. harzianum* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants (T₄) and seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* + *P. fluorescens* 2.5 kg/250 kg FYM during sowing (T₆) were on par with each other but significantly different from untreated control (34.85%) (Table 28).

4.8.1.4 Per cent reduction in disease

The highest reduction in dry root rot (77.60%) disease incidence was observed in the treatment combination of seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants (Table 28). Seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants showed per cent reduction of 71.50% for dry root rot. It was comparatively less in other treatments viz., Seed treatment with *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* @ 2.5 kg/250 kg of FYM during sowing (29.32%), seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg (33.68%) and seed treatment with *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water (33.80%) (Table 28).

4.8.2 Yield data on yield and test weight

Mean yield data of dry root rot of chickpea during *rabi*, 2015-16 and 2016-17 indicated that the treatment combination seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants recorded significantly highest seed yield of 12.26 q/ha and test weight of 38.90 g which is significantly different from rest of the treatment combinations evaluated. Next best treatment was seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants recorded yield and test weight of 11.20

Table 28. Integrated management of dry root rot of chickpea during *rabi*, 2015-16 and *rabi*, 2016-17

Sl. No.	Treatments	Disease incidence (%)			
		2015-16	2016-17	Pooled Mean	Reduction over control
1	T ₁	21.90 (27.26)	23.55 (29.03)	22.27	36.09
2	T ₂	24.27 (29.51)	25.00 (30.0)	24.63	29.32
3	T ₃	7.70 (16.11)	8.25 (16.69)	7.97	77.60
4	T ₄	22.59 (28.38)	20.59 (26.99)	21.59	38.04
5	T ₅	26.00 (30.66)	25.15 (26.67)	25.55	33.80
6	T ₆	22.33 (28.20)	20.47 (26.90)	21.40	38.59
7	T ₇	9.57 (18.04)	10.30 (18.72)	9.93	71.50
8	T ₈	22.10 (28.04)	24.12 (29.41)	23.11	33.68
9	T ₉	33.45 (34.11)	36.25 (33.38)	34.85	
S.Em±		3.12	3.08		
CD (P=0.05)		9.06	8.94		

- T₁ Seed treatment with *T. harzianum* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* @ 2.5 kg/250 kg of FYM during sowing
- T₂ Seed treatment with *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* @ 2.5 kg/250 kg of FYM during sowing
- T₃ Seed treatment with mancozeb 50% + carbendazim 25% WS (Sprint) @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/lit water to infected and surrounding plants
- T₄ Seed treatment with *T. harzianum* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/lit water to infected and surrounding plants
- T₅ Seed treatment with *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/lit water to infected and surrounding plants
- T₆ Seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* + *P. fluorescens* @ 2.5 kg/250 kg of FYM during sowing
- T₇ Seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS (Sprint) @ 3 g/lit water to infected and surrounding plants
- T₈ Seed treatment with mancozeb 50% + carbendazim 25% WS (Sprint) @ 3.5 g/kg
- T₉ Untreated control

Table 29. Yield and economics of integrated disease management trial on chickpea as influenced by dry root rot disease during *rabi*, 2015-16 and 2016-17

Sl. No.	Treatment	Test weight (g)	Mean yield q/ha	Cost of cultivation (₹ ha ⁻¹)	Treatment cost (₹ ha ⁻¹)	Total cost of cultivation (₹ ha ⁻¹)	Gross returns (₹ ha ⁻¹)	B:C ratio
1	T ₁	24.0	7.96	25,000	450	25,450	55,720	2.22
2	T ₂	26.19	8.05	25,000	360	25,360	56,350	2.25
3	T ₃	38.90	12.26	25,000	1,225	26,225	85,820	3.40
4	T ₄	20.00	9.25	25,000	675	25,675	64,750	2.01
5	T ₅	16.90	7.20	25,000	660	25,660	50,400	2.39
6	T ₆	29.13	10.06	25,000	810	25,810	70,420	2.81
7	T ₇	37.50	11.20	25,000	1,110	26,110	75,600	3.02
8	T ₈	19.34	7.76	25,000	300	25,300	54,320	2.17
9	T ₉	15.60	5.74	25,000	0	25,000	40,180	1.60
	S.Em±	1.11	0.47					
	CD @ 0.05%	3.34	1.42					

- *T. harzianum* @ ₹ 150 /kg, *P. flurecense* @ ₹ 150 /kg and Sprint @ ₹ 1225/kg



T₃- Seed treatment with Mancozeb 50% + Carbendazim 25% WS (Sprint) @ 3.5 g/kg followed by soil drenching of Mancozeb 50% + Carbendazim 25% WS (Sprint) @ 3 g/lit water to infected and surrounding plants



T₉- Untreated control



Overall field view

Plate 19. Integrated management of dry root rot of chickpea

q/ha and 37.50 g, respectively. Further, seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* + *P. fluorescens* 2.5 kg/250 kg FYM during sowing recorded the seed yield and test weight of 10.06 q/ha and 29.13 g, respectively. However, the lowest seed yield and test weight was observed in untreated control (5.74 q/ha and 15.60 g) (Table 29).

4.8.3 Benefit cost ratio (B:C)

The economics of integrated disease management trial was worked out by taking into consideration the total cost of cultivation, cost of treatment and gross returns. The highest benefit (3.40) was obtained in seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants. The treatment i.e., seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants recorded BC ratio of 3.02 and other treatment combinations seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* + *P. fluorescens* 2.5 kg/250 kg FYM during sowing also showed little bit more BC ratio (2.81) when compared to rest of the treatments (Table 29). Least BC ratio of 1.60 was in untreated control.

Discussion

V. DISCUSSION

Chickpea (*Cicer arietinum* L.) is the world's second-most widely grown legume after beans (*Phaseolus vulgaris*). It is rich in dietary proteins and good for human consumption; moreover, its ability to form nitrogen-fixing nodules *via* interaction with rhizobia adds to its uniqueness (Ferguson *et al.*, 2010). Dry root rot of chickpea caused by *R. bataticola* (Taub.) Butler [Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid] is one of the major constraints for chickpea production in many countries in warm and humid areas, especially where chickpea cultivation in rainfed condition and/or expanding to high temperature areas.

It has been reported that loss of production in chickpea is because of diseases that attack the crop. The yield of chickpea is low due to several phytopathogens. Among them dry root rot caused by one of the serious soil borne sclerotial pathogen, *R. bataticola* (Tassi) Goid., which belongs to the anamorphic ascomycetes, became a serious menace in chickpea growing areas and causing severe yield loss.

Rhizoctonia as presently understood as an important soil borne fungus causing different types of diseases on a wide variety of plants (>500 spp.) over a large part of the world under more diverse environmental conditions, than any other plant pathogenic species. For example, diseases like root rot, charcoal rot, black scurf of potato, damping off and stem rot *etc.* are the diverse type of diseases with different *Rhizoctonia* species occurring on important crops like cotton, chilli, paddy, pulses *etc.*, (Ogoshi, 1984).

The present investigations on various aspects of survey on disease incidence and collection of pathogen isolates, diversity among isolates with respect to cultural, morphological, and molecular characteristics, influence of environmental factors like temperature, moisture and carbon dioxide on disease development, identification of resistant sources for dry root rot in chickpea and finally management of the disease by using biological and chemical tactics.

5.1 Symptoms of the disease

The first symptom was yellowing and partial or complete wilting of the stem or one or more branches. In advanced stage of the disease, affected plants became straw coloured, later lower leaves and stems showed brown discoloration. The tap root appeared

black, rotten and devoid of most of the lateral and fine roots. The dead root became quite brittle and shows shredding of bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil when plants are uprooted, these symptoms are most commonly observed in chickpea in field condition. In severe cases there was no formation of the seeds. Similar field diagnostic symptoms were recorded by Nene *et al.* (1991) who observed that drooping of petioles and leaflets at top of the plant, and leaves and stem of affected plant usually straw coloured but in some cases, the lower portion of the stem was brown. When the plant was uprooted, lower portion of the tap root remains in the soil. It was devoid of lateral and finer roots. Dark minute sclerotial bodies can be seen on the roots exposed inside the wood.

The dry root rot symptoms are most commonly observed in chickpea during post-flowering stage as gradual yellowing and drying of leaves followed by loss of vigour and premature death. The bark of such plants was easily peeled off along with extensive shredding of affected bark. The tap root turns black with signs of rotting and is devoid of most of the lateral and finer roots (Sharma *et al.*, 2016).

Srinivas (2016) observed the dry root rot symptoms on chickpea plants as bronzing of the leaves on one or more of the lower branches, leaves became yellow to brown in plants showing advanced disease symptoms. In such plants, the affected branches and leaf stalks were stiff, turned upwards and the leaflets stand more or less vertically and were shed prematurely. The terminal part of the tap root and lateral roots became brown to black and shriveled. The tap root without any lateral roots was also observed.

5.2 Pathogen:

The necrotrophic fungus *R. bataticola* exists in an anamorph (sclerotial) stage in soil and crop residues. *R. bataticola* produced grey cottony mycelial growth on potato dextrose agar medium and the colony morphology was compact or fluffy or velvety. The characteristic features of *R. bataticola* are right angle branching of the mycelium and constriction of the branch near the point of origin. The young hyphae are thin, hyaline, septate and dichotomously branched and the sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes. Similar symptoms were recorded by several workers (Byadgi, 1979; Sinclair 1982; Yang and Owen 1982; Kolte 1985; Singh 1989; Veerendra Kumar 2004 and Gowdra Nagamma and Saifulla, 2011b).

Sharma *et al.* (2012a) reported that the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later produce typical black sclerotia. The fungus may produce abundant aerial mycelium, as high as to touch the cover of the culture plates, or mycelium is found to be completely or partially suppressed. Further, growth and microsclerotial bodies production studies on PDA indicated that the mycelium was pale white in color in the initial stages of the growth but, later turned to dark brown to black as and when microsclerotial bodies started to form. Hypha showed right angled branching. The microsclerotia varied in shape and size with growth pattern of scattered to clustered (Manjunatha, 2014).

5.3 Isolation, identification and pathogenecity of *R. bataticola*

The fungus was successfully isolated on PDA by tissue isolation from diseased sample of infected root collected from chickpea field. The fungus isolated from infected root was confirmed as *R. bataticola* based on its cultural, morphological and formation of microsclerotia. Similarly, Aghakhani and Dubey (2009b) isolated 23 isolates of *R. bataticola* from chickpea plants showing characteristic dry root rot symptoms from different chickpea growing states of India. Ali and Dennis (1992) isolated *R. bataticola* from the roots of diseased plants of field peas on PDA medium.\

In the present study, it is also observed that the mycelium was initially white in colour which was later converted to dark brown to black in colour. Production of aerial mycelium was also observed in some isolates. The vegetative mycelium was characterized by the formation of barrel shaped cells and the formation of septum near the origin of branch of the mycelium. Branching occurred at right angle to parent hyphae, but branching at acute angles was also observed. Similar study was carried out by Shekhar *et al.* (2006a) who isolated seven isolates of *R. bataticola* from chickpea and reported that colony pigmentation was grayish white, blackish gray and dark black in centre with creamish periphery. Sharma *et al.* (2012a) reported that the fungus grows rapidly on potato dextrose agar (PDA) and produces brown to grey coloured mycelium that become darker with age. The young hyphae are thin, hyaline, aseptate and dichotomously branched and later produce typical black sclerotia.

In pathogenecity, initial symptoms of yellowing and partial or complete wilting of one or more branches was observed in inoculated pots. Drying and shredding of leaves

were observed near the soil surface and wilting of entire plants at later stage. Uprooting of the affected plants was easy and root portion was showed typical rotting as well as shredding. Symptoms due to dry root rot in the pots inoculated with *R. bataticola* culture were similar to dry root rot affected plants in the field. Veerendra Kumar (2004) and Kataria *et al.* (2007) assessed nine isolates of *R. bataticola*, the incitant of dry root rot of chickpea and revealed that all nine isolates were found pathogenic on C-235 chickpea cultivar in both sterilized and unsterilized soils. Further, Sharma *et al.* (2012) also proved the pathogenicity of 50 isolates of *R. bataticola* using chickpea cultivar BG 212 and reported the dry root rot disease severity ranging between 4 and 9 (on the basis of 1 to 9 severity rating scale).

5.4 To survey for incidence of dry root rot of chickpea in North Karnataka

The survey on the incidence of dry root rot of chickpea was carried out to know the prevalence and distribution of the disease in eight districts of North Karnataka viz., Bidar, Kalaburgi, Yadgir, Raichur, Koppal, Ballari, Dharawad and Vijayapura representing chickpea growing rainfed situations. Survey of the disease over a period of time provides information about the intensity with which it affects the yield. In addition, it is a source of information about severity of disease in relation to soil environment and edaphic factors. In the present study, the data on survey revealed that, the dry root rot incidence varied from locality to locality (Fig. 1). This may be due to varieties grown, environmental conditions viz., relative humidity, temperature and soil moisture content, cropping pattern like maize-chickpea, bajra-chickpea and chickpea and chickpea and build up of inoculum. Even, it could also be attributed to the existence of variability or pathogenic diversity present in the fungus. This is in agreement with Nene *et al.* (1991) and Sharma *et al.* (2016).

The overall mean incidence of dry root rot (38.72%) was more during *rabi*, 2015-16 compared to *rabi*, 2016-17 (32.73%). This may be due to variations in climatic conditions prevailed during the year at different locations. In recent years, dry root rot is becoming an important disease in North Karnataka and overall mean incidence was maximum in Kalaburgi district (32.93%) followed by Raichur (29.87%) and least incidence was recorded in Yadgir (3.35%) (Fig. 2 and 3).

In *rabi* 2015-16, the incidence of dry root rot was maximum (45.55%) in Bhusnur village of Aland taluk, Kalaburgi district followed by Huvinahadgi village of Devadurga

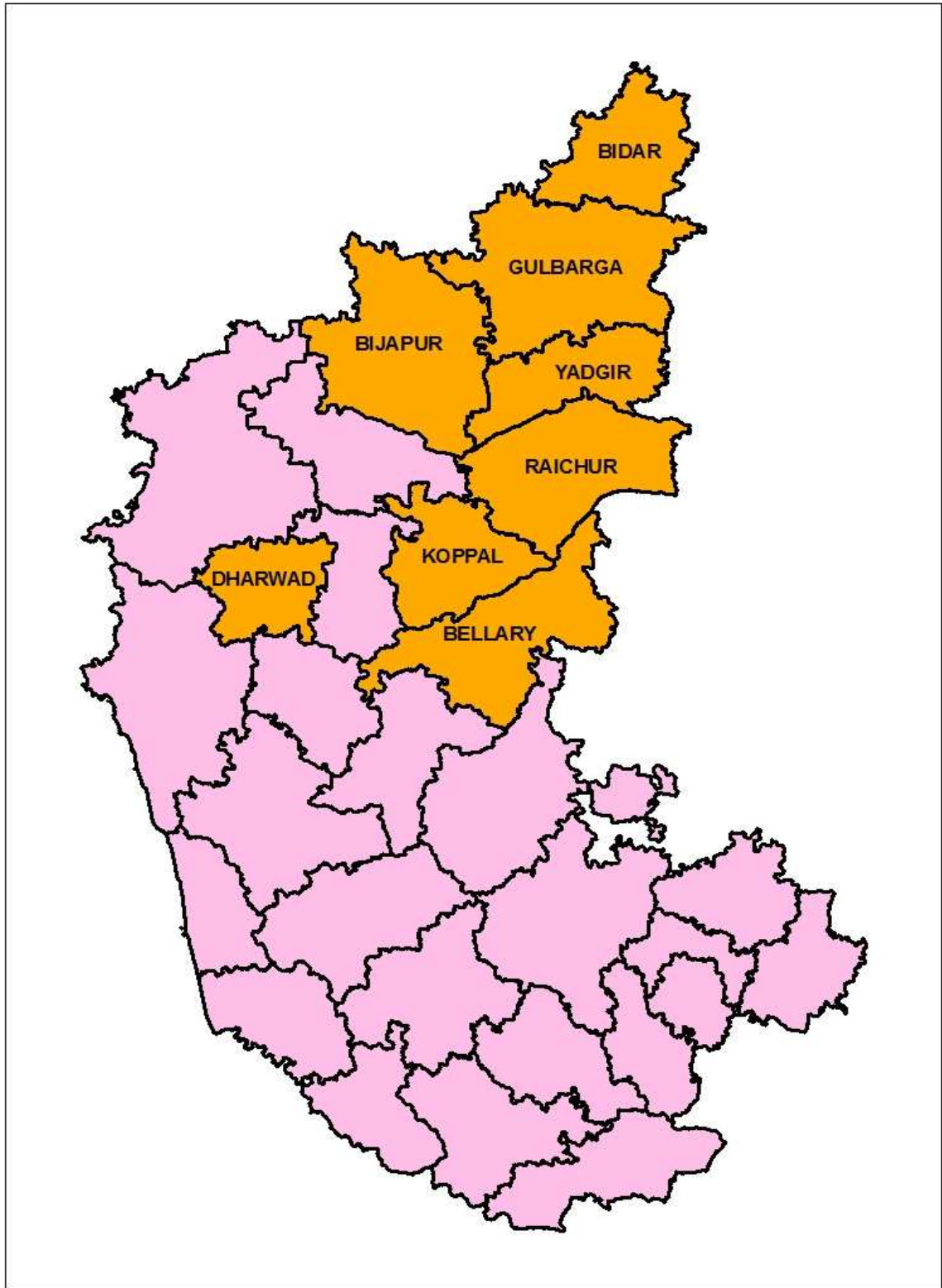


Fig. 1: Survey districts in North Karnataka

taluk, Raichur district, recorded highest dry root rot (45.23%). The least was noticed in Aldahal (1.00%) and Munderga village (1.00%) of Yadagiri taluk. Similarly, Veerendra Kumar (2004) noticed an incidence of 9.97 per cent dry root rot of chickpea in Gulbarga followed by Raichur (5.16%) and Bidar (4.28%). The highest incidence was 12.24 per cent in Chittapur of Gulbarga and least was in one of the locations of Raichur (3.21%) (Fig. 2 and 3).

In *rabi* 2016-17, the incidence of dry root rot was maximum (43.20%) in Huvinahadgi village of Devadurga taluk, Raichur district followed by Sannur village of Kalaburgi taluk, Kalaburgi district, recorded highest dry root rot (43.00%). The least was noticed in Mailapur (1.00%) and Saidapur village (1.00%) of Yadagiri taluk. The similar observations were supported by Manjunatha *et al.* (2011) during *rabi*, 2008 in the farmers' fields at Raichur, Gulbarga and Bidar districts in Karnataka. The maximum dry root rot incidence in chickpea was noticed in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18%) in Bidar. The overall incidence of dry root rot ranged from 1-19% across the districts. This is mainly attributed to variation in soil type (black and red soil) and cultivars grown (Annigeri and JG 11) under rain fed condition. Gurha and Trivedi (2008) surveyed chickpea fields in Shimoga, Raichur and Bangalore districts in Karnataka and reported the prevalence of dry root rot. They also observed the predominance of *R. bataticola* infecting 60-70% of the plants in the fields of Gulbarga, Manchalapur, Ekklapur, Raichur and Tengri.

The high incidence in Bhusnur, Huvinahadgi and Sannur village may be attributed to the monocropping of A-1 variety in larger area and high temperature prolonged dry spell, which might be responsible for existence of aggressive isolate of pathogen. The least incidence in Aldahal, Mailapur, Munderga and Saidapur village of Yadagiri taluk could be attributed to the usage of other improved varieties and irrigation facility, which might have shown unfavorable condition for pathogen growth and the disease. The high incidence was generally noticed in black soil compared to red and loamy type soil, it may be due to the high organic content and soil pH level helpful for the growth of the pathogen.

Knowledge on the irrigation facility and soil moisture status helps in determining the percentage of disease incidence. Such variations in disease incidence in different locations is usually attributed to environmental and/or cultivar variations. However, the

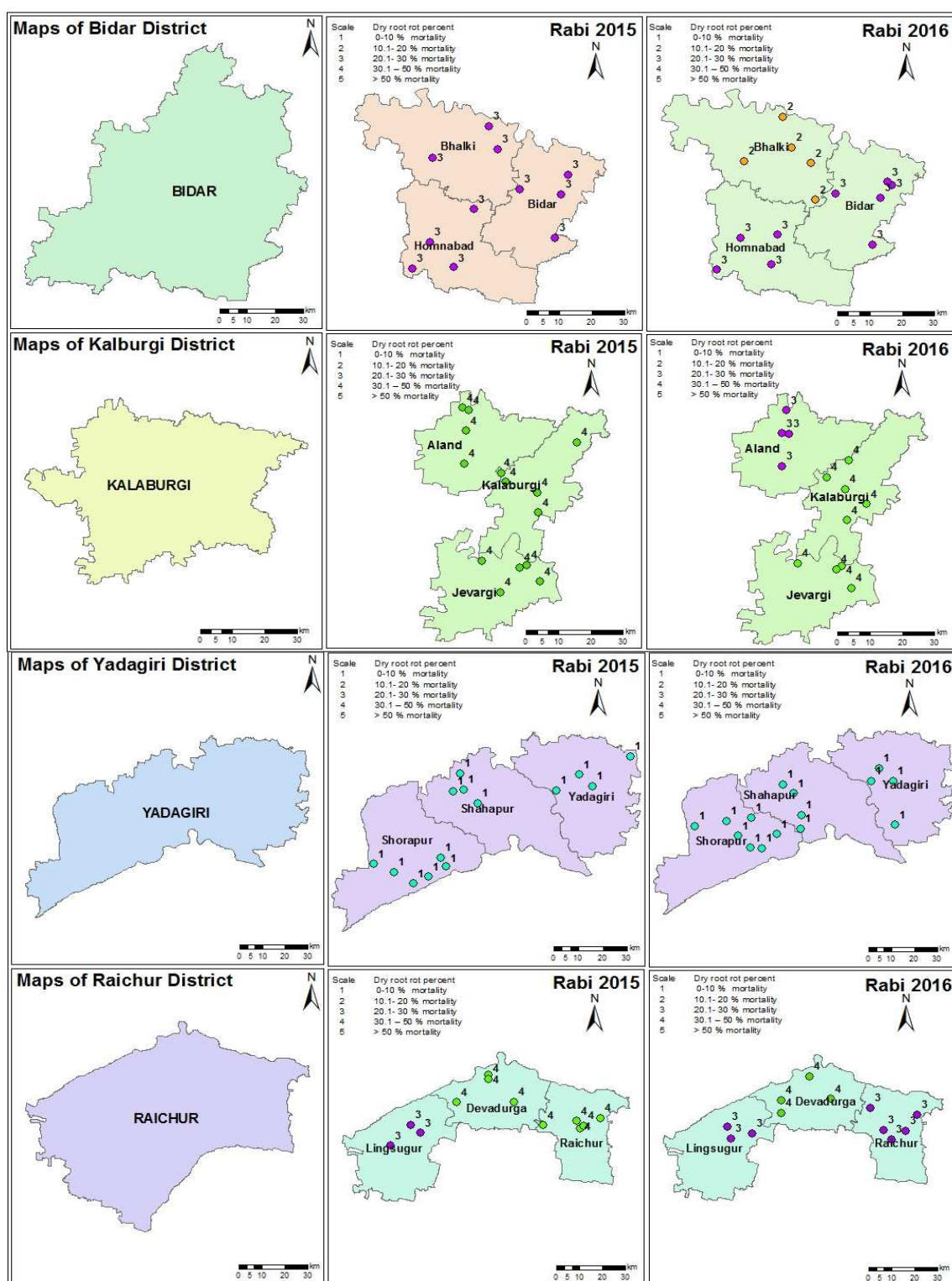


Fig. 2: GIS map showing incidence of dry root rot of chickpea in Bidar, Kalaburgi Yadgiri and Raichur district during *rabi* 2015-16 and 2016-17

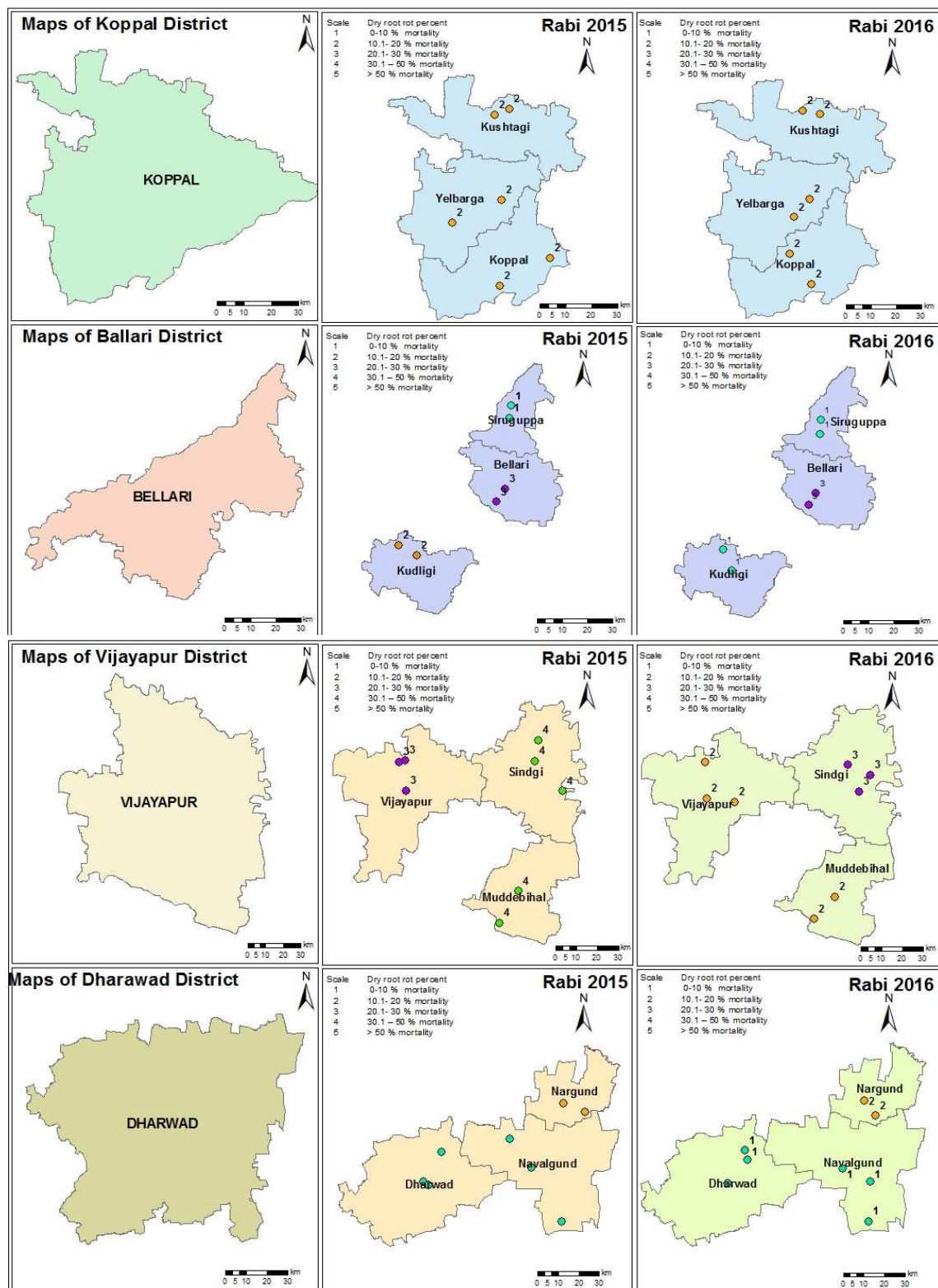


Fig. 3: GIS map showing incidence of dry root rot of chickpea in Koppal, Ballari, Vijayapur and Dharwad district during *rabi* 2015-16 and 2016-17

variations may also be attributed to the presence of variability in pathogenic fungus and also due to variation in weather parameters which affected growth and development of disease. The average temperature of 30-40 °C and prolonged dry spell (Rainfall 20 mm) from flowering to harvesting stage was observed in Kalaburgi during Nov-Feb 2015 and 2016, which was resulted in enhanced disease. These observations corroborate the results of Nobel and Richardson (1968) and Bajpal *et al.* (1999). Mono-cropping with closer spacing and also drought and high temperature increased the vulnerability of the crop and enhanced the aggressiveness of the pathogen. The present findings are also supported by Garrett *et al.* (2006); Kaur *et al.* (2012) and Sharma *et al.* (2016). Srinivas (2016) reported through survey that percent disease incidence was ranged from 6.67 to 15.33 per cent in Karnataka and 0.67 to 3.33 per cent in Andhra Pradesh at different locations. Rajkrishan *et al.* (1999) studied the effect of edaphic factors and moisture regime on the incidence of *Macrophomina* root rot in sesame. Among the various soil types, sandy soil supported maximum disease (78.33%) as compared with clay soil (51.56%). Ghosh *et al.* (2013) conducted survey during *rabi* 2010-11 to obtain information on the distribution and incidence of chickpea diseases in respect to soil type, cultivar and seed treatment in central and southern parts of India. Dry root rot was found at all the sites and the per cent incidence was ranged from 8.9-10.3% irrespective of cultivar type and location.

5.5 Studies on cultural, morphological and molecular variability of pathogen

5.5.1 Collection of *R. bataticola* isolates infecting chickpea

Totally sixty dry root rot infected diseased specimens of chickpea plants showing typical symptoms were collected from different locations of North Karnataka viz., Bidar, Kalaburgi, Yadgir, Raichur, Koppal, Ballari, Dharawad and Vijayapura districts during *rabi*, 2015 and 2016. The fungus was obtained from infected plants by tissue isolation on potato dextrose agar medium. The identity of isolates was confirmed as *R. bataticola* based on morphological and cultural characters as described by previous workers (Vinod Kumar *et al.*, 2007; Konde *et al.*, 2008; Sharma *et al.*, 2016 and Srinivas, 2016).

Similar work was reported by Aghakhani and Dubey (2009b) who collected 23 isolates of *M. phaseolina* from chickpea plants showing characteristic dry root rot symptoms from different chickpea growing states of India and isolated the associated

fungus from the roots of diseased plants on PDA medium. Manjunatha and Naik (2011) collected thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions from Karnataka.

5.5.2 Cultural variability of *R. bataticola*

In nature, new strains may arise by mutation, hybridization, differential cytoplasmic inheritance (Worthington *et al.*, 2011), heterokaryosis (Milgroom *et al.*, 2009) and by parasexual life cycle (Rosada *et al.*, 2010). Study of pathogenic variability is essential for breeding for disease resistance in crop improvement programme. A potential pathogen is often blessed with biodiversity within its population. Basically, variation in pathogen is desirable trait for its existence in nature. This variability among the pathogens underlies their diverse nature and ability to withstand the host environment. Variability of pathogens was studied with respect to cultural behavior to focus on the existence of variation in *R. bataticola* collected from different locations.

In the present investigation, cultural characters studied on potato dextrose agar at room temperature showed variation among the *R. bataticola* isolates. On the basis of radial growth of mycelium, colony morphology, aerial mycelia and colony pigmentation, sixty isolates were categorized into different groups. Each group included different number of isolates from different locations.

Diversity in cultural characters such as radial growth, colony morphology, aerial mycelia and colony pigmentation were closely observed in sixty isolates of *R. bataticola*. Based on radial growth, isolates were categorized into three groups: very fast growing, moderate and slow growing. Data after seven days incubation the results revealed that Group I comprised 55 isolates which were fast growing with an average growth rate (75-90 mm), Group II comprised of two moderate growing isolates with an average growth rate (60-75 mm) and Group III comprised of three isolates slow growing with an average growth rate of 45-60 mm. The findings are in accordance with Manjunatha and Naik (2011) who reported that thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions were highly variable in their cultural characters. Among these, twelve isolates were fast in growth, another twelve were moderate and the remaining six isolates were slow in growth. Aghakhani and Dubey, (2009) studied the variability of twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states of India

and reported that all isolates were highly variable in their cultural characters. After 48 hours of incubation, highest radial growth (90 mm) in Rb1 (Bangalore) followed (89 mm) in Rb4 (Faridkot) was observed. Rb3 (New Delhi) and Rb5 (Faridkot) were next best with similar radial growth of (83.5 mm).

Based on colony morphology, isolates were categorized into three groups: fluffy and appressed and velvety in the present study. Group I comprised of 17 isolates which showed appressed growth, Group II colonies of 13 isolates were velvety growth and Group III includes 30 isolates showed fluffy growth. Similar variation was observed by Dhingra and Sinclair (1978) who noticed fluffy growth in stem isolates, partially fluffy growth in root isolates and appressed growth in all other plant part isolates of *M. phaseolina* on soybean. Edraki and Banishashemi (2010) studied sixty isolates of *M. phaseolina* collected from different parts of Iran. Isolates were grouped in to four phenotypes viz., fluffy with abundant sclerotia, fluffy with few sclerotia, partially fluffy and appressed growth.

Based on aerial mycelia, isolates were categorized into two groups: presence and absence of aerial mycelia in the present study. Group I comprised of 28 isolates which showed presence of aerial mycelial growth and Group II colonies of 32 isolates absence of aerial mycelia. These results are supported by Fernandez *et al.*, (2006) who studied the diversity pattern of 15 isolates from Mexico and 15 isolates from other countries of *R. bataticola* from beans based on presence or absence of aerial mycelium. They observed 15 isolates had intermediate and 15 isolates had poor aerial mycelium. Eight from Mexican isolates and 7 isolates of non Mexican isolates had intermediate aerial growth.

Sixty isolates exhibited variability with respect to pigmentation of mycelia, in this study isolates were categorized as black with grey aerial mycelia (15 isolates), black (15 isolates), grey (25 isolates) and white pista colour (5 isolates). Similar results with respect to variation in colony pigmentation of *R. bataticola* were observed by Manjunatha and Naik (2011) who categorized isolates into three groups, the first group comprised of all isolates from Bidar exhibited light brown pigmentation. The second group comprised of all isolates from Raichur, Gulbarga and Bidar exhibited black colony pigmentation. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed white colony pigmentation.

5.5.3 Morphological variability of *R. bataticola* isolates

Morphological characters studied on potato dextrose agar at room temperature showed variation among the *R. bataticola* isolates. On the basis of morphological characters viz., width of mycelia, production of sclerotia, number of days to produce sclerotia, number of sclerotia produced, shape of sclerotia and size of sclerotia. Isolates of *R. bataticola* were categorized into four to five groups. Each group included different number of isolates from different locations.

There was much variation in the width of the hyphae, among the sixty isolates, highest width was found in the isolate Rb-58 (175.03 μm) followed by Rb-18 (171.24 μm) and Rb-57 (168.00 μm). Least width of the hyphae was recorded in isolate Rb-16 (86.56 μm). Similar results were also reported by Gupta *et al.* (2012) observed variation in size of hyphal width of forty *R. bataticola* isolates from chickpea. Eighteen isolates were classified under very thin (5.2-6.5 μm), 19 isolates under thin (6.6 -7.9 μm) and only three isolates from Chhattisgarh and Gujrat exhibited (8.0 - 9.3 μm) thick hyphal width.

Among the sixty isolates collected production of the microsclerotia was observed under electronic microscope. Out of sixty isolates fifty one isolates were found to produce sclerotia, but in nine isolates (Rb-5, Rb-6, Rb-16, Rb-21, Rb-22, Rb-24, Rb-43, Rb-54, Rb-55) there was absence of sclerotia which indicated that there is variation among the isolates. Absence of microsclerotia could be due to differences in nutritional requirements and genetic characteristics. Low temperature, osmotic and matric potential in the growth medium can influence the formation of reproductive structures by soilborne fungi (Cook and Duniway, 1980). The present investigations are supported by Kaur *et al.* (2013). Maruthi (2016) isolated 24 isolates of *Rhizoctonia bataticola* from pigeonpea plants in which microsclerotial production was recorded in 18 isolates and the size ranged from 12.33 to 57.66 per microscopic field. Further, they also reported that six isolates viz., Rb-8, Rb-11, Rb-12, Rb-16, Rb-17 and Rb-19 did produce any microsclerotia under PDA.

Grouping of *R. bataticola* isolates was made on the basis of number of days taken to produce sclerotial bodies and sixty isolates were categorized into three groups. Twenty nine isolates produced within 1-2 days, nineteen isolates were taken 3-4 days, and rest of the three isolates took 5-6 days. This fact has been observed by earlier workers

(Vishwadar and Sarabhay, 1993 and Shekhar *et al.*, 2006a). Purkayastha *et al.* (2004) observed variability in time taken for production of sclerotia of 55 isolates of *R. bataticola* from cluster bean. Among 55 isolates, sixteen isolates produced sclerotia in one day, thirteen isolates in three days and the rest of isolates taken five days to produce sclerotia.

The results also indicated that there is variation with respect to number of microsclerotial bodies produced by isolates of *R. bataticola* per microscopic field (10x) and the isolates were categorized into three groups. Twenty one isolates produced 0-25 sclerotial bodies, twenty six isolates produced 26-50 sclerotial bodies and four isolates produced 51-100 sclerotial bodies. Similar results were observed by Gupta *et al.* (2012) who reported that maximum number of sclerotia (22.3 sclerotia/ microscopic field) were produced by Betul isolate of Madhya Pradesh and isolate from Dhule, Maharashtra exhibited minimum number of sclerotia (8.1 sclerotia/ microscopic field).

Morphological variations with respect to shape of sclerotia were also recorded during present investigation and the isolates were categorized as round, ovoid and irregular groups based on shape of sclerotia. Round shaped sclerotia were observed in twenty while ovoid shaped sclerotia were observed in thirteen isolates and irregular shape of sclerotia was observed in eighteen isolates. Gupta *et al.* (2012) evaluated diversity in isolates of *R. bataticola* causing dry root rot in chickpea collected from four major chickpea growing states of central India. The isolates were highly variable in their sclerotial morphology, two groups of isolates were formed, one with oblong shape having irregular edges and the other being round with regular edges.

Variation in size of the microsclerotia was also observed in sixty isolates. Among them maximum size was recorded in isolates Rb-33 (3.20 x 2.71 mm) followed by Rb-34 (3.05x1.47 mm) and Rb-1 (2.75x2.00 mm). Least size of the microsclerotia was recorded in isolates Rb-27 (1.31x0.94 mm) and Rb-26 (1.09x1.00 mm). Aghakhani and Dubey (2009) who observed variation in size of twenty three isolates of *R. bataticola* causing dry root rot of chickpea. The sclerotia formed in different isolates were varied in size from 40-600 µm. The isolates from Bangalore (Rb1) and Faridkot (Rb 5) produced the largest size of sclerotia which ranged from 200 – 600 µm and 100 – 400 µm, respectively. Dumka (Rb19) isolate produced small sized sclerotia which ranged from

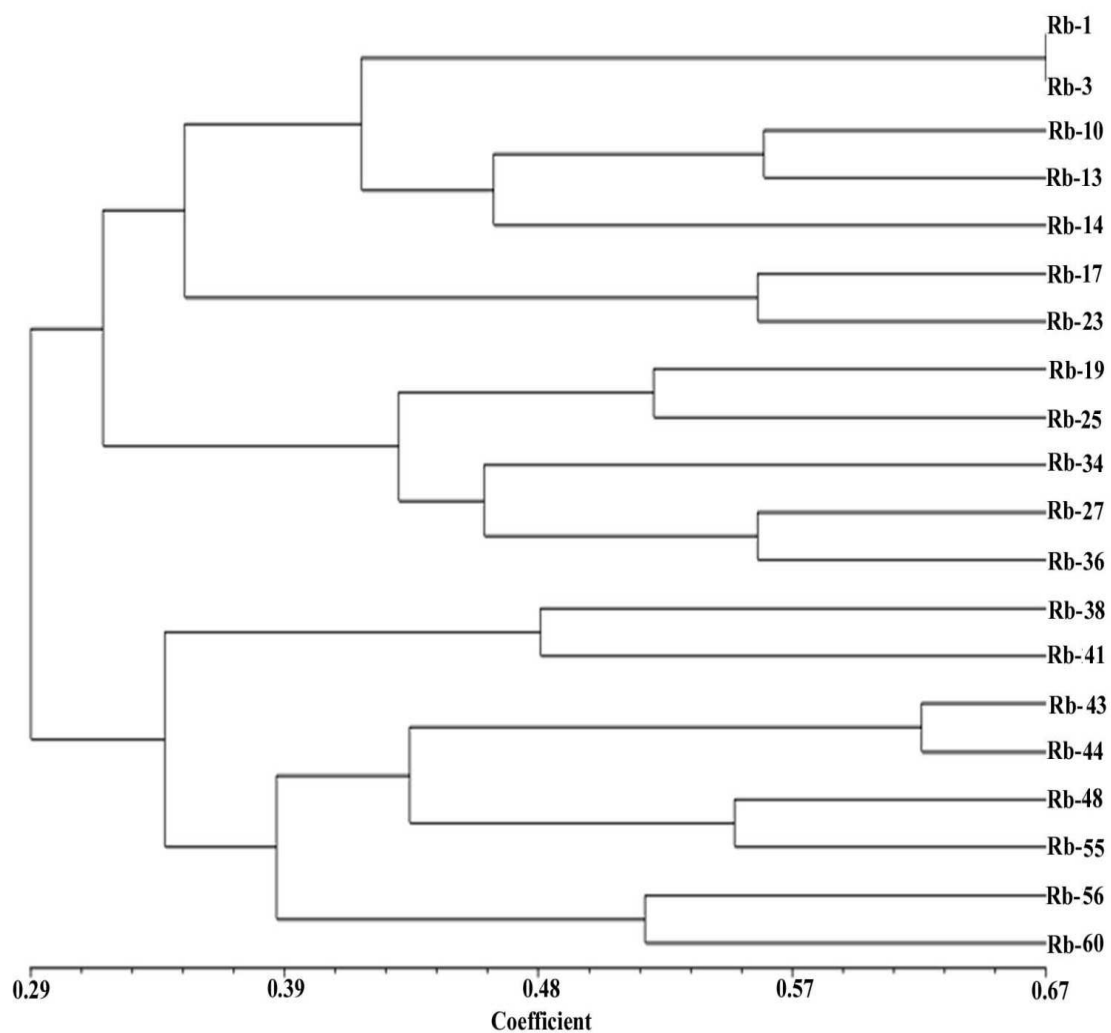


Fig. 4: Hierarchical horizontal dendrogram showing clustering of 20 *R. bataticola* isolates by DARwin V.6 software

43.9- 81 μm . Forty isolates of *R. bataticola* from chickpea were studied by Gupta *et al.* (2012), they observed sclerotial sizes varied from 72.7- 117.2 x 57.1-106.5 μm . Based on sclerotial size, the isolates were divided into three groups viz. minute size (72.7 - 87.5 x 57.1-73.0 μm), medium size (87.6 - 103.2 x 73.5 - 90.0 μm) and bold (103.3 -117.2 x 90.1-106.5 μm) size of sclerotia groups.

5.5.4 Molecular variability of *R. bataticola* isolates

5.5.4.1 Identification of *R. bataticola* isolates by PCR

Amplification of the ITS region produced an approximately 550–600 bp fragment which is specific to *R. bataticola*. In the present study, all twenty isolates gave the same size of the fragment that is 550–600 bp. The BLAST data results revealed that the *R. bataticola* species matched with the reference strains of NCBI results and identified as *Rhizoctonia bataticola*. Out of them, eight isolates were sequenced and the results indicated that all the isolates were distinct from one another in the cluster but identical to each other in homology and divergence with different nodes which are represented in the phylogenetic tree (Fig.4). The results are in agreement with scientists Atkins and Clark (2004) and Brunner *et al.* (2007).

Aghakhani and Dubey (2009b) determined genetic diversity among six isolates of *R. bataticola* representing five different ITS groups with respect of their nucleotide sequences of ITS1, 5.8S and ITSII regions. The ITS of Karnataka isolate (Rb1) was largest (670 bp), where as it was smallest (499 bp) in Haryana (Rb21) isolate. After sequencing the ITS sequences of the isolates were deposited to Gen-Bank nucleotide database. The phylogeny tree constructed from the nucleotide sequence, similarity of these six isolates along with 27 other ITS sequences of fungi which showed 96–100% similarity during blast analysis clustered them into three major groups.

5.5.4.2 Molecular diversity among *R. bataticola* isolates by using RAPD primers

Among the twenty four isolates, four random primers viz., OPC-06, OPG-15, OPP-14 and OPU-07 generated reproducible polymorphism. Amplified products with all the primers showed polymorphic and distinguishable banding pattern indicating the genetic diversity among all the isolates of *R. bataticola*. A total of 389 reproducible and scorable polymorphic bands ranging approximately as low as 150 bp to as high as 2500 bp was generated with four primers in twenty isolates.

Based on the results obtained, all the twenty isolates were grouped into two main clusters indicating there is genetic diversity in the isolates of *R. bataticola*. Cluster I contained twelve isolates, main cluster divided into two sub clusters, sub cluster I had seven (Rb-1, Rb-3, Rb-14, Rb-27, Rb-25, Rb-60 and Rb-41) and sub cluster II had five isolates (Rb-36, Rb-48, Rb-10, Rb-13 and Rb-34). Main cluster II has two sub clusters, sub cluster I consisted of two isolates (Rb-55 and Rb-43) and in sub cluster II six isolates (Rb-19, Rb-23, Rb-17, Rb-38, Rb-44 and Rb-56) were grouped. Genetic similarity coefficient was found based on RAPD analysis of 20 isolates of *R. bataticola*. Similarity coefficient ranged from 14 to 92 per cent. The maximum genetic similarity of 92 was between Rb-1 and Rb-3 whereas least genetic similarity (14 per cent) was observed between Rb-17 and Rb-27 (Fig. 4).

The results are also in confirmation with the studies conducted by Shekhar *et al.* (2006b) where they analyzed seven isolates of *R. bataticola*, incitant of maize rot through RAPD marker for genetic diversity. They observed that the most closely related isolates were Hyderabad and Delhi with an affinity percentage of 75.5 followed by Udaipur and Bangalore isolates with 62.9 per cent similarity. Similarly, Pancheswar *et al.* (2012) studied the genomic DNA of 21 isolates of *R. bataticola* of soybean from Jabalpur were subjected for diversity analysis with 8 randomly selected decamer primers which amplified 64 RAPD marker loci. Out of these 64 bands, 29 bands (45.3%) were polymorphic. Similarity coefficient values for 21 accessions of *R. bataticola* in RAPD marker was 0.69–0.98. Kumar and Gaur (2011) studied virulence and genetic diversity among 14 isolates of *M. phaseolina* using PCR-RAPD profiles. A total of 85 amplicon levels were obtained with 7 primers with an average of 12.1 bands per primer were available for analysis, of which 83 were polymorphic (97.74%). It was possible to discriminate all the isolates with any of the 7 primers employed. The UPGMA clustering of data indicated the isolates shared varied levels of genetic similarity within a range of 0.069 to 0.65 similarity coefficient index.

5.6 Effect of climate change variables (temperature, moisture and carbon dioxide) on dry root rot disease

5.6.1 Effect of temperature on pathogen and disease severity

There was very significant relation between the temperature and the development of the pathogen and diseases. Present studies were conducted to know the optimum

temperature requirements for the growth of three isolates of *R. bataticola*. Results indicated that three isolates (Rb 13, Rb 17 and Rb 34) were grown at eight different temperatures viz., 15°C, 20°C, 25°C, 30°C, 35°C, 40°C, 45°C and 50°C. Among the three different isolates tested, the maximum colony growth (18.15, 61.00, 75.50, 86.50, 90.00 and 18.00 mm) and disease severity (1.7, 3.1, 7.4, 8.2, 9.0 and 3.0) was observed in Rb 34 and least colony growth (10.20, 34.20, 37.00, 47.70, 80.00 and 15.00 mm) and disease severity (1.0, 2.9, 6.7, 7.2, 8.5 and 2.5) was observed in Rb 17 at 15°C, 20°C, 25°C, 30°C, 35°C and 40°C temperature levels (Fig. 5). No growth was observed at 45°C and 50°C. This was due to variability present in the isolates collected from the different regions. Similar results were observed by Srinivas (2016) that significant difference in the radial growth among the isolates of *R. bataticola* ranging from 17.7 mm to 80.0 mm at 72h after incubation. Isolate Rb 14, Rb 17, Rb 22, Rb 26, Rb 49 and Rb 54 showed significantly highest colony growth (80 mm). The least colony diameter was observed in the isolate Rb 20 (17.7 mm). Further, Veerendra Kumar, (2004) reported that the maximum growth of the fungus was 90 mm of radial growth at 30 °C and 35 °C followed by 40 °C (87.3 mm) in isolate RB5.

Among the eight different temperatures tested, the maximum colony growth (90.0, 80.0 and 90.0 mm) and disease severity (9.0 8.5 and 9.0) was observed at 35 °C followed by 30 °C with colony growth (85.0, 47.70 and 86.50 mm) and disease severity (8.0, 7.2 and 8.2) and these temperatures were considered as optimum for the growth of pathogen (Fig. 6). Optimum temperature (35 °C and 30 °C) influenced on the rate of growth, multiplication and metabolism process of the pathogen and high soil temperature may lead to the selection of hyphae and microsclerotia that are adapted to better growth at higher temperatures. Similarly, the growth and sclerotial production of *R. bataticola* causing root rot in mulberry at different temperatures *in vitro* were studied by Chowdary and Govindaiah (2007) who recorded maximum growth of 90 mm at 30°C and 35°C and minimum of 40 mm growth at 20°C. Sclerotial production was observed in all the temperatures except at 20 °C. Savary *et al.* (2011) observed the effect of temperature on increasing incidence of dry root rot at various locations over the years which suggests a strong influence of rising temperature (above 30 °C) on *R. bataticola*. Further, influence of seven temperatures regimes (15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C and 45 °C) were tested on growth of *R. bataticola* isolates representing Madhya Pradesh, Karnataka,

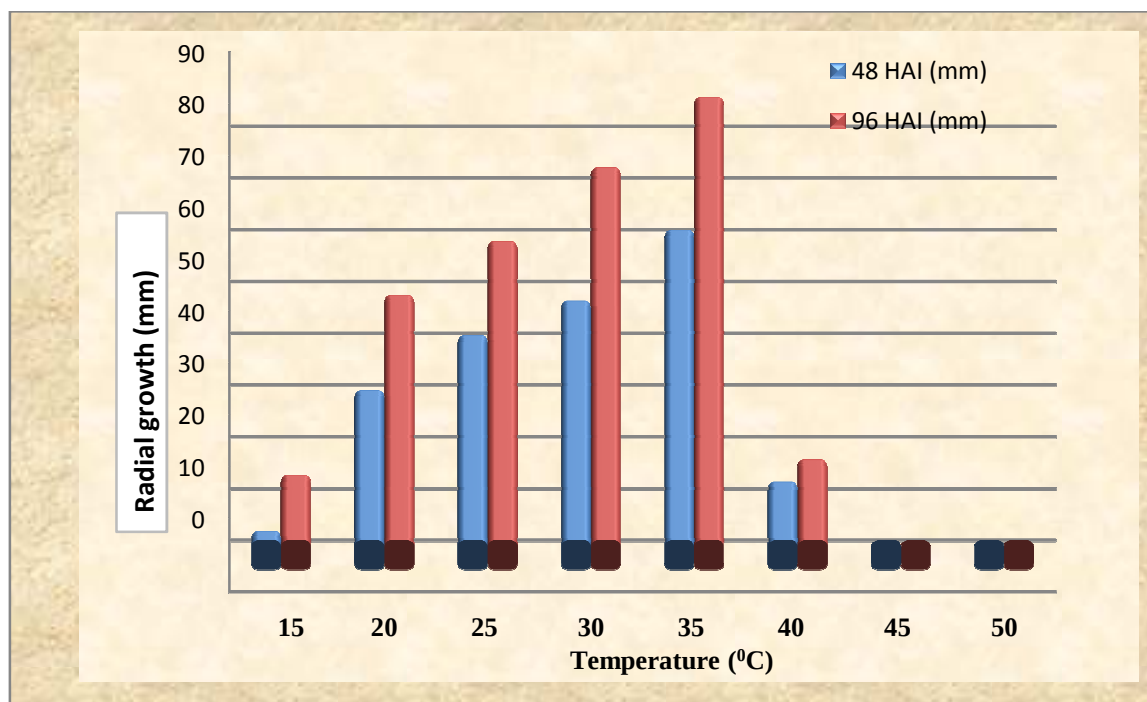


Fig. 5: Effect of temperature levels on radial growth of *R. bataticola* at 48 and 96 HAI

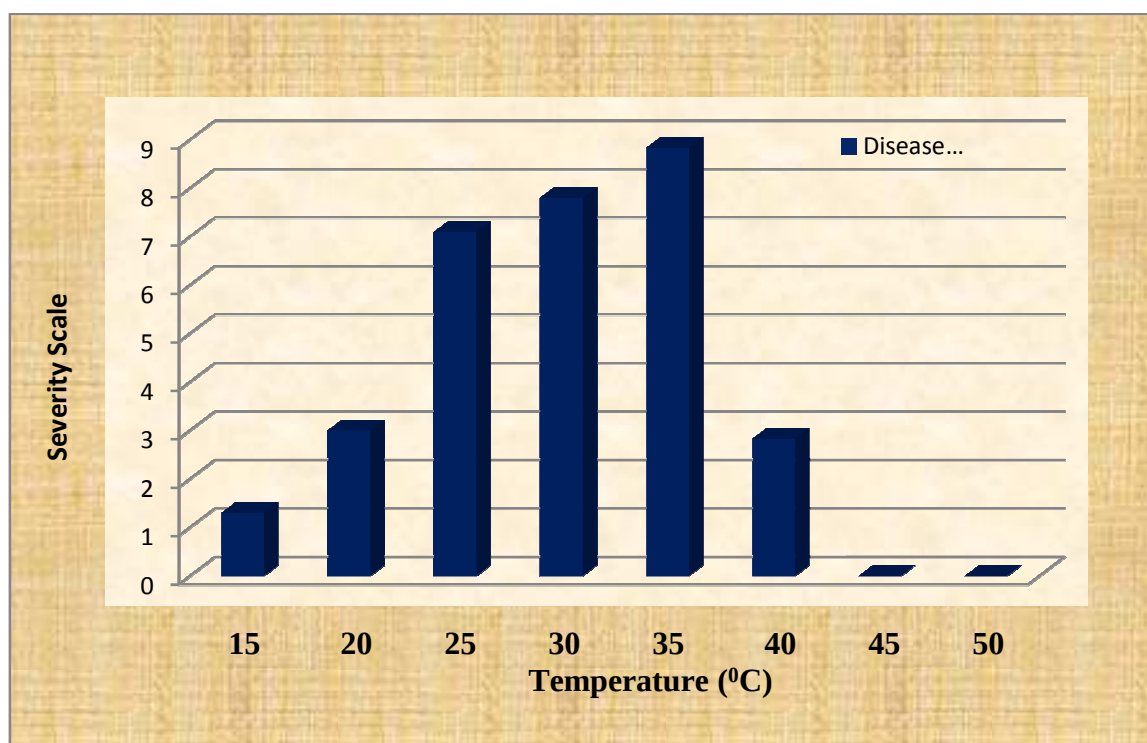


Fig. 6: Effect of temperature levels on dry root rot disease in chickpea

Maharashtra, Andhra Pradesh and Telangana states of India. The maximum mycelial growth was observed at 35 °C followed by 30 °C in all the isolates (Srinivas *et al.*, 2017).

The results also indicated least growth of pathogen isolates was at 15 °C followed by 20 °C and 40 °C. However, there was no growth at 45 °C and 50 °C because of very high temperature. The results were supported by Khan *et al.* (2012) as they also observed pathogen growth over a wide range of temperature from 10 °C to 40 °C, but the optimum temperature for its growth was found to be 30 °C. The next best temperature for its growth was recorded at 35 °C. Statistically the growth of the pathogen gradually decreased both at below 30 °C and above 35°C. Sharma and Pande (2013) reported that the disease incidence of dry root rot was significantly affected by high temperature. Out of five temperature levels viz., 15 °C, 20 °C, 25 °C, 30 °C and 35 °C tested, chickpea predisposed to dry root rot early and severity was more at 35 °C.

5.6.2. Effect of soil moisture

5.6.2.1 Effect of soil moisture on growth parameters

Among the seven soil moisture regimes tested against the dry root rot and its effect on plant root length, shoot length and total biomass. The maximum mean root length (4.97 cm and 5.29 cm), shoot length (29.56 cm and 29.13 cm) and total biomass (3.82 g and 3.81 g) was recorded in both black soil and red soil, respectively at 100 per cent soil moisture followed by 90 per cent with root length (4.45 cm and 4.80 cm), shoot length (29.48 cm and 28.85 cm) and total biomass (3.45 g and 3.35 g) and at 80 per cent with root length (4.16 cm and 4.60 cm), shoot length (25.21 cm and 27.14 cm) and total biomass (1.32 g and 1.39 g). This may due to lesser growth of pathogen and lesser disease incidence in high soil moisture. Moreover, soil moisture plays a vital role in development and growth of plants. At 40%, 50% and 60% soil moisture regimes, least root length, shoot length and total biomass was observed in plants because at lower moisture condition, higher disease severity and sudden mortality of the plants was observed (Fig. 7 and 8). Mayek *et al.* (2002) reported that the stress prone and infected plants had poor growth compared to healthy and irrigated plants. Drought stress showed higher negative effects coupled with *R. bataticola* which attack in vegetative growth of the plant which decreased leaf area and dry weight of all vegetative structures significantly. Srinivas, (2016) studied different soil moisture levels on growth of chickpea and results indicated that 100% and 90% soil moisture levels recorded the highest root length, shoot length and

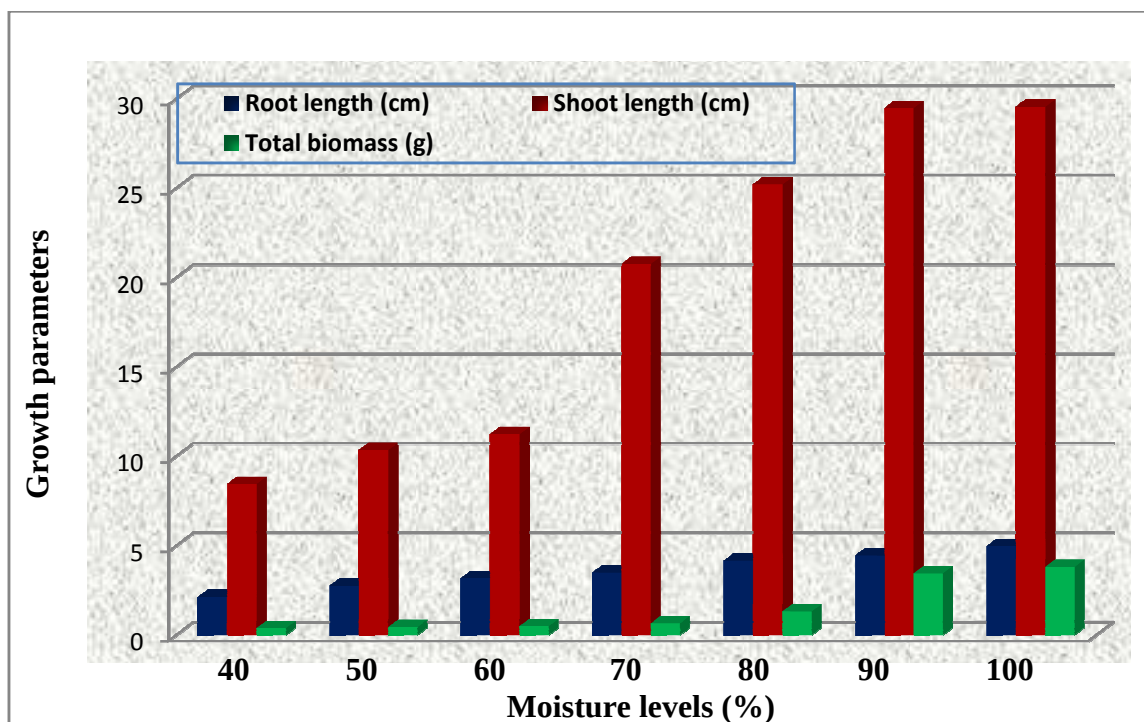


Fig. 7: Effect of moisture levels on growth parameters of chickpea in black soil inoculated with *R. bataticola*

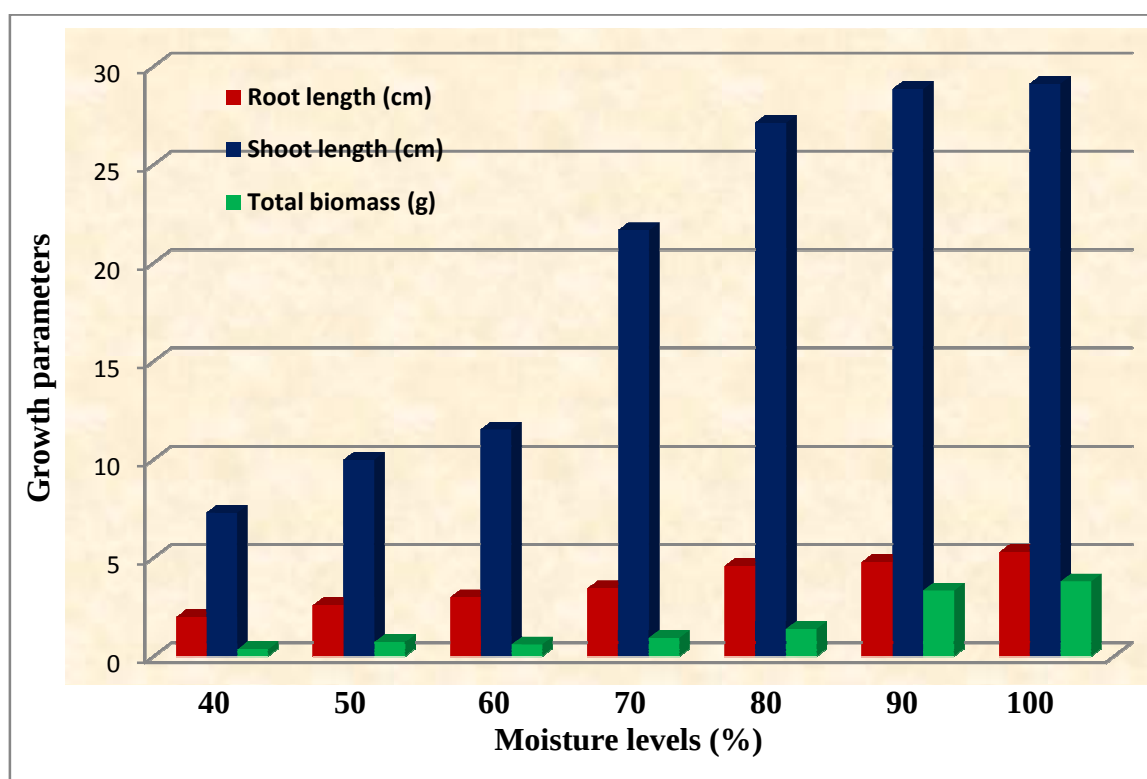


Fig. 8: Effect of moisture levels on growth parameters of chickpea in red soil inoculated with *R. bataticola*

total dry weight of chickpea plants in both inoculated and uninoculated soils, while at 40%, 50% and 60% soil moisture levels recorded least growth parameters.

5.6.2.2 Effect of soil moisture on disease incidence

Among seven moisture levels tested, there was early development of disease symptoms at 40%, 50% and 60% soil moisture. The plants became complete drying and lacking of the lateral roots and root system became completely black within twenty days of sowing. The plants recorded the maximum disease severity scale of 9.0 in case of both the soils because of moisture stress condition favoured the growth of pathogen and disease. Least disease severity was recorded at 100% (3.30 and 3.56) soil moisture levels in black soil and red soil, respectively, the disease decreased with respect to the increase in soil moisture because high moisture condition was not suitable for pathogen growth (Fig. 9). Similar observations were recorded by Ratnoo *et al.* (1997), they observed that disease development was low in flooded soil compared to drier soils (40 to 60% moisture). Srinivas (2016) also observed difference in the dry root rot incidence with change in soil moisture content. The plants grown in 50%, 60% soil moisture recorded highest disease severity scale compared to the 80%, 90% and 100% soil moisture. Further, Sharma and Pande (2013) reported that plants exposed to 40% and 60% soil moisture, dry root rot severity was maximum, showed higher mortality as compared to 80% and 100%. Disease progressed slowly at 80% and 100% soil moisture had least disease severity. Patel and Anahosur (2001) conducted an experiment with chickpea cultivars ICC 4951 (susceptible) and Bheema (resistant) in earthen pots filled up with artificially infected soil with *R. bataticola* at four levels of soil moisture viz., 25, 50, 75 and 100%. They observed that the infection was maximum at 25% and minimum at 100% soil moisture. ICC 4951 developed higher disease than Bheema at 25% soil moisture. Both cultivars showed maximum disease incidence at the lowest soil moisture level compared to high moisture.

5.6.3 Effect of carbon dioxide

5.6.3.1 Effect of carbon dioxide on growth parameters

Carbon dioxide is one of the factor which influenced plant disease severity and spread of disease, increased atmospheric CO₂ combined with other climatic factor like high temperature causes the increased severity of disease and reduced yield. The increase

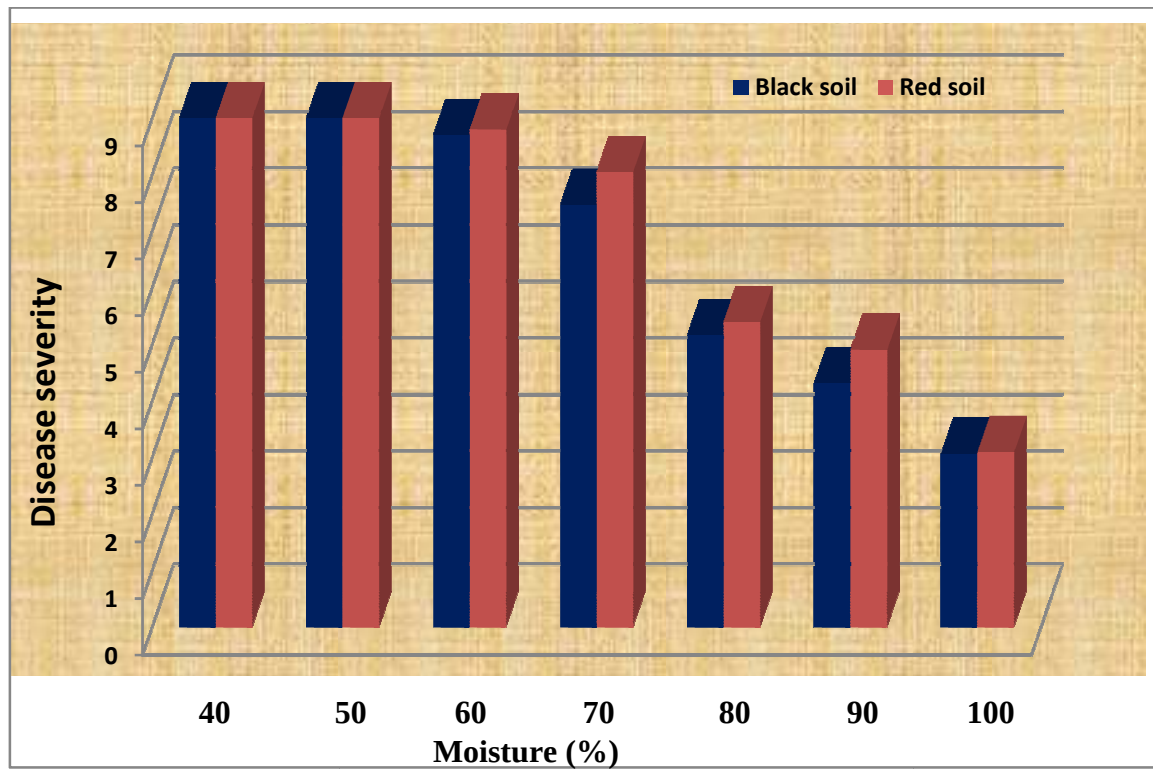


Fig. 9: Effect of soil moisture levels on disease severity in chickpea in black soil and red soil inoculated with *R. bataticola*

in CO₂ and the concentration of other greenhouse gases has already resulted in an increase in the global average temperature of 0.6–0.7 °C over the last century (Mann *et al.* 1998; Walther *et al.* 2002 and Benvenuti, 2009). Effect of carbon dioxide on growth parameters viz., shoot length, root length, no. of branches and total biomass were observed under the open top chambers during present investigation.

The maximum shoot length, root length, number of branches and total biomass was observed in open plot (T₅) followed by in ambient CO₂ @ 390 ± 25 ppm (T₄) and ambient CO₂ @ 390 ± 25 ppm with 2 °C rise in temperature (T₃) and least growth was observed in elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂). This trend of results may be due to the low level of CO₂ and low temperature in (T₅), (T₄) and in (T₃) which helped for plant growth and reduced the disease severity. In T₂ the elevated CO₂ combined with high temperature which increased the disease severity extremely. Hence, the reduction in the shoot length, root length, number of branches and total biomass. These results are supported with Cheng *et al.* (2009) who conducted a pot experiment in four controlled environmental chambers with two levels of CO₂ (ambient, 380 ppm; and elevated, 680 ppm) and two levels of night temperature (22 and 32°C). They observed increased dry root rot disease severity and the reduced growth of plants at elevated carbon dioxide of 680 ppm with increased night temperature (Fig. 10). Horie *et al.* (2000) observed that the whole plant and stem dry weight were decreased by both elevated CO₂ and high night temperature. Further, Jagadish *et al.* (2007) found reduction in the weight of grains of pigeonpea under increased temperature with high levels of CO₂. Baker (2004) also studied the effect of elevated CO₂ (700 ppm) under different temperature regimes (24 °C, 28 °C, 32 °C, 36 °C and 40 °C), found that no increase in grains weight of chickpea under enriched CO₂ combined with high temperature.

5.6.3.2 Effect of carbon dioxide on disease severity

Effect of carbon dioxide on dry root rot of disease was studied in open top chambers (OTC). Among the five sets of treatments, elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂) showed higher disease severity scale of 9.00 with early infection showing drying of leaves and wilting of entire plant and roots were completely rotten. This is due to the controlled condition of elevated level of CO₂ and increased temperature by 2 °C, as earlier stated that *R. bataticola* grows and multiplies well in

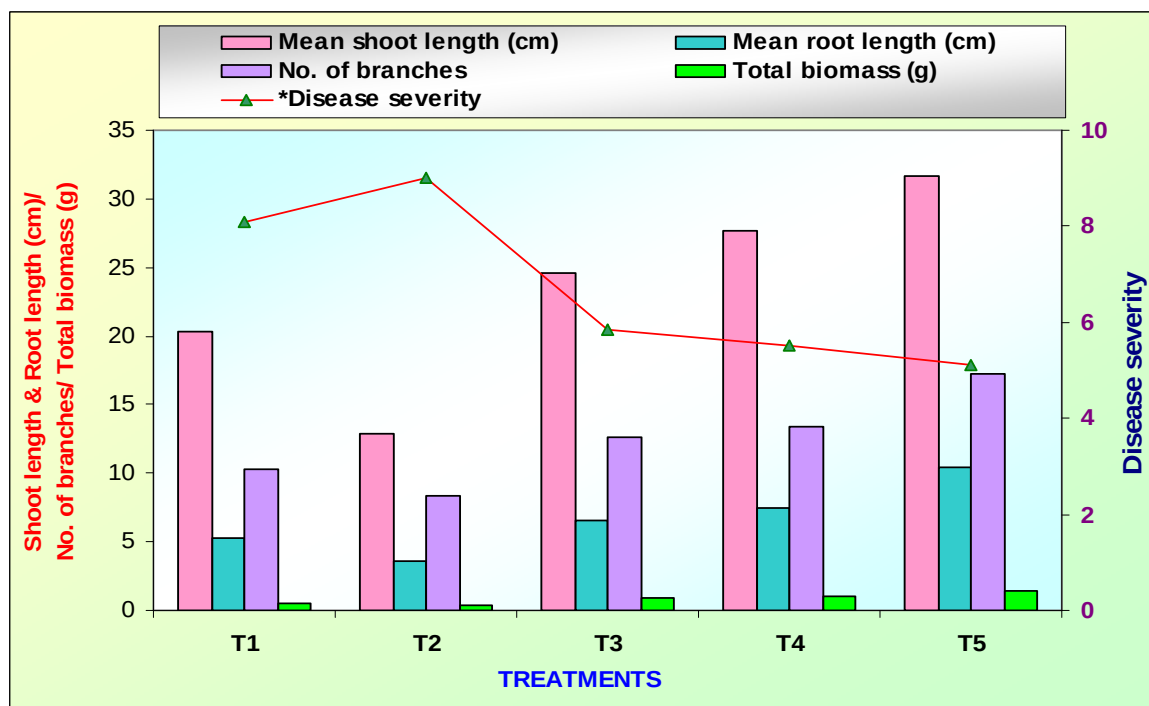


Fig. 10. Effect of elevated carbon dioxide and temperature on growth parameters and disease severity of dry root rot of chickpea

TREATMENT DETAILS

T₁ - Elevated CO₂ @ 550 ± 25 ppm alone

T₂ - Elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature

T₃ - Ambient CO₂ @ 390 ± 25 ppm with 2 °C rise in temperature

T₄ - Ambient CO₂ @ 390 ± 25 ppm

T₅ - Open plot

increased temperature compare to the normal temperature (Fig. 10). Similarly, Jagadish *et al.* (2007) studied effect of carbon dioxide on root rot of pigeonpea grown in elevated CO₂ condition of 550 ± 25 ppm with rised temperature which increased disease incidence in pigeonpea. Further, Chakraborty and Datta (2003) studied the effects of increasing atmospheric CO₂ (increasing CO₂ from 365 ppm to 550 ppm) on plant–pathogen interactions, found that disease incidence of root rot was increased and reduced the yields of pulses in controlled condition.

5.7 Identification of dry root rot resistant sources of chickpea

Effective method for managing soil borne disease of crop plants is by using resistant varieties which is most economical, inexpensive and eco-friendly for poor farmers in comparison to chemicals. The cost of cultivation with respect to disease resistant varieties is always one of the best ways and will go a long way not only in reducing loss due to disease but also in avoiding fungicidal toxicity and soil pollution.

5.7.1 Preliminary screening of chickpea germplasm lines against dry root rot of chickpea under field condition

Rhizoctonia bataticola is having wide host range, prolonged saprophytic survival ability and the sclerotia can survive upto 10 months even in the absence of the host can make any resistant cultivar to become susceptible with in a less time. Therefore, development of resistant cultivars for dry root rot disease is becoming challenge for both breeders and pathologists. Out of one hundred germplasm lines screened, 14 germplasm lines (IC83736, IC 83526, IC 83538, IC83798, IC95135, EC498827, IC83358, IC83357, IC83487, IC83500, IC83501, IC83506, IC83514, IC 555217) showed resistance reaction to dry root rot with disease reaction of 0-10 per cent. Five germplasm lines showed moderately resistance reaction with 10.1 to 20 per cent. Three germplasm lines were moderately susceptible (20.1-30%). Twelve germplasm lines showed susceptible reaction of 30.1 to 50 per cent and rest of the sixty six germplasm lines showed highly susceptible reaction of more than 50 per cent.

The present findings are well supported by Saifulla *et al.* (2011a) who screened chickpea entries for dry root rot pathogen under field condition and found that 20 entries viz., GNG 1861, Phule G 07112, BGD1056, RSG 931, Phule G 07101, Vijay, RSG 888,

GJG 0825, PG064, GNG 1947, JG 1307, GBC 6, GNG 2002, GNG 1936, CSJ 313, JG 18, HIR 60, JG 11, ICCV 07107 and ICCV 08323 were resistant and 36 entries were moderately resistant to dry root rot out of 196 genotypes. Further, Nagamma and Saifulla (2012) screened sixty four Kabuli genotypes against dry root rot, out of which six genotypes viz., Phule G 04305, IPCK 07-62, RVSSG 12, HK 08-212, Phule G 09305 and AKG 2002-1K showed resistant reaction, nine genotypes viz., HK 94-134, GNG 1888, HK 06-152, GNG 2112, HK 06-171, Phule G 09316, CSJK 74, JGK 13 and CSJK 70 showed moderately resistant reaction.

Similarly, Gupta *et al.* (2012) screened one hundred seventy chickpea accessions against dry root rot disease, out of which 68 genotypes exhibited resistant reaction (<10 per cent mortality), of which 26 were the promising lines viz., (JG1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas) from JNKVV, Jabalpur. These lines were further evaluated for their performance in sick field for three consecutive years which revealed that six lines viz., JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas as resistant exhibiting <10 per cent mortality. Further, Gangwar *et al.* (2002), Om Gupta and Anita Babbar (2006) and Al Shareef *et al.* (2009) also reported resistant genotypes, moderate resistant genotypes, moderate susceptible genotypes, susceptible genotypes and highly susceptible genotypes for dry root rot.

4.7.2 Advanced screening of promising entries of chickpea against dry root rot by phenotyping technique

Nineteen promising (Resistant and moderately resistant) germplasm lines selected in preliminary screening in field condition were subjected to screening under phenotyping technique for confirmation of their disease reaction recorded in the field condition. The results indicated that among nineteen germplasm lines screened against dry root rot, only four genotypes namely IC83798, IC95135, IC83506 and IC83500 remained as resistant and showed stable performance in both field and controlled condition. Further, five germplasm lines viz., IC83736, IC83526, IC83358, IC83501 and IC83514 were moderately resistant and rest of the lines showed either moderately susceptible and susceptible reaction with more than 30-50 per cent disease incidence. This may be due to

the heavy inoculum load and favorable conditions like high temperature and moisture stress for growth and multiplication of the rhizoctonia pathogen under phenotypic technique conditions. Hence, the chance of escaping from disease was restricted. Pande *et al.* (2004) tested 47 lines of chickpea against the dry root rot in advanced screening and reported that only one germplasm accession (ICC 14395), a cultivar (ICCV 2) and an advanced breeding line (ICCX830203-BH-BH-11H) were resistant to dry root rot and the remaining 22 lines moderately resistant, 19 susceptible and 3 highly susceptible. Jayalakshmi *et al.* (2008) also tested 12 genotypes against dry root rot disease and found only four genotypes as resistant.

5.8 Integrated management of dry root rot of chickpea

Management of soil borne diseases only with host plant resistance and cultural methods is difficult in chickpea. Thus, there is an urgent need to formulate effective disease management module by using chemicals and bioagents as a first barrier for plant disease development by treating the seeds with chemicals while sowing. Therefore, experiments were conducted to find out the best management strategy using combination of chemicals and biocontrol agents for effective management of dry root rot complex of chickpea caused by *R. bataticola*.

Field experiments conducted during *rabi*, 2015-16 and *rabi*, 2016-17 indicated that the treatment containing seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants was significantly superior in recording least disease incidences of dry root rot from rest of the treatment combinations. The next best treatments was seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants (Fig. 11). Khan *et al.* (2012) tested eight fungicides against dry root rot fungus, among them mancozeb, carbendazim, copper-oxy-chloride and benomyl completely inhibited the growth of the fungus compared to control. Further, in fungicidal trails on management of dry root rot of chickpea caused by *R. bataticola*, carbendazim (0.2 per cent) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 per cent highest grain yield of 192 q/ha respectively (Vijay Mohan *et al.*, 2006).

Paramasivan *et al.* (2007) tested the fungal and bacterial antagonists against dry root rot pathogen in which *T. virens* resulted in maximum growth inhibition followed by *T. harzianum* and least growth inhibition by *P. fluorescens*. Further, Out of the ten isolates tested against *R. bataticola*, *Trichoderma* isolate-7 collected from chickpea rhizosphere showed highest inhibition percentage (83.33) (Veena *et al.*, 2013). Singh *et al.* (2003) also reported the efficacy of *T. harzianum* in controlling *R. bataticola* causing dry root rot in chickpea under *in vitro* and in field experiment conducted in Kanpur, Uttar Pradesh.

Average yield data of dry chickpea obtained during *rabi*, 2015-16 and 2016-17 indicated that the treatment combination of seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants recorded highest seed yield and significantly different from rest of the treatment combinations evaluated. Next best treatment is seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants recorded little higher yields and test weight (Fig. 11).

In integrated disease management trial, economics was worked out by taking into consideration of the total cost of cultivation, cost of treatment and gross returns. The highest benefit (3.40) was obtained in seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants. The other effective treatment was seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants recorded BC ratio of 3.02.

Nagamani *et al.* (2011) conducted experiment on dry root rot of chickpea and reported that seed treatment with carbendazim @ 2g/kg of seed+ seed treatment with *T. viride* @ 4 g/kg of seed + soil application of FYM fortified with *T. viride* recorded least disease incidence with highest yield and BC ratio. Further, Integrated management study of root rot disease of chickpea under field condition showed that soil application of *T. harzianum* @ 5 kg in 500 kg neem cake/ha in furrow 5 days prior to sowing resulted in

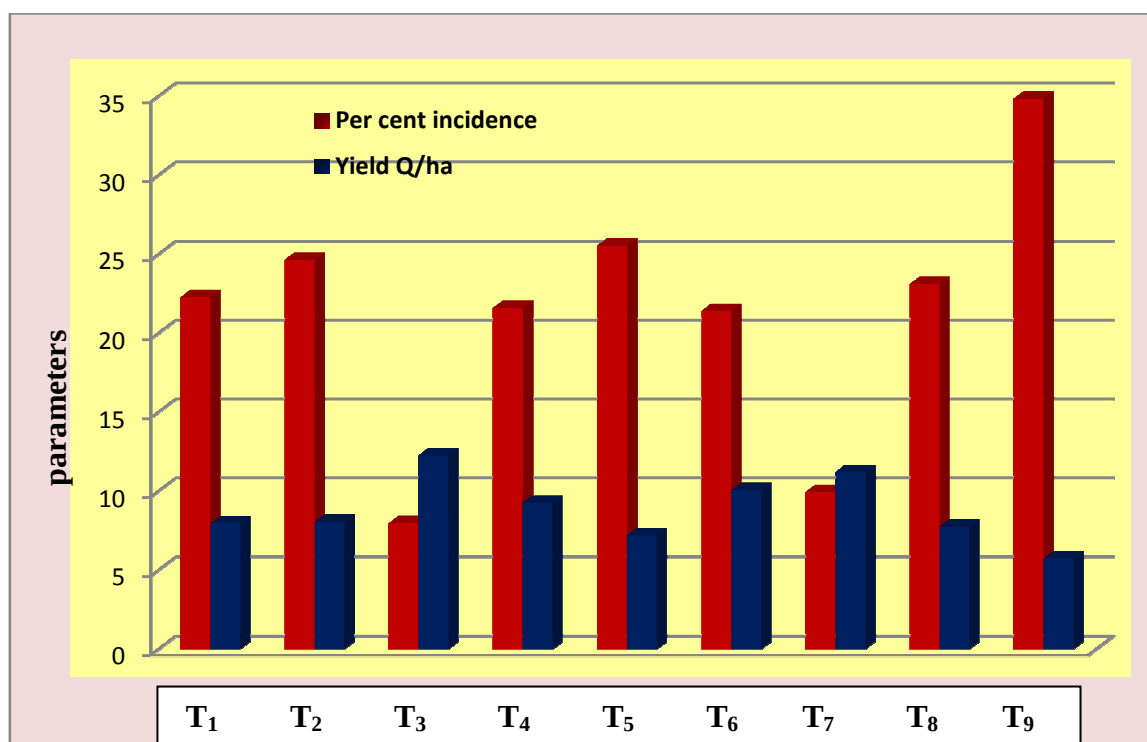


Fig. 11: Integrated management of dry root rot of chickpea

LEGEND

- T₁ Seed treatment with *T. harzianum* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* @ 2.5 kg/250 kg of FYM during sowing
- T₂ Seed treatment with *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* @ 2.5 kg/250 kg of FYM during sowing
- T₃ Seed treatment with mancozeb 50% + carbendazim 25% WS (Sprint) @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants
- T₄ Seed treatment with *T. harzianum* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants
- T₅ Seed treatment with *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants
- T₆ Seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil application of enriched *T. harzianum* + *P. fluorescens* @ 2.5 kg/250 kg of FYM during sowing
- T₇ Seed treatment with *T. harzianum* @ 5 g/kg seed + *P. fluorescens* @ 5 g/kg seed followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/lit water to infected and surrounding plants
- T₈ Seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg
- T₉ Untreated control

higher seed germination lowest disease incidence and gave highest yield (Dhingani and Solanky, 2016).

Sprint is a combination of contact and systemic fungicide, having both curative and preventive activity. Mixture partner mancozeb has a contact action and preventive fungicide with multisite mode of action on fungal cell wall. It remains on the seed surface in treated seeds. It is fungitoxic when exposed to air, converted into isothiocynate, which inactivates the sulphahydral group of enzymes in fungi, causing disturbance in fungal enzyme functioning. Other mixture partner carbendazim is systemic in action acts as preventive and curative. It acts by disrupting the spindle formation during cell division in fungi.

Trichoderma harzianum acts on degradation of chitin in the cell wall of *R. bataticola* by chitinases of *Trichoderma* spp. facilitate their penetration of *R. bataticola* mycelium (Manjula *et al.*, 2004). Since, sclerotia are the perennating bodies, which carry *R. bataticola* from one season to the other; inhibition of sclerotial production can reduce the overall available inoculum of *R. baticola* in field. *P. fluorescens* acts as antifungal agent by occupying the same sites in the root as the fungus and competing for nutrients. These findings are similar to that of other researchers (Shekhar and Kumar, 2010; Nagamma *et al.*, 2012; Kumari *et al.*, 2012 and Manjunatha *et al.*, 2013).

Summary and Conclusions

VI. SUMMARY AND CONCLUSIONS

Rhizoctonia bataticola has long been recognized as a destructive pathogen on a wide variety of crops throughout the world. Dry root rot of chickpea caused by *R. bataticola* once considered a minor disease has become a major problem in successful production of chickpea in South India. The present investigation included survey on disease incidence of dry root rot, collection of pathogen isolates, diversity among isolates with respect to cultural, morphological and molecular variability, influence of environmental factors on growth and disease development, screening of chickpea entries to identify resistance sources and management of dry root rot under field condition using seed treatment and soil application of chemicals and bio-agents. Results of the investigations are summarized here under.

The dry root rot showed typical symptoms of yellowing and partial or complete wilting of the stem or one or more branches. In advanced stage of the disease, affected plants became straw coloured, later lower leaves and stems showed brown discoloration. The tap root appeared black, rotten and devoid of most of the lateral and fine roots. The dead root became quite brittle and shows shredding of bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil when plants are uprooted.

Rhizoctonia bataticola produced grey cottony mycelial growth on potato dextrose agar medium and the colony morphology was compact or fluffy or appressed. The characteristic features of *R. bataticola* are right angle branching of the mycelium and constriction of the branch near the point of origin. The young hyphae are thin, hyaline, septate and dichotomously branched and the sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes.

Standard tissue isolation was followed to obtain *R. bataticola* culture from diseased sample of infected root with typical symptoms of rotting and shredding of the bark on root portion, collected from chickpea field. The fungus isolated from infected stem was confirmed as *R. bataticola* based on their cultural, morphological and formation of microsclerotial bodies.

Pathogenecity test was carried out on chickpea seedlings (susceptible variety Annigeri-1) and Koch's postulates were successfully proved.

In survey, Bhusnur village of Aland taluk recorded highest average dry root rot incidence of 45.55% followed by Huvinahadgi village of Devadurga taluk (45.23%) and Bableshtar village of Sindagi taluk (45.00%). The least disease incidence was noticed in Aldahal and Munderga villages of Yadagiri taluk (1.00%) during *rabi* 2015-16. During *rabi* 2016-17, the incidence of dry root rot was maximum in Huvinahadgi village of Devadurga taluk (43.20%) followed by Sannur village of Kalaburgi taluk (43.00%) and Jevargi village of Jevargi taluk (41.00%). The least was noticed in Mailapur and Saidapur village of Yadagiri taluk (1.0%).

Over two seasons, the maximum mean incidence was recorded in Devadurga taluk (35.70%) followed by Kalaburgi taluk (33.81%) and Jevargi taluk (33.40%). The least incidence was in Yadagiri taluk (3.25%). Among the districts mean incidence was maximum in Kalaburgi (32.93%) followed by Raichur (29.87%) and Vijayapur district (24.84%). The least incidence was recorded in Yadgiri district (3.35%).

The overall mean incidence of dry root rot (38.72%) was more during *rabi*, 2015-16 compared to *rabi*, 2016-17 (32.73%) and the North Karnataka recorded overall mean incidence of 35.72%.

Isolation of as many as sixty isolates of *R. bataticola* infecting chickpea was carried out by tissue isolation and the identity of all the isolates was confirmed based on the cultural, morphological and formation of microsclerotial bodies.

Cultural characters studied on potato dextrose agar showed variation among *R. bataticola* isolates. On the basis of **cultural characteristics viz.**, radial growth of mycelium, colony morphology, sclerotial production, colony pigmentation of sixty isolates, they were categorized into four to five groups. Each group included different number of isolates from different locations.

Based on radial mycelial growth, 55 isolates which were fast growing with an average growth rate (75-90 mm), two intermediate growth with an average growth rate (60-75 mm) and three isolates slow growing with an average growth rate of 45-60 mm. With respect to colony morphology, 17 isolates showed appressed growth, 13 isolates velvety growth and 30 isolates were fluffy. In case of presence and absence of aerial mycelia, 28 isolates showed presence of aerial mycelial growth and in 32 isolates aerial mycelia was absent. Sixty isolates exhibited variability with respect pigmentation of

mycelia, black with grey aerial mycelia (15 isolates), black, (15 isolates) grey (25 isolates) and white pista colour (5 isolates).

Morphological characters studied on potato dextrose agar showed variation among the *R. bataticola* isolates. Morphological variability such as sclerotial production, number of days to initiate sclerotia, number of sclerotia produced, shape of the sclerotia and width of the hyphae of sixty isolates was categorized into four to five groups. Each group included different number of isolates from different locations.

Based on production of the microsclerotia, 51 isolates were found to produce sclerotia, but in 9 isolates microsclerotia were absent. With respect to shape of sclerotia, 20 isolates were round shaped while 13 isolates had ovoid shaped sclerotia and 18 isolates produced irregular shape of sclerotia. Among sixty isolates, twenty one isolates produced 0-25 sclerotial bodies per microscopic field (10x), twenty six isolates produced 26-50 sclerotial bodies, four isolates produced 51-100 sclerotial bodies.

Identification of *R. bataticola* isolates by PCR indicated that the primers, ITS-1 and ITS-4 produced amplified product size of 550- 600 bp in all the twenty isolates indicating that isolates belong to *Rhizoctonia bataticola*. Further, eight isolates *R. bataticola* were selected from eight districts (each isolate from one district) and nucleotide sequencing was done for ITS region of 18S rRNA. The BLAST data results revealed that the *R. bataticola* species matched with the reference strains of NCBI results and identified as *Rhizoctonia bataticola*. Out of eight isolates, all the isolates were distinct from one another in the cluster but identical to each other in homology and divergence with different nodes.

In studies on molecular diversity, four random primers viz., OPC-06, OPG-15, OPP-14 and OPU-07 generated reproducible polymorphism. Amplified products with all the primers have showed polymorphic and distinguishable banding pattern indicating the genetic diversity among all the isolates of *R. bataticola*. A total of 389 reproducible and scorable polymorphic bands ranging approximately as low as 150 bp to as high as 1000 bp was generated with four primers among the twenty isolates studied.

The twenty isolates were grouped into two main clusters indicating there is genetic diversity in the isolates of *R. bataticola*. Cluster I contained twelve isolates, main cluster divided into two sub clusters, sub cluster I had seven isolates (Rb-1, Rb-3, Rb-14,

Rb-27, Rb-25, Rb-60 and Rb-41) and sub cluster II had five isolates (Rb-36, Rb-48, Rb-10, Rb-13 and Rb-34). Main cluster II has two sub clusters, sub cluster I consisted of two isolates (Rb-55 and Rb-43) and in sub cluster II six isolates (Rb-19, Rb-23, Rb-17, Rb-38, Rb-44 and Rb-56) were grouped.

Similarity coefficient ranged from 14 to 92 per cent. The maximum genetic similarity of 92 was between Rb-1 and Rb-3 whereas, least genetic similarity (14%) was observed between Rb-17 and Rb-27.

The maximum colony growth and disease severity was observed at 30 °C and 35 °C which is considered as optimum temperature for the growth of pathogen and disease incidence. The maximum colony growth and disease severity was observed in Rb 34 and least growth was observed in Rb 17 among three isolates.

Different soil moisture levels influenced the root length, shoot length, total biomass and disease severity of dry root rot. Maximum dry root rot and lesser root length, shoot length and total biomass were recorded at 40%, 50% and 60% soil moisture regimes because of stress condition as a result sudden mortality of the plants was observed. There was no effect of soil type on disease severity as well as growth parameters at different soil moisture regimes.

With respect to effect of carbon dioxide and temperature on disease severity and shoot length, root length, number of branches and total biomass, elevated CO₂ @ 550 ± 25 ppm with 2 °C rise in temperature (T₂) showed higher disease severity scale as well as reduced growth parameters. The least disease severity and higher growth parameters were observed in open plot (T₅).

Preliminary screening 100 germplasm lines over two seasons resulted in the identification of fourteen resistant and five are moderately resistant germplasm lines. Out of nineteen genotypes which were promising in the field were subjected to advanced screening under phenotyping technique for confirmation of their disease reaction recorded in the field condition and it has yielded only four genotypes namely IC83798, IC95135, IC83506 and IC83500 remained as resistant and five genotypes namely IC83736, IC 83526, IC83358, IC83501 and IC83514 were moderately resistant by showing stable performance under both field and controlled conditions.

In the integrated disease management, the treatment combination *i.e.*, seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil

drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants was highly effective by recording least disease incidence of dry root rot (7.79%) with highest seed yield (12.26 q/ha), test weight (38.90 g) and highest benefit cost ratio (3.40).

Conclusions:

- The dry root rot typical symptoms are drooping and chlorosis of petioles and leaflets, the tap root turns black with signs of rotting and is devoid of most of the lateral and finer roots. The dead roots are quite brittle and show shredding of the bark. The tip of the root is easily broken leaving the lower portion of the tap root in the soil when plants are uprooted.
- *Rhizoctonia bataticola* produced grey cottony mycelial growth, right angle branching of the mycelium and constriction of the branch near the point of origin. The young hyphae are thin, hyaline, aseptate and dichotomously branched and the sclerotia formed are black, smooth, varying from spherical through oblong to irregular shapes.
- Standard tissue isolation was followed to obtain *R. bataticola* culture from diseased sample of infected root. The fungus isolated from infected root was confirmed as *R. bataticola* based on their cultural, morphological and formation of microsclerotial bodies.
- Pathogenicity test was positive for the fungus indicating the proof of Koch's postulates.
- In survey, Bhusnur village of Aland taluk recorded highest average dry root rot incidence of 45.55% followed by Huvinahadgi village of Devadurga taluk (45.23%) and Bableshtar village of Sindagi taluk (45.00%). The least disease incidence was noticed in Aldahal village of Yadagiri taluk (0.50%).
- Maximum mean incidence was in Kalaburgi district (32.93%) followed by Raichur district (29.87%) and Vijayapur district (24.84%). The least incidence was recorded in Yadgiri district (3.35%).
- The overall mean incidence of dry root rot (38.72%) was more during *rabi*, 2015-16 compared to *rabi*, 2016-17 (32.73%) and the North Karnataka state recorded overall mean incidence of 35.72%.

- Sixty isolates of *R. bataticola* infecting chickpea were isolated by tissue isolation and the identity of all the isolates was confirmed based on the cultural, morphological and formation of microsclerotial bodies.
- Cultural characters showed variation among *R. bataticola* isolates. On the basis of radial growth of mycelium, colony morphology, sclerotial production, color of the mycelia of sixty isolates was categorized culturally into four to five groups. Each group included different number of isolates from different locations.
- Morphological characters showed variation among the *R. bataticola* isolates, such as sclerotial production, number of days to initiation of sclerotia, number of sclerotia produced, shape of the sclerotia and width of the hyphae.
- Identification of *R. bataticola* isolates by ITS– PCR indicated that both the primers, ITS-1 and ITS-4 produced amplified product size of 550- 600 bp of ITS region of rRNA clusters in all the twenty isolates indicating that all the isolates belonged to genus *Rhizoctonia bataticola*.
- The maximum colony growth of *R. bataticola* and disease severity was observed at 30-35°C which is considered as optimum temperature.
- Maximum dry root rot and lesser root length, shoot length and total biomass were recorded at 40%, 50% and 60% soil moisture regimes with no effect of both soil types on disease severity as well as growth parameters.
- Elevated CO₂ @ 550 ± 25 ppm with 2°C rise in temperature (T₂) showed higher disease severity scale as well as reduced growth parameters. The least disease severity and higher growth parameters were observed in open plot (T₅).
- Preliminary screening of 100 germplasm lines resulted in the identification of fourteen resistant and five moderately resistant germplasm lines.
- Advanced screening of 19 genotypes under phenotyping technique yielded only four genotypes resistant (IC83798, IC95135, IC83506 and IC83500) and five moderately resistant (IC83736, IC 83526, IC83358, IC83501 and IC83514) genotypes.

- In the integrated disease management, seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg followed by soil drenching of mancozeb 50% + carbendazim 25% WS @ 3 g/l water to infected and surrounding plants was highly effective by recording least disease incidence of dry root rot (7.79%) with highest seed yield (12.26 q/ha), test weight (38.90 g) and highest benefit cost ratio (3.40).

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VII. REFERENCES

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**“STATUS, VARIABILITY AND MANAGEMENT OF DRY ROOT ROT OF
CHICKPEA CAUSED BY *Rhizoctonia bataticola* [(Taub.) Butler] WITH
CHANGING CLIMATE”**

DEEPA

2017

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ABSTRACT

Chickpea (*Cicer arietinum* L.) is one of the most important food legumes being cultivated in almost all over the world. Dry root rot disease caused by *Rhizoctonia bataticola* is recently becoming serious disease.

Symptom was yellowing and partial or complete wilting of the stem or one or more branches. In advanced stage of the disease, affected plants are became straw coloured, but in some cases the lower leaves and stems showed brown discolouration. In survey, Bhusnur village of Aland taluk recorded highest average dry root rot incidence of 45.55%. The least disease incidence was noticed in Aldahal village of Yadagiri taluk (0.50%). Cultural and morphological characters showed variation among *R. bataticola* isolates. Identification of 20 *R. bataticola* isolates by PCR produced amplified product size of 550- 600 bp. In studies on molecular diversity, Amplified products with all the primers have showed polymorphic and distinguishable banding pattern indicating the genetic diversity among all the isolates of *R. bataticola*. The maximum colony growth of *R. bataticola* and disease severity was observed at 30-35°C which is considered as optimum temperature. Maximum dry root rot and lesser root length, shoot length and total biomass were recorded at 40%, 50% and 60% moisture Elevated CO₂ @ 550 ± 25 ppm with 2°C rise in temperature showed higher disease severity and reduced growth parameters. The least disease severity and higher growth parameters were observed in open plot. Preliminary screening of 100 germplasm lines resulted in the identification of fourteen resistant and five moderately resistant germplasm lines. Advanced screening of 19 genotypes under phenotyping technique yielded only four genotypes resistant and five moderately resistant genotypes. In the integrated disease management, seed treatment with mancozeb 50% + carbendazim 25% WS @ 3.5 g/kg - soil drenching water to infected and surrounding plants was highly effective by recording least disease incidence of dry root rot (7.79%) with highest seed yield (12.26 q/ha), test weight (38.90 g) and highest benefit cost ratio (3.40).